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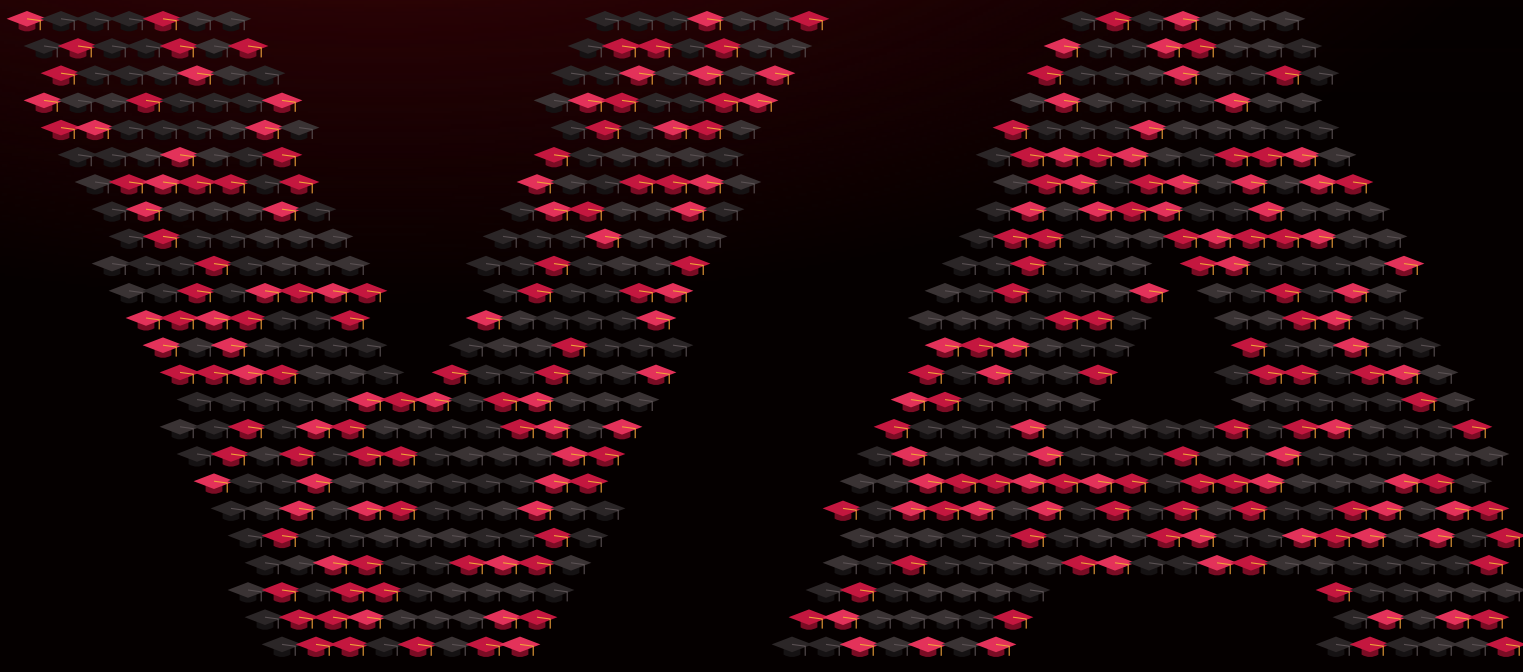
HAI Stanford University Human-Centered Artificial Intelligence

NOVEMBER 2026

The 2026 Virginia Talent Profile

Emerson Johnston
Amy Zegart

The bright caps, **39 in every 100**, mark the international share of Virginia's STEM master's degrees.



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Stanford University
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Artificial Intelligence

The 2026 Virginia Talent Profile

By Emerson Johnston and Amy Zegart

NOVEMBER 2026

A publication jointly produced by the Hoover Institution and the Institute for Human-Centered Artificial Intelligence at Stanford University.

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Data come from the Integrated Postsecondary Education Data System (IPEDS), the Department of Homeland Security's SEVIS by the Numbers, the National Assessment of Educational Progress, NSF's Survey of Graduate Students and Postdoctorates and Survey of Earned Doctorates, the Department of Labor's Office of Foreign Labor Certification, the USCIS H-1B Employer Data Hub, the Census Bureau's American Community Survey and USPTO patent data via PatentsView, processed in Python; every figure and table is generated from those sources, and each one names its source.

About This Profile

This profile is a companion to *The 2026 US Talent Report: Benchmarking America's STEM Pipeline Amid the Great Talent Competition*. It applies the report's measures to Virginia: the students the state develops at home, those it draws from abroad, and how both compare with the nation and with the states closest to it in size.

Degrees, enrollment, prices, outcomes and faculty come from IPEDS, the federal census of U.S. colleges and universities, harmonized across 1980–2025. Countries of citizenship come from SEVIS by the Numbers, the Department of Homeland Security's monthly count of active F-1 and M-1 students. Each figure names its source. Peer states are the four closest in total fall enrollment.

How to Cite this Profile

Emerson V. Johnston and Amy Zegart, “The 2026 Virginia Talent Profile,” Technology Policy Accelerator, Hoover Institution, and the Institute for Human-Centered Artificial Intelligence, Stanford University, Stanford, CA, November 2026.

Public Data and Tools

The data behind every chart come from federal sources: the Integrated Postsecondary Education Data System (IPEDS), published by the National Center for Education Statistics, as the backbone, together with NAEP, NSF's surveys of graduate students, doctorates and research funding, USAspending.gov, DHS and ICE SEVIS data, the Department of Labor, USCIS, the Census Bureau's American Community Survey and Post-Secondary Employment Outcomes, and USPTO PatentsView. Each figure names its source, and the Sources page lists all of them. An interactive version of this profile accompanies the report.

To share any charts and figures, proper attribution is required. This profile is licensed under Attribution-NoDerivatives 4.0 International (CC BY-ND 4.0). Please cite as: The 2026 Virginia Talent Profile.

AT A GLANCE

What Virginia's numbers say

1. **International students earn 39% of Virginia's STEM master's degrees,** 12.7 points below the U.S. (52%) in 2025, ranking 31st of 51 on dependence. The share is 41% for STEM research doctorates (U.S. 43%) and 4.0% for STEM bachelor's (U.S. 6.6%).
2. **Virginia awarded 2.07K STEM master's degrees to international students in 2025,** 1.8% of the U.S. total and 19th of 51 among states. Since 2015 the count changed 3.1×, against 2.2× nationally.
3. **The heaviest dependence is in computer & info sciences,** where international students earned 59% of master's degrees (2.52K degrees; U.S. 65%). One STEM field with at least 50 master's graduates is majority-international.
4. **Among states of similar size,** Virginia ranks 5th of 5 on the international share of STEM master's: Virginia 39%, North Carolina 49%, Georgia 44%, Ohio 58%, Pennsylvania 54%.
5. **75% of Virginia's first-time students in fall 2024 were state residents** and 2.0% came from abroad. More students arrived from other states (17.6K) than residents left (16.3K), a net gain of 1.3K; the top destination for those leaving was North Carolina.

39%

of the STEM master's degrees Virginia awards go to international students (U.S. 52%).

SCORECARD

Nine numbers against the nation

575K #11
Students enrolled, 2024
U.S. **19.7M** · +0.9% since 2015

2.07K #19
International STEM master's, 2025
U.S. **117K** · 3.09× since 2015

39% #31
International share of STEM master's, 2025
U.S. **52%** · +16.4 pp since 2015

8.4% #42
International share of grad students, 2024
U.S. **17%** · +0.7 pp since 2015

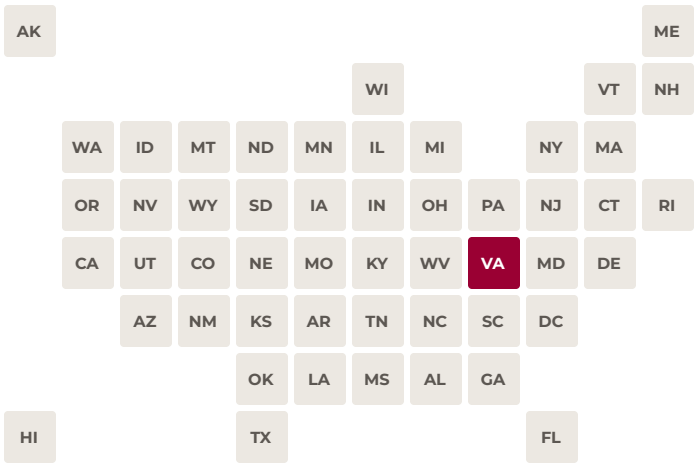
24% #21
STEM share of bachelor's, 2025
U.S. **24%** · +3.7 pp since 2015

68% #11
Six-year graduation rate, 2024
U.S. **65%** · +2.4 pp since 2015

$11.3K #22
Appropriations per student, 2024
U.S. **$11.4K** · +51% since 2015

75% #20
First-time students from in state, 2024
−1.5 pp since 2016

$106K #16
Average faculty salary, 2024
U.S. **$106K** · +1.3% since 2016



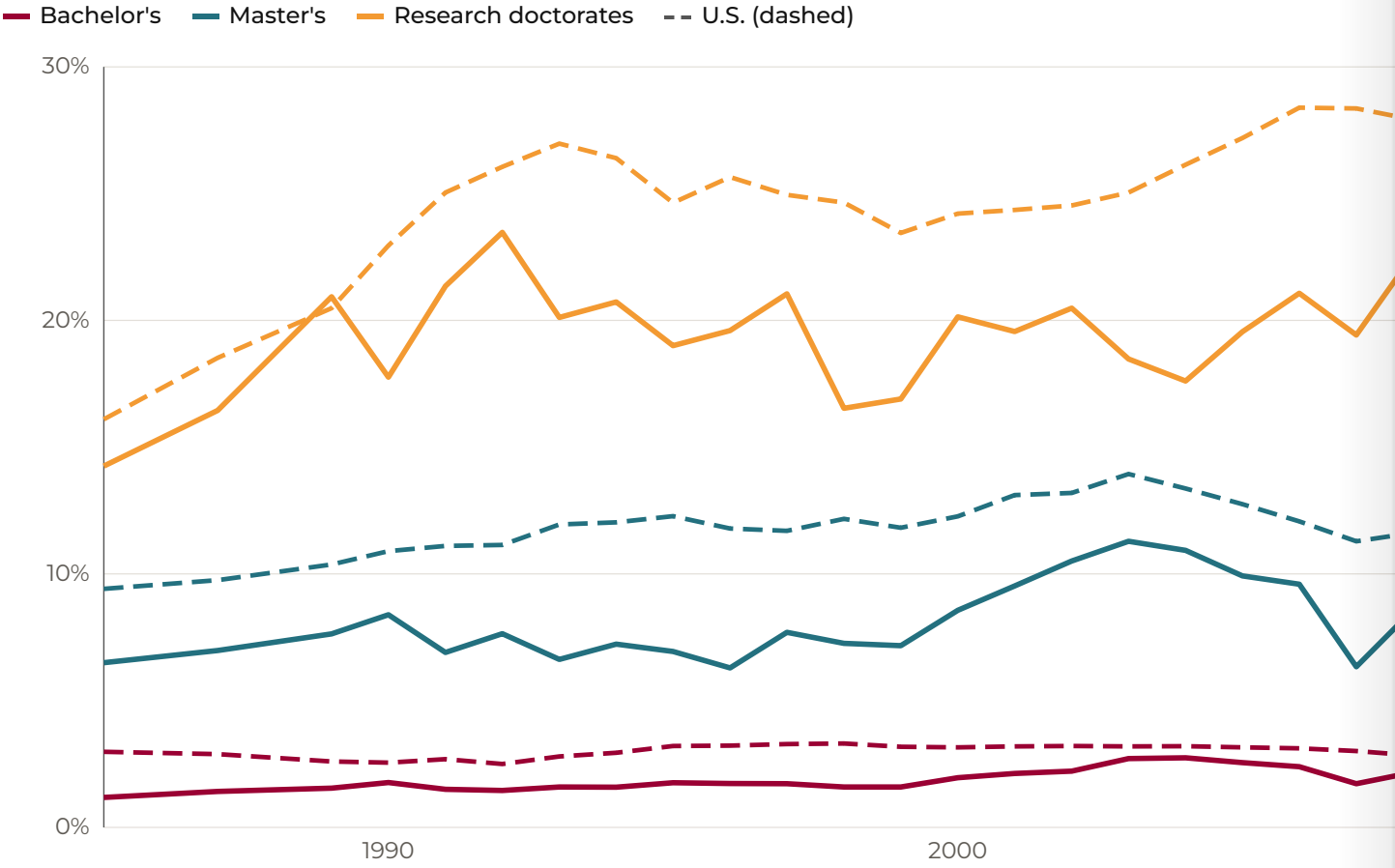
Badges give Virginia's rank among the 50 states and DC (1 = highest). Changes are since 2015.

SECTION 1

International Dependence

Virginia ranks 42nd of 51 on the international share of graduate enrollment, at 8.4% in 2024 against 17% for the U.S.

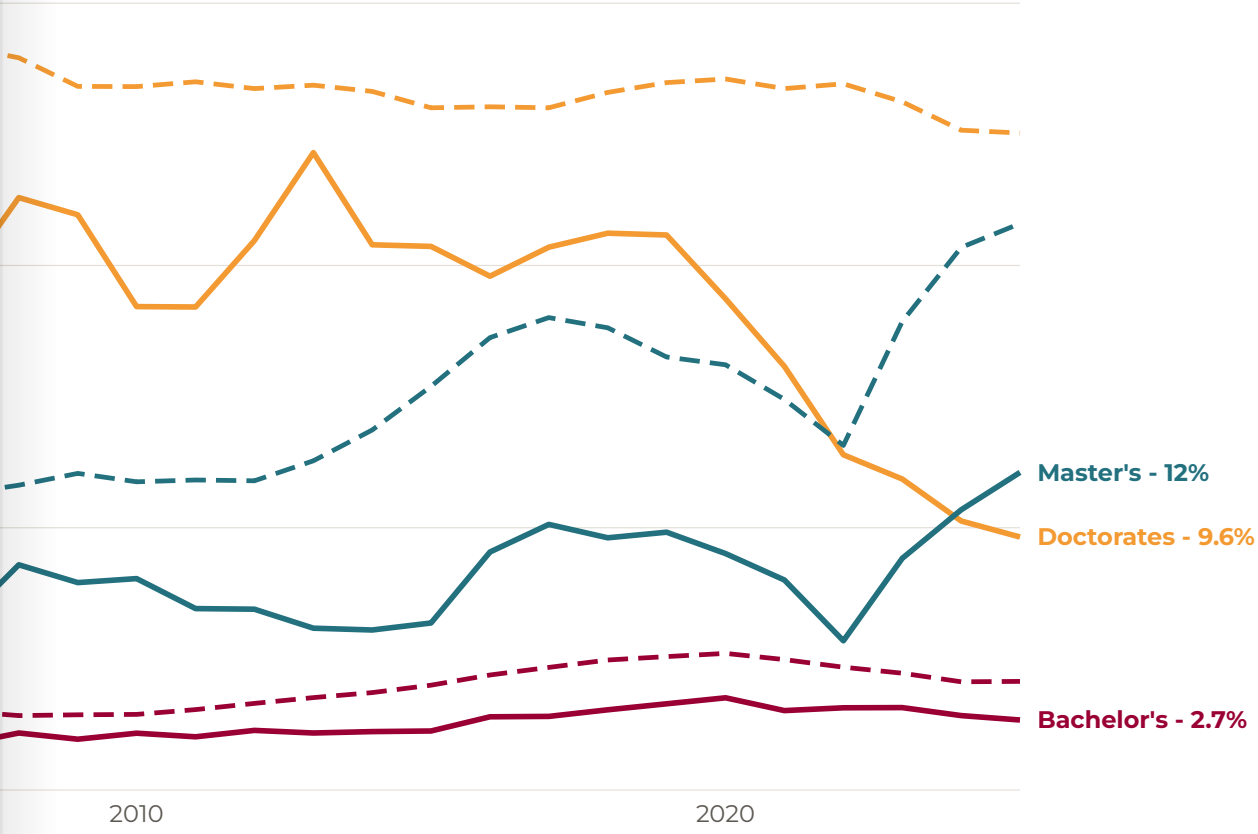
Figure 1 What Share of Virginia's Degrees Go to International Students?



International share of degrees awarded, all fields; state solid, U.S. dashed. Data from NCES, IPEDS Completions, 1985–2025

International students make up 8.4% of Virginia's graduate enrollment and 2.7% of its undergraduates. Dependence climbs with each degree: 2.7% of bachelor's, 12% of master's and 9.6% of research doctorates go to international students.

In STEM the shares are higher still: 4.0% of STEM bachelor's, 39% of STEM master's and 41% of STEM doctorates. The state awards 1.8% of all U.S. international STEM master's, 19th of 51.



5.

8.4%

Int. share of grad students, 2024

U.S. **17%** · **42nd** of 51 · +0.7 pp since 2015

12%

Int. share of all master's, 2025

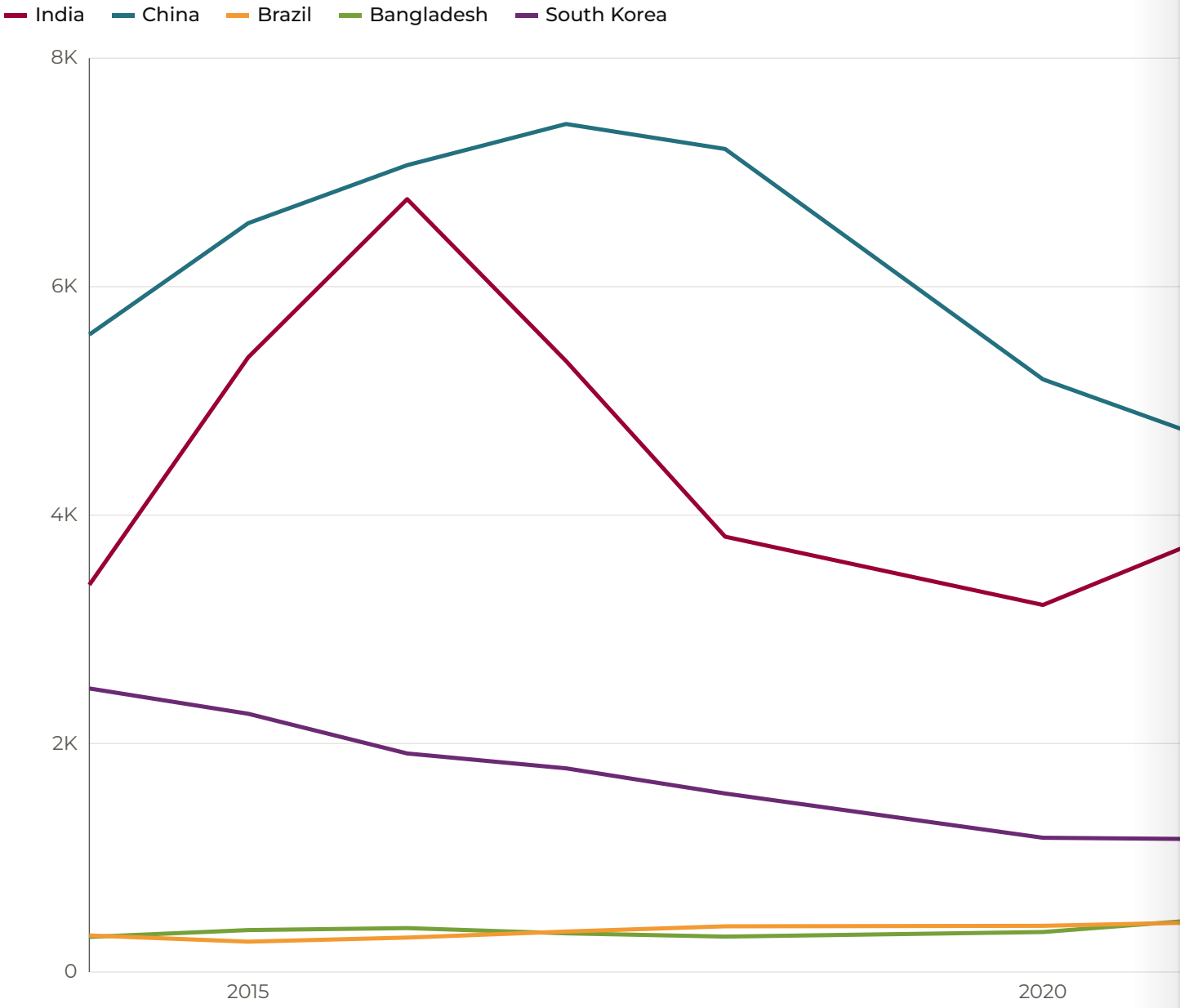
U.S. **22%** · **32nd** of 51 · +5.7 pp since 2015

39%

Int. share of STEM master's, 2025

U.S. **52%** · **31st** of 51 · +16.4 pp since 2015

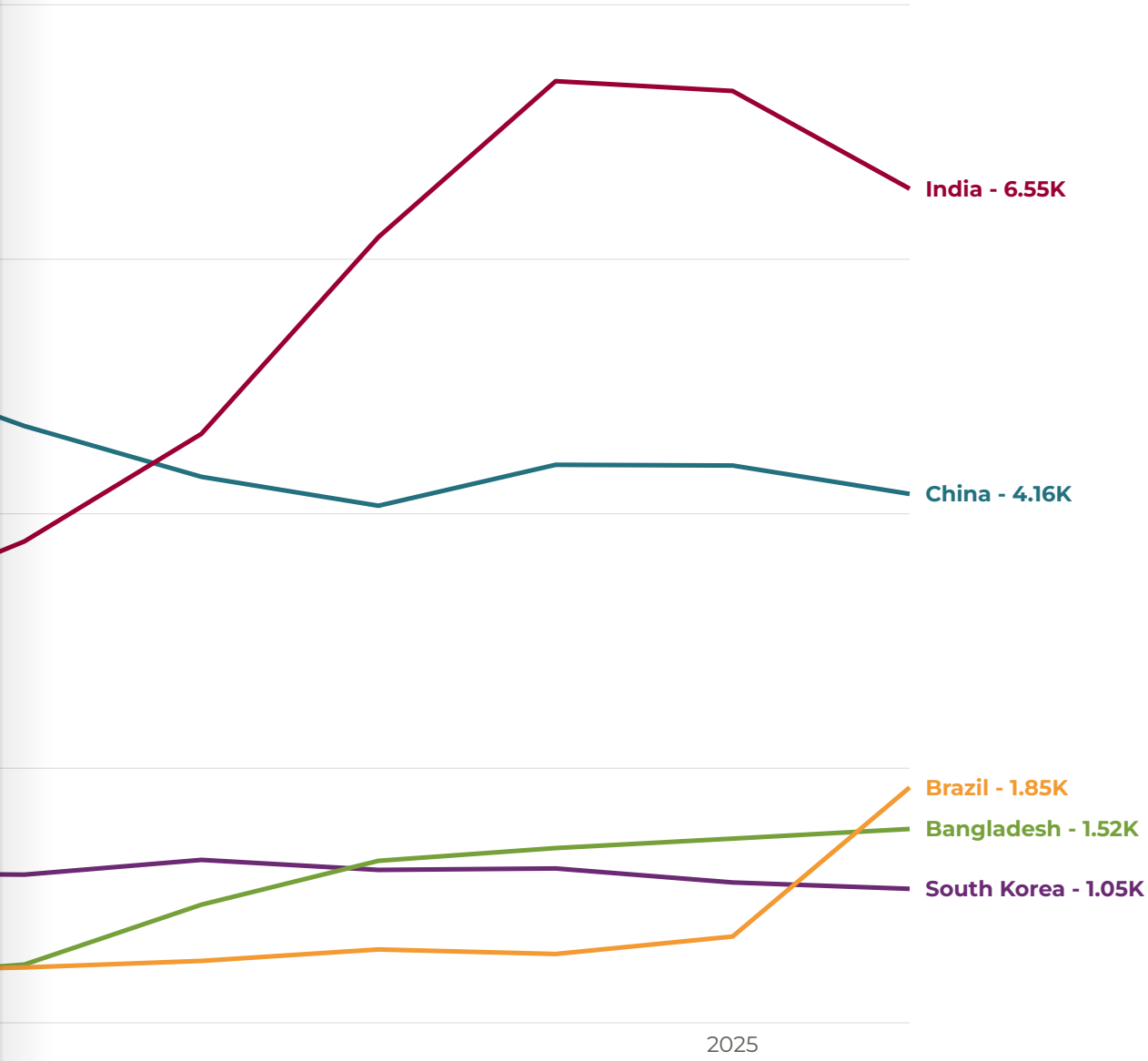
Figure 2 Where Do Virginia's International Students Come From?



Active F-1 and M-1 students by country of citizenship, the five largest in October 2026; one fall snapshot per year (the mor

In October 2026, Virginia had 29.8K active international students in SEVIS, 13th of 51. India accounted for 22% of them and China for 14%, against 27% and 18% nationally.

Students came from 182 countries, but the two largest supplied 36%. The state's total is up 13% since 2015, when it stood at 26.4K.



nth closest to November), October 2014–October 2026. All education levels. Data from DHS, SEVIS by the Numbers.

29.8K
Active international students, October 2026
13th of 51 · U.S. **1.19M**

22%
From India
U.S. **27%**

66%
In STEM fields
U.S. **63%**

SECTION 1 · INTERNATIONAL DEPENDENCE

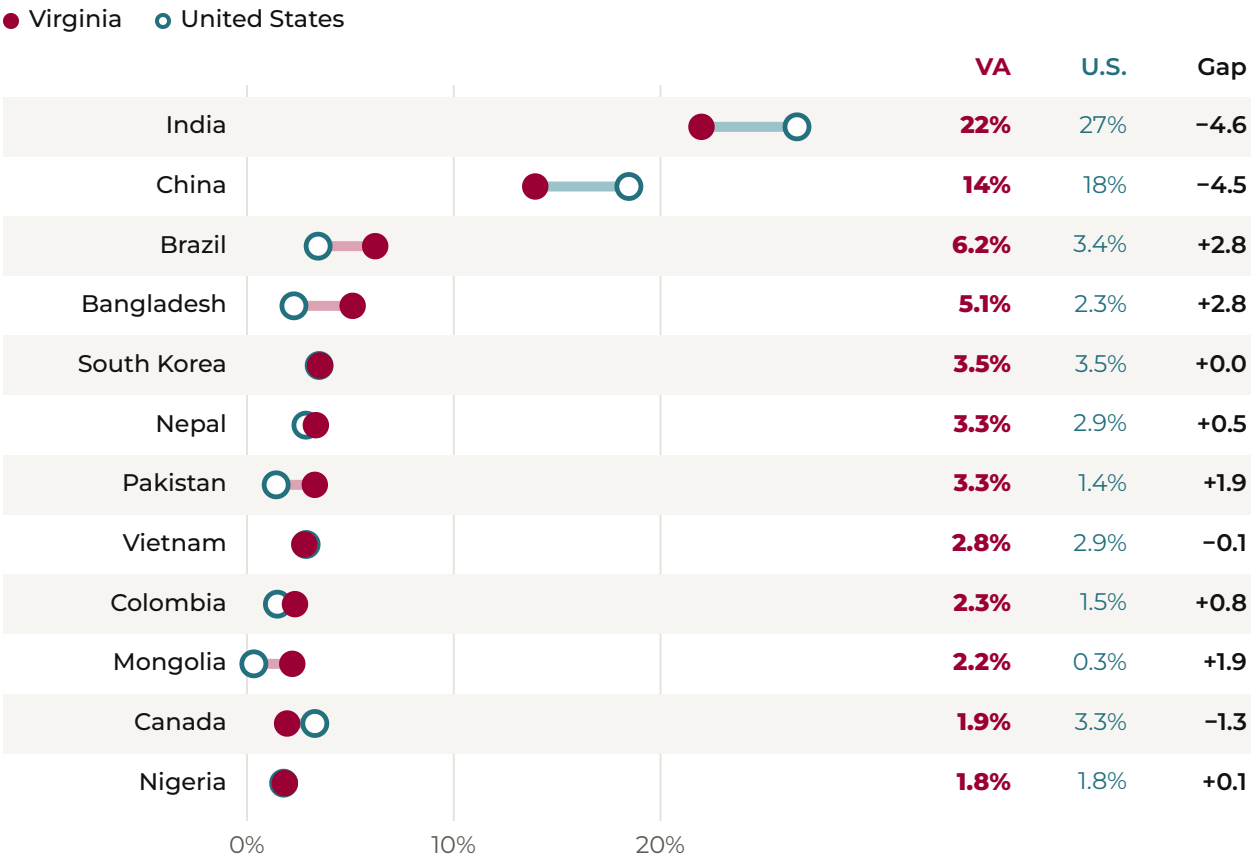
How Does Virginia's Mix of Countries Compare With the Nation's?

In October 2026, Virginia had **29.8K** active F-1 and M-1 students, 13th among the 50 states and DC and **2.5%** of the U.S. total. They came from 182 countries.

India supplied **22%** of them, against 27% nationally, and China **14%**, against 18%. Brazil (6.2%), Bangladesh (5.1%), and South Korea (3.5%) followed.

Together the two largest countries supplied **36%** of the state's international students (U.S. 45%), 29th among the states, against 45% in 2015.

Figure 3 Is Virginia's Mix of Countries Like the Nation's?



Share of active F-1 and M-1 students, all levels. Data from DHS, SEVIS by the Numbers, October 2026.

29.8K
Active international students, October 2026
13th of 51 · U.S. **1.19M**

22%
From India
U.S. **27%**

14%
From China
U.S. **18%**

182
Countries represented
U.S. **219**

The state's count is up 1.1% from 2025 and up 13% since 2015, when it stood at 26.4K. Its share of all U.S. international students is up 0.3 points since 2015, to **2.5%**.

Since 2015, the number of students from India went from 5.38K to 6.55K, China's from 6.55K to 4.16K, and Brazil's from 266 to 1.85K.

Over the same years, the number from Bangladesh went from 368 to 1.52K and South Korea's from 2.26K to 1.05K. The table gives each country's count for the first, previous and latest years.

Table 1 International Students in Virginia, 2015–2026

	2015	2025	2026	Change since 2015	U.S. 2026	Rank
ALL INTERNATIONAL STUDENTS						
Active F-1 and M-1 students	26.4K	29.5K	**29.8K**	+13%	1.19M	13th
Share of U.S. active F-1/M-1 students	2.2%	2.3%	**2.5%**	+0.3 pp	–	13th
Countries of citizenship represented	180	177	**182**	+1.1%	219	11th
Share from the two largest countries of origin	45%	40%	**36%**	−9.3 pp	45%	29th
LARGEST COUNTRIES OF CITIZENSHIP (STUDENTS)					**U.S. SHARE**	**STATE SHARE**
India	5.38K	7.32K	**6.55K**	+22%	27%	22%
China	6.55K	4.38K	**4.16K**	−37%	18%	14%
Brazil	266	678	**1.85K**	+595%	3.4%	6.2%
Bangladesh	368	1.45K	**1.52K**	+314%	2.3%	5.1%
South Korea	2.26K	1.1K	**1.05K**	−53%	3.5%	3.5%

Active F-1 and M-1 students, one fall snapshot per year (the month closest to November; 2019 has none; the latest year is October 2026). Rank among the 50 states and DC. Data from DHS, SEVIS by the Numbers.

Figure 4

Which Countries Supply Virginia's International STEM Students?

19.6K of Virginia's international students were in STEM fields in October 2026, **66%** of the total against 63% nationally, up 81% since 2015.

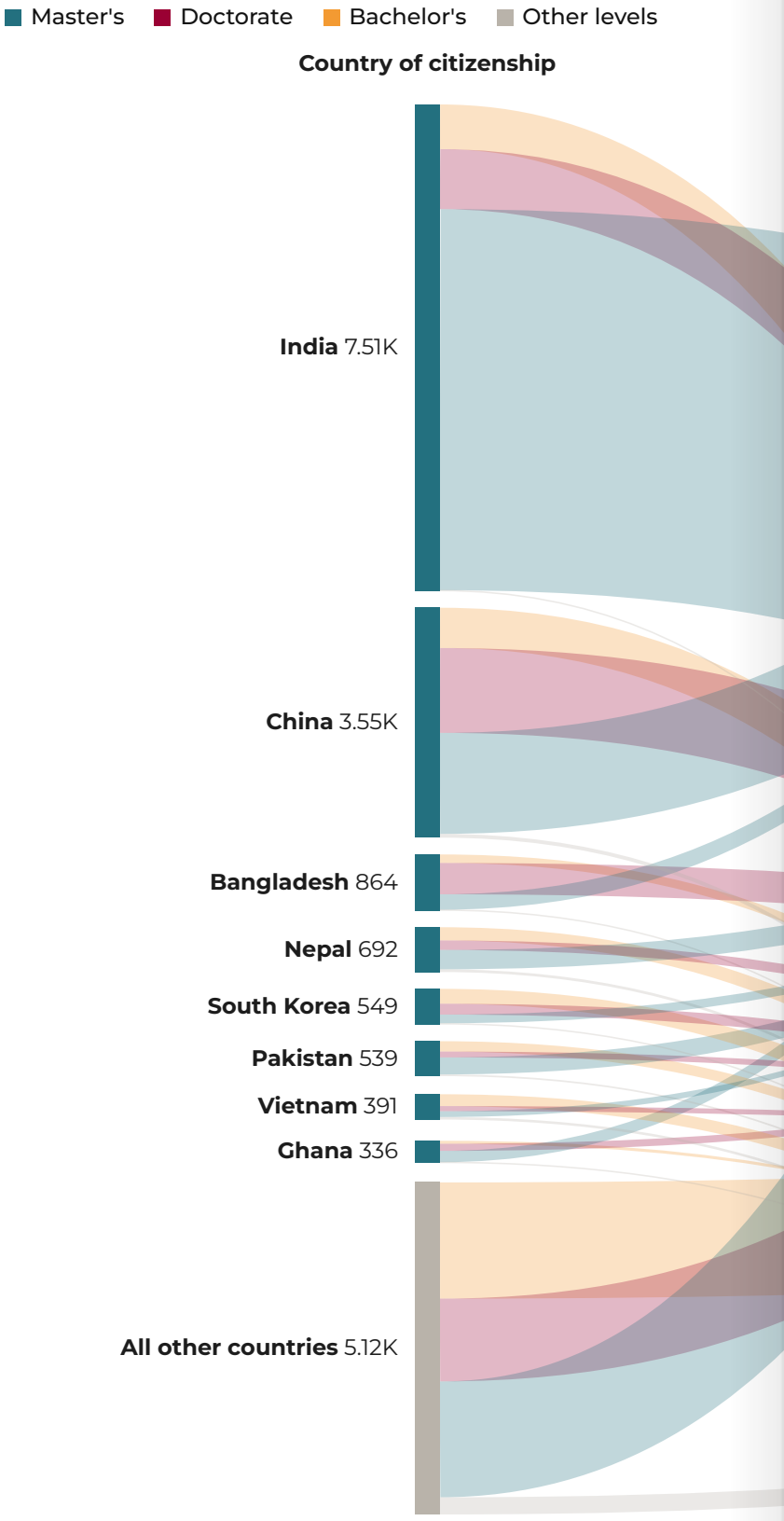
Master's students were the largest group, **10.4K**. India supplied 56% of them and China 15%.

STEM doctoral students went from 2.57K in 2015 to **4.59K**, and STEM master's students from 4.99K to 10.4K.

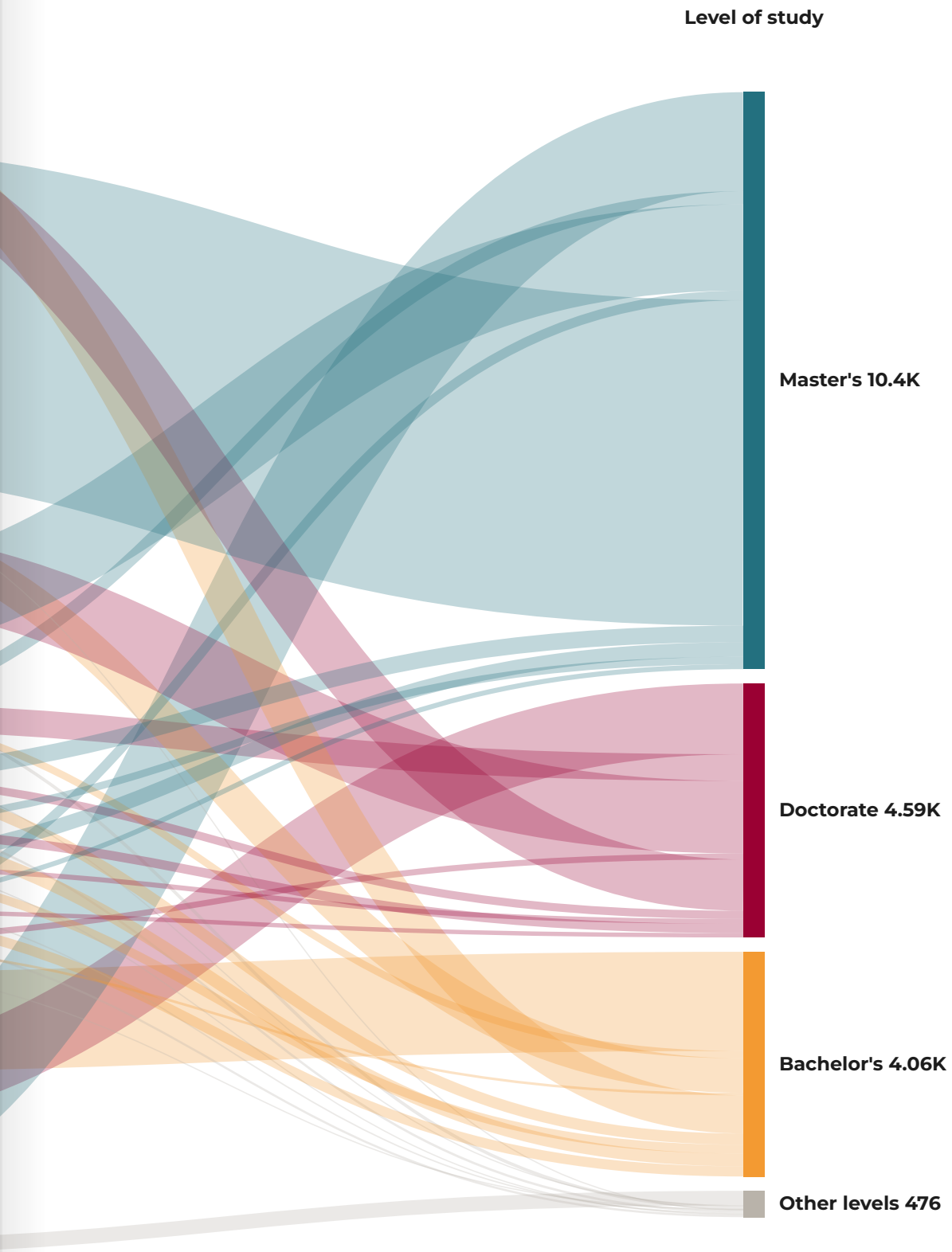
19.6K
Active STEM F-1/M-1 students, 2026
U.S. **749K** · **13th** of 51 · +81% since 2015

66%
Share in STEM fields
U.S. **63%**

10.4K
Active STEM master's students, 2026
U.S. **393K** · **12th** of 51 · 2.09× since 2015



Active F-1 and M-1 students in STEM fields by country of citizenship an
basis as its all-student pages, so the STEM count here exceeds the all-



d level of study, October 2026; the eight largest countries are shown. DHS does not draw its STEM pages on the same evel count for India (7.51K against 6.55K). Data from DHS, SEVIS by the Numbers, October 2026.

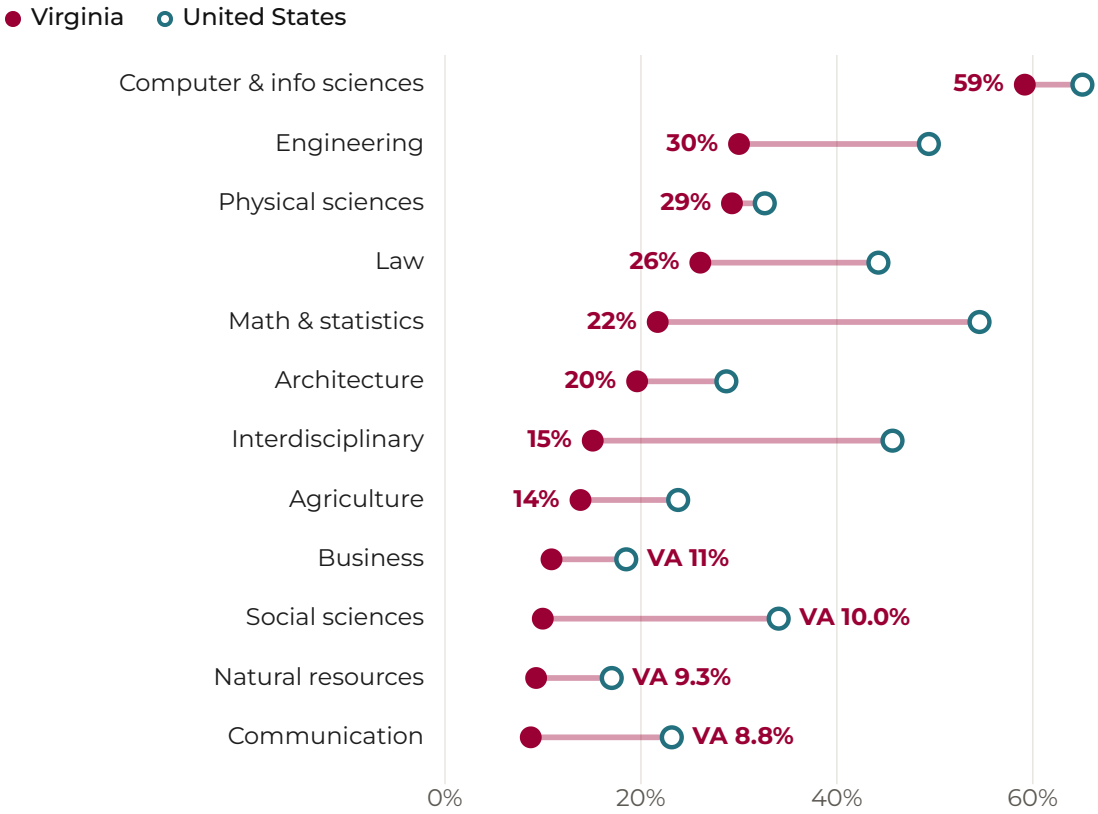
SECTION 1 · INTERNATIONAL DEPENDENCE

How International Are Virginia's Master's Fields?

Each row is a master's field with at least 30 graduates in Virginia in 2025, ranked by the share of degrees earned by international students (U.S. nonresidents on temporary visas). The filled dot is the state's share; the open dot is the national share for the same field.

Highest international shares. International students earned the largest shares of master's degrees in computer & info sciences (59%, against 65% for the U.S.), engineering (30%, against 49% for the U.S.), and physical sciences (29%, against 33% for the U.S.).

Figure 5 Which Master's Fields Depend Most on International Students?



International share of master's degrees by field, 2025; fields with 30+ graduates. Data from NCES, IPEDS Completions.

12%
Int. share of all master's, 2025
U.S. **22%** · **32nd** of 51

39%
Int. share of STEM master's, 2025
U.S. **52%** · **31st** of 51

59%
Int. share of CS master's, 2025
U.S. **65%** · **29th** of 47

The table gives the international share of degrees in each STEM field at three levels: Virginia first, the United States after the slash. The master's column is highlighted. A dash means the state awarded no degrees in the field at that level.

Highest international share among STEM master's fields. International students earned 59% of the master's degrees in computer & info sciences, against 65% for the U.S.

Fields with an international majority. International students earned more than half of the master's degrees in one STEM field.

Table 2 How International Is Each STEM Field?

Field	Bachelor's	Master's	Doctorates
Computer & info sciences	5.0% / 8.6%	**59% / 65%**	56% / 56%
Engineering	4.3% / 7.2%	**30% / 49%**	49% / 55%
Engineering technology	1.5% / 4.5%	**2.5% / 54%**	– / 42%
Math & statistics	10% / 17%	**22% / 55%**	51% / 53%
Physical sciences	2.0% / 6.0%	**29% / 33%**	33% / 38%
Biological sciences	2.4% / 3.4%	**6.8% / 23%**	18% / 23%
Natural resources	0.8% / 2.0%	**9.3% / 17%**	28% / 27%
Architecture	7.3% / 7.3%	**20% / 29%**	– / 55%

International share in Virginia / U.S., 2025. Data from NCES, IPEDS Completions.

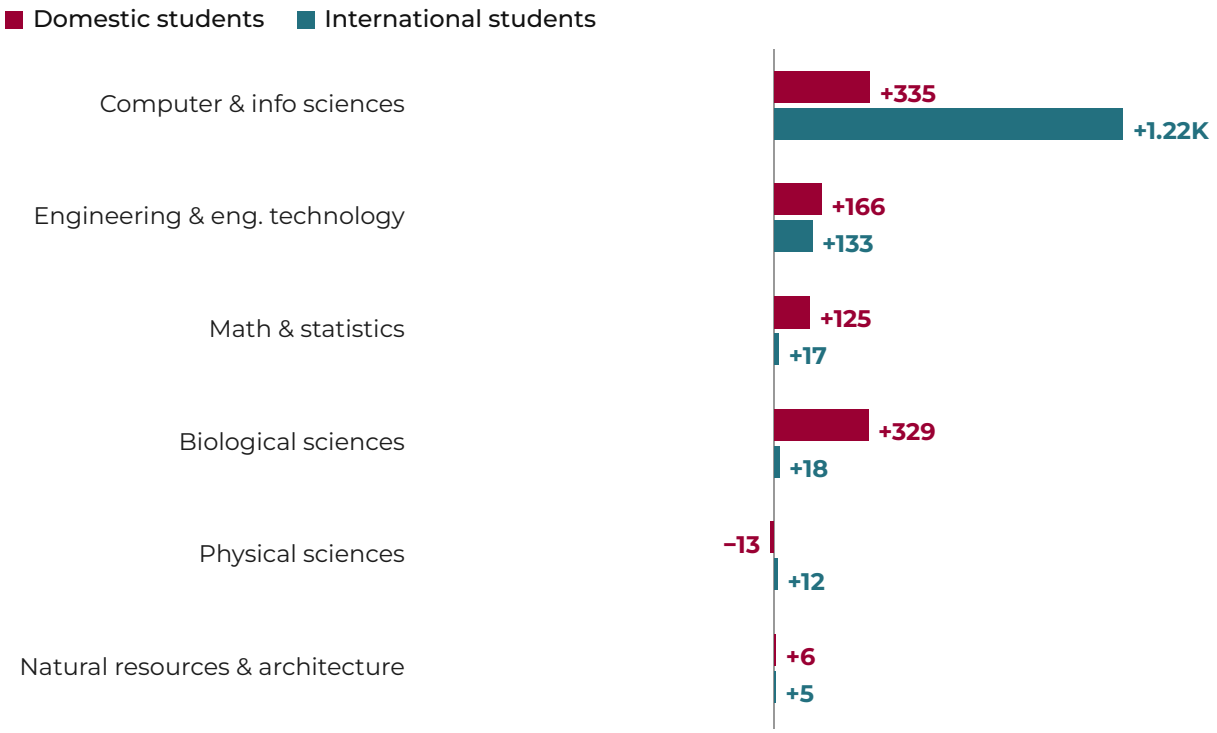
SECTION 1 · INTERNATIONAL DEPENDENCE

How Has Virginia's STEM Master's Output Grown?

Each pair of bars is the change in the number of STEM master's degrees awarded in Virginia between 2015 and 2025 in one field: to domestic students (U.S. citizens and permanent residents) and to international students. Bars to the left of the axis are declines.

STEM master's degrees. The state awarded 5.27K in 2025 and 2.92K in 2015, a change of +1.4K for international students and +948 for domestic students.

Figure 6 Where Did the STEM Master's Growth Come From?



Change in STEM master's degrees by field, 2015–2025. Data from NCES, IPEDS Completions.

2.07K
International STEM master's, 2025
U.S. **117K** · **19th** of 51

19%
STEM share of master's, 2025
U.S. **23%** · **29th** of 51

1.8%
Share of U.S. international STEM master's, 2025
19th of 51

5.7
STEM master's per STEM research doctorate, 2025
U.S. **6.4** · **24th** of 51

The table gives the number of STEM master's degrees awarded in Virginia in each field in 2015 and 2025, the change between the two years, the number awarded to international students in 2025, and international students' share of the field's degrees in each year.

STEM share of master's degrees. The share was 19% in 2025, against 23% for the U.S., 29th among the states, up 2.0 points from 17% in 2024 and up 6.5 points since 2015, when it was 12%.

Table 3 How Many STEM Master's Degrees Does Each Field Award?

Field	2015	2025	Change	International 2025	Int. share 2015	Int. share 2025
Computer & info sciences	968	**2.52K**	+160%	1.49K	28%	59%
Engineering & eng. technology	1.09K	**1.39K**	+27%	396	24%	28%
Math & statistics	116	**258**	+122%	56	34%	22%
Biological sciences	284	**631**	+122%	43	8.8%	6.8%
Physical sciences	141	**140**	−0.7%	41	21%	29%
Natural resources & architecture	314	**325**	+3.5%	46	13%	14%

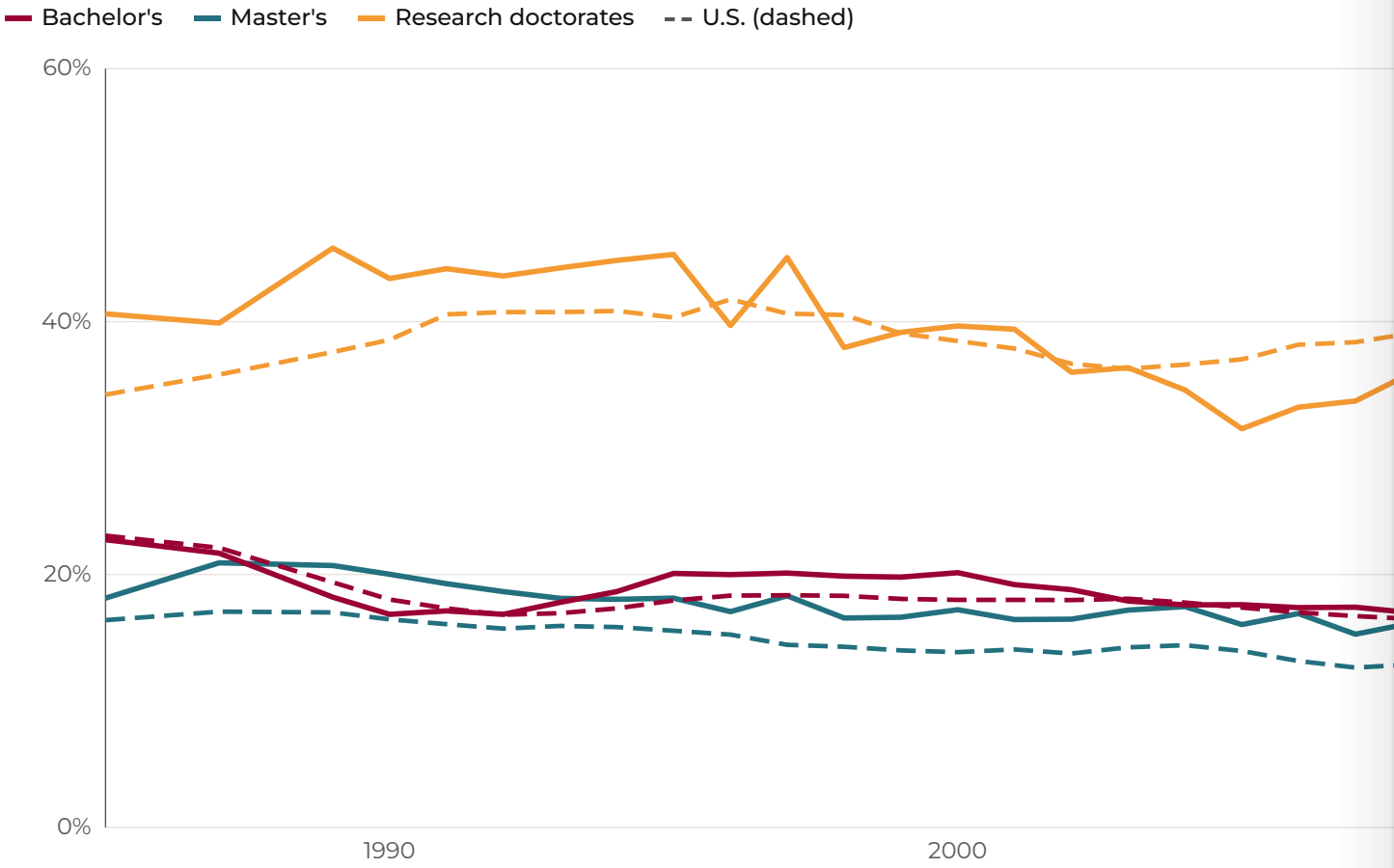
STEM master's degrees awarded in Virginia, all students and international students, 2015 and 2025. Data from NCES, IPEDS Completions.

SECTION 2

STEM Degrees

Virginia ranks 50th of 51 on the STEM share of research doctorates, at 15% in 2025 against 39% for the U.S.

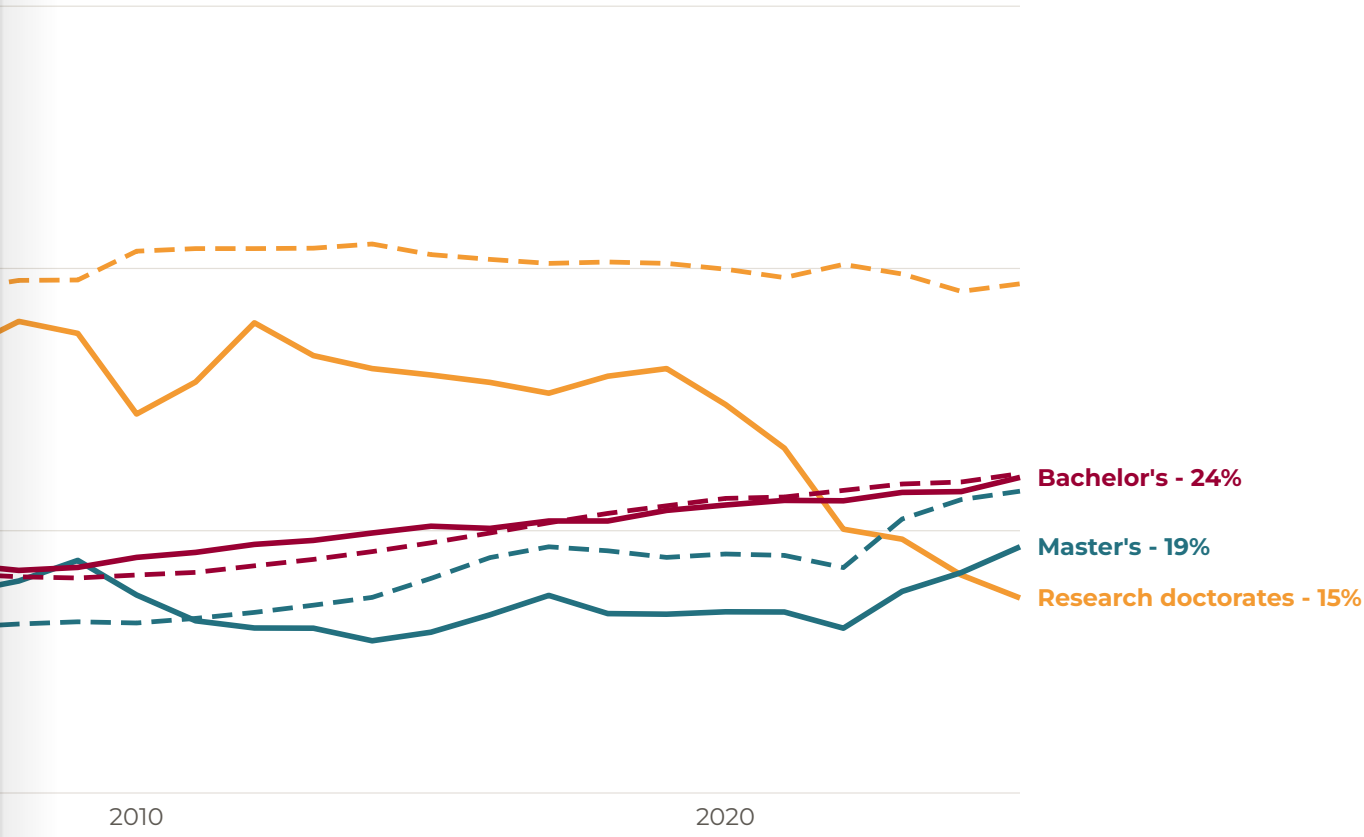
Figure 7 How Much of Virginia's Degree Output Is STEM?



STEM share of degrees by level; state solid, U.S. dashed. Data from NCES, IPEDS Completions.

STEM fields account for 24% of Virginia's bachelor's degrees, 19% of its master's and 15% of its research doctorates (U.S. 24%, 23% and 39%).

The STEM share of master's degrees is up 6.5 points since 2015, and the state ranks 29th of 51 on it. On the STEM share of bachelor's degrees it ranks 21st of 51.



24%

STEM share of bachelor's, 2025

U.S. **24%** · **21st** of 51 · +3.7 pp since 2015

19%

STEM share of master's, 2025

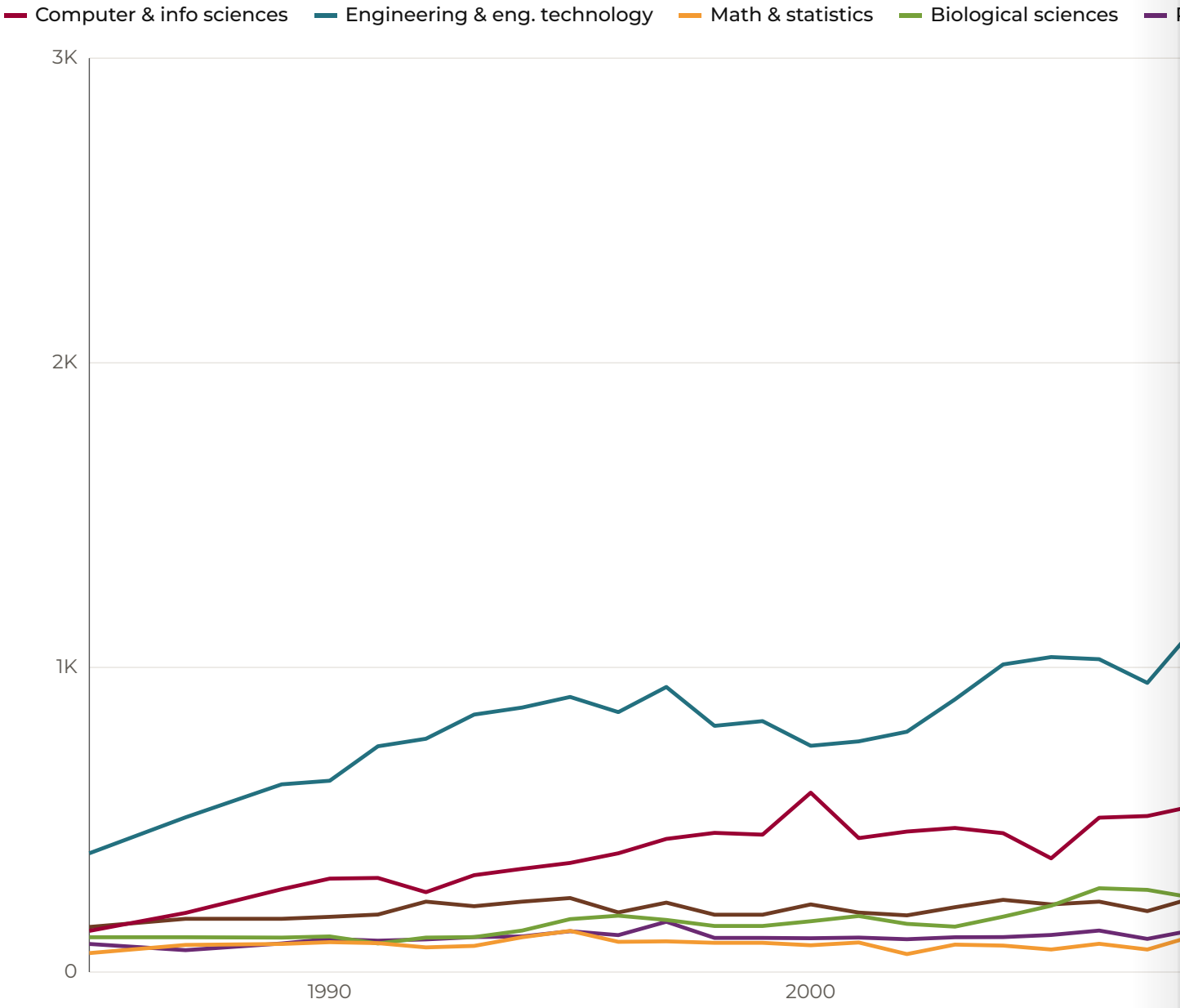
U.S. **23%** · **29th** of 51 · +6.5 pp since 2015

15%

STEM share of research doctorates, 2025

U.S. **39%** · **50th** of 51 · −17.0 pp since 2015

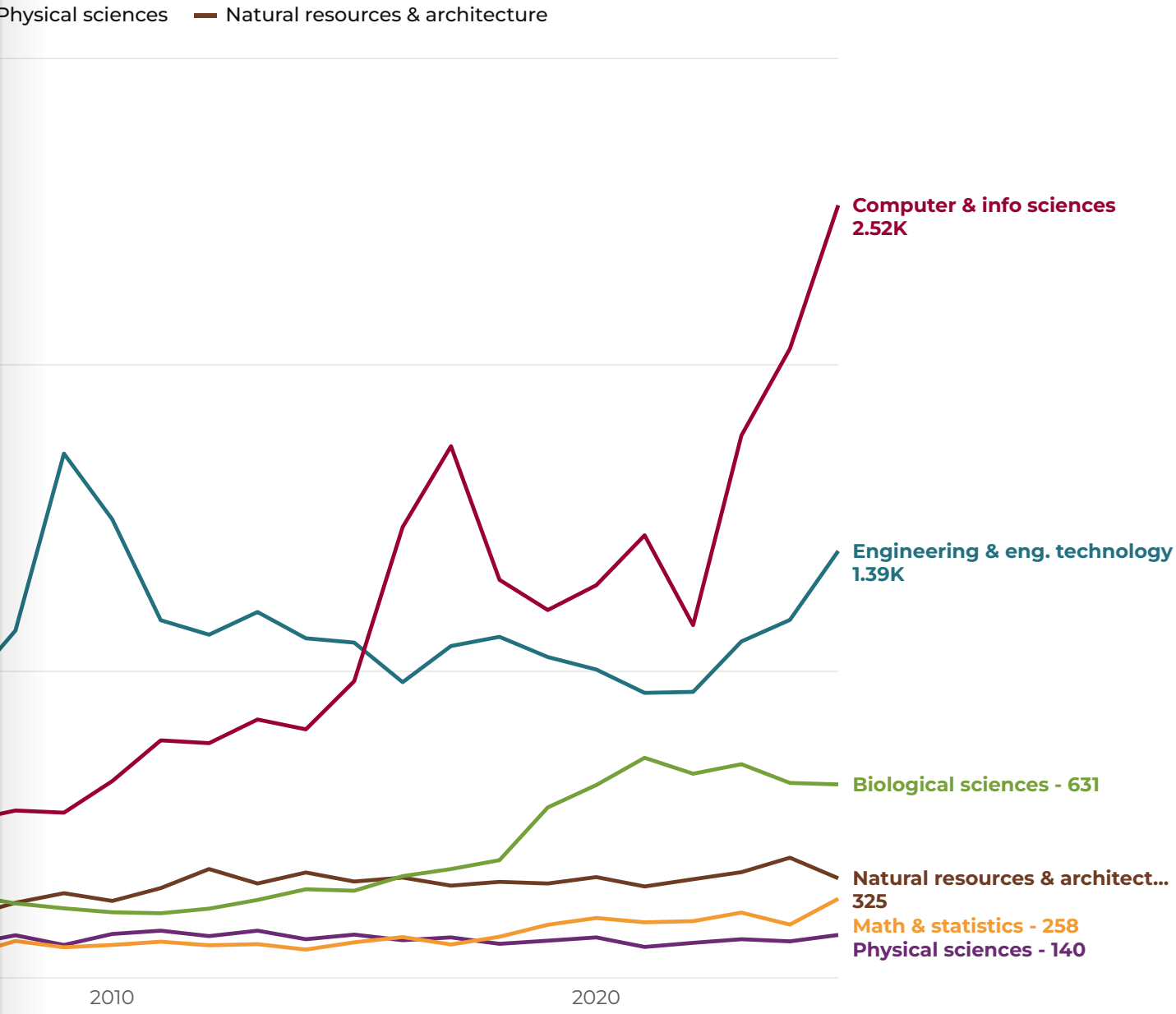
Figure 8 Which STEM Fields Produce Virginia's Master's Degrees?



STEM master's degrees by field, all students. Data from NCES, IPEDS Completions.

Computer & info sciences produced the most STEM master's degrees in 2025, 2.52K, followed by engineering & eng. technology (1.39K) and biological sciences (631). Master's degrees in computer & info sciences are up 160% since 2015.

International students earned 2.07K of the state's STEM master's degrees in 2025, 1.8% of the national total (19th of 51).



2.07K

International STEM master's, 2025

U.S. **117K** · **19th** of 51 · 3.09× since 2015

1.8%

Share of U.S. international STEM master's, 2025

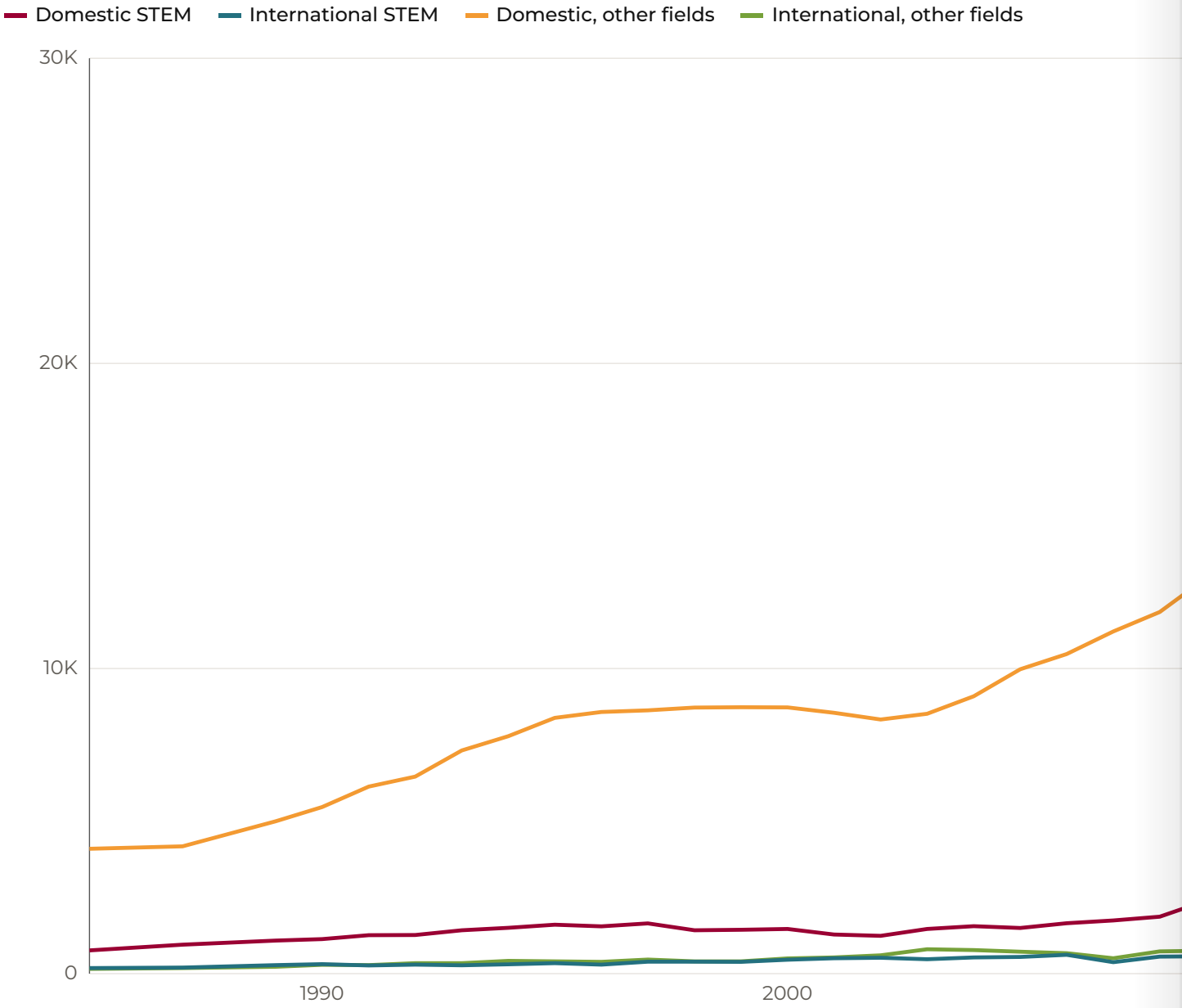
19th of 51 · +0.5 pp since 2015

59%

Int. share of CS master's, 2025

U.S. **65%** · **29th** of 47 · +30.9 pp since 2015

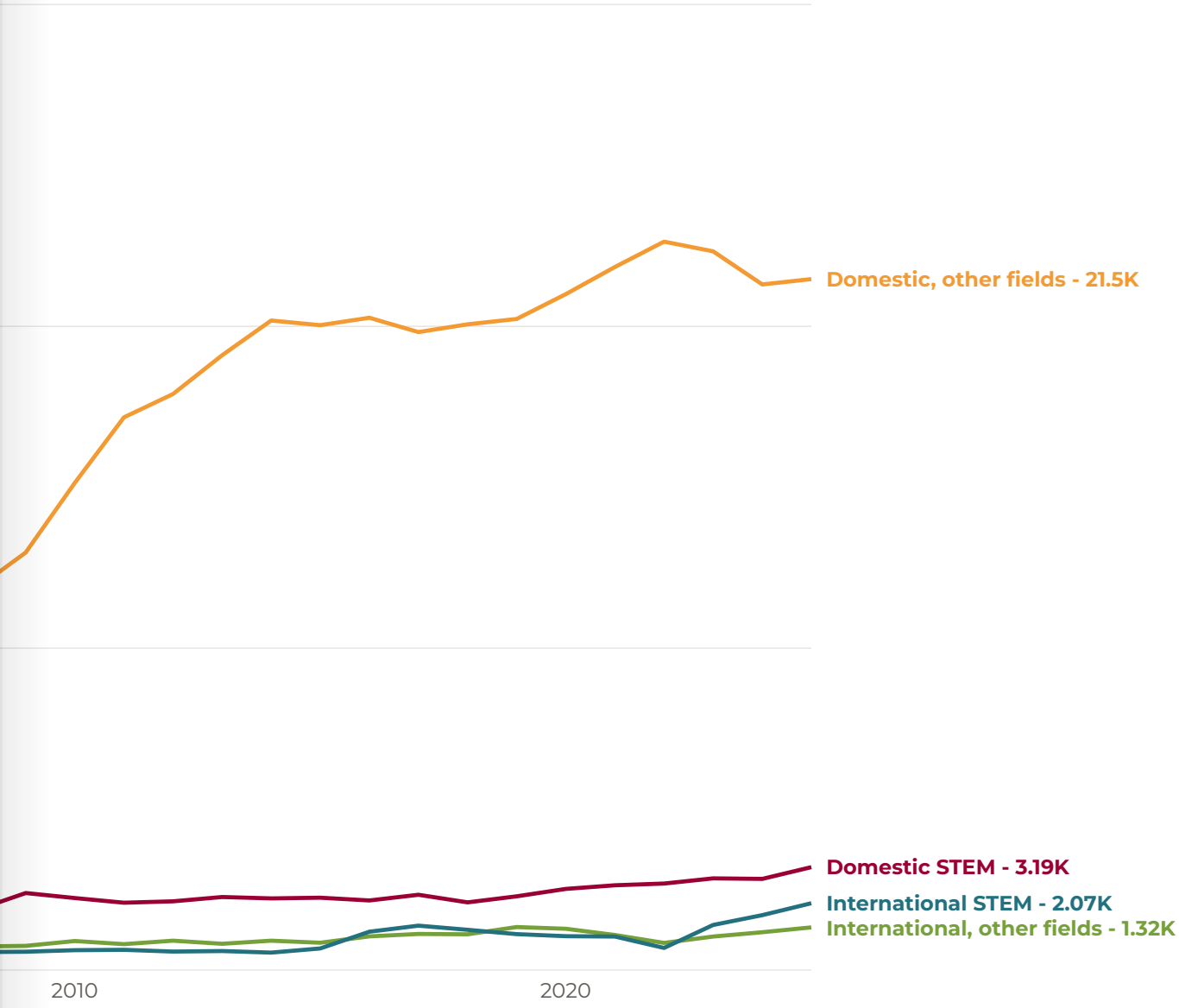
Figure 9 Who Earns Virginia's Master's Degrees?



Master's degrees by student residency and field group. Data from NCES, IPEDS Completions.

Between 2015 and 2025, 1.4K more STEM master's degrees went to international students and 948 more to domestic students.

International students earn 39% of Virginia's STEM master's but 5.8% of master's in other fields (U.S. 52% and 12%).



12%

Int. share of all master's, 2025

U.S. **22%** · **32nd** of 51 · +5.7 pp since 2015

39%

Int. share of STEM master's, 2025

U.S. **52%** · **31st** of 51 · +16.4 pp since 2015

5.7

STEM master's per STEM research doctorate (IPEDS), 2025

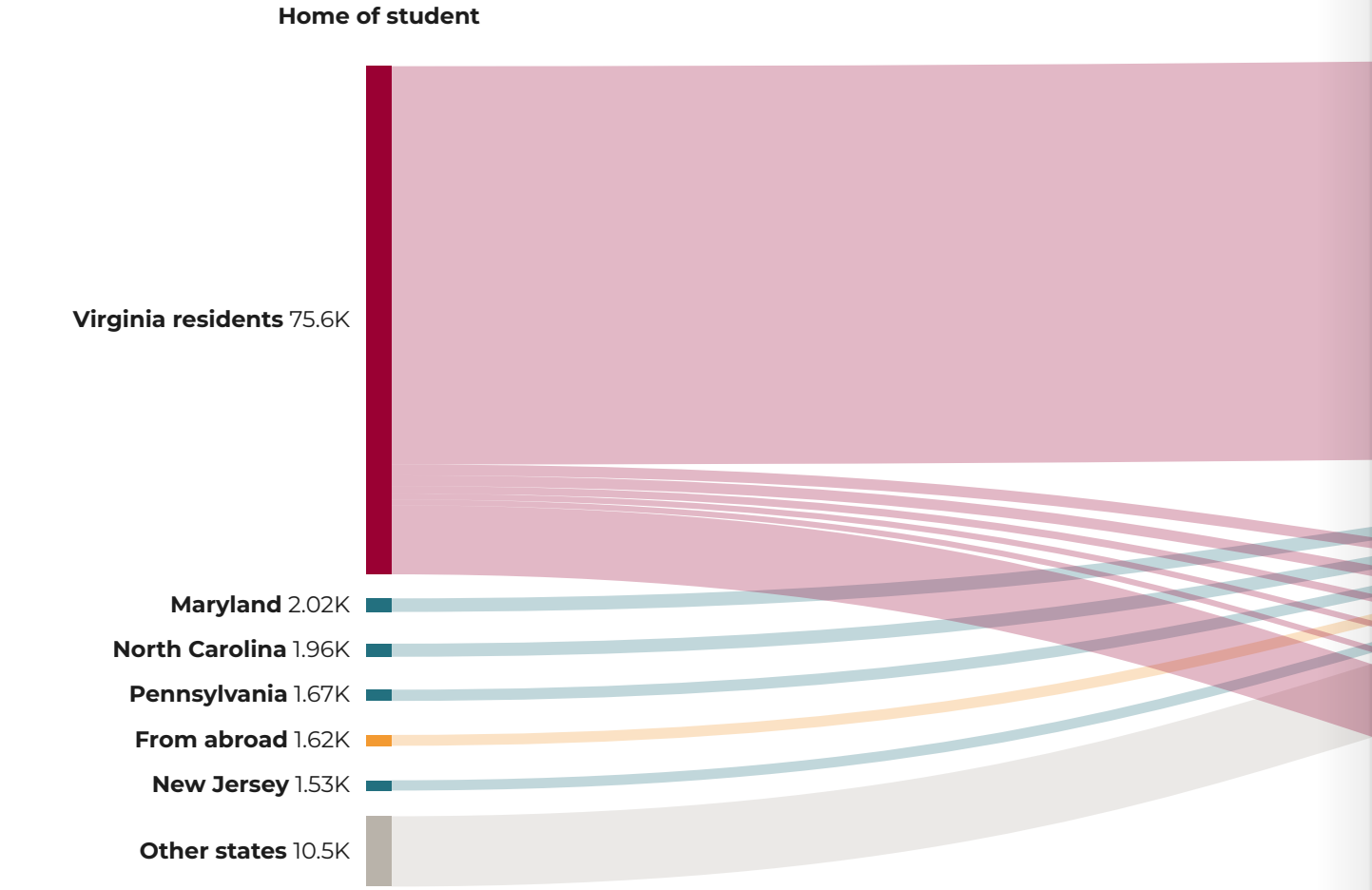
U.S. **6.4** · **24th** of 51 · +1.8 since 2015

SECTION 3

Where Students Come From

Virginia ranks 35th of 51 on the share of first-time students whose home is outside the US, at 2.0% in 2024 against 3.1% for the U.S.

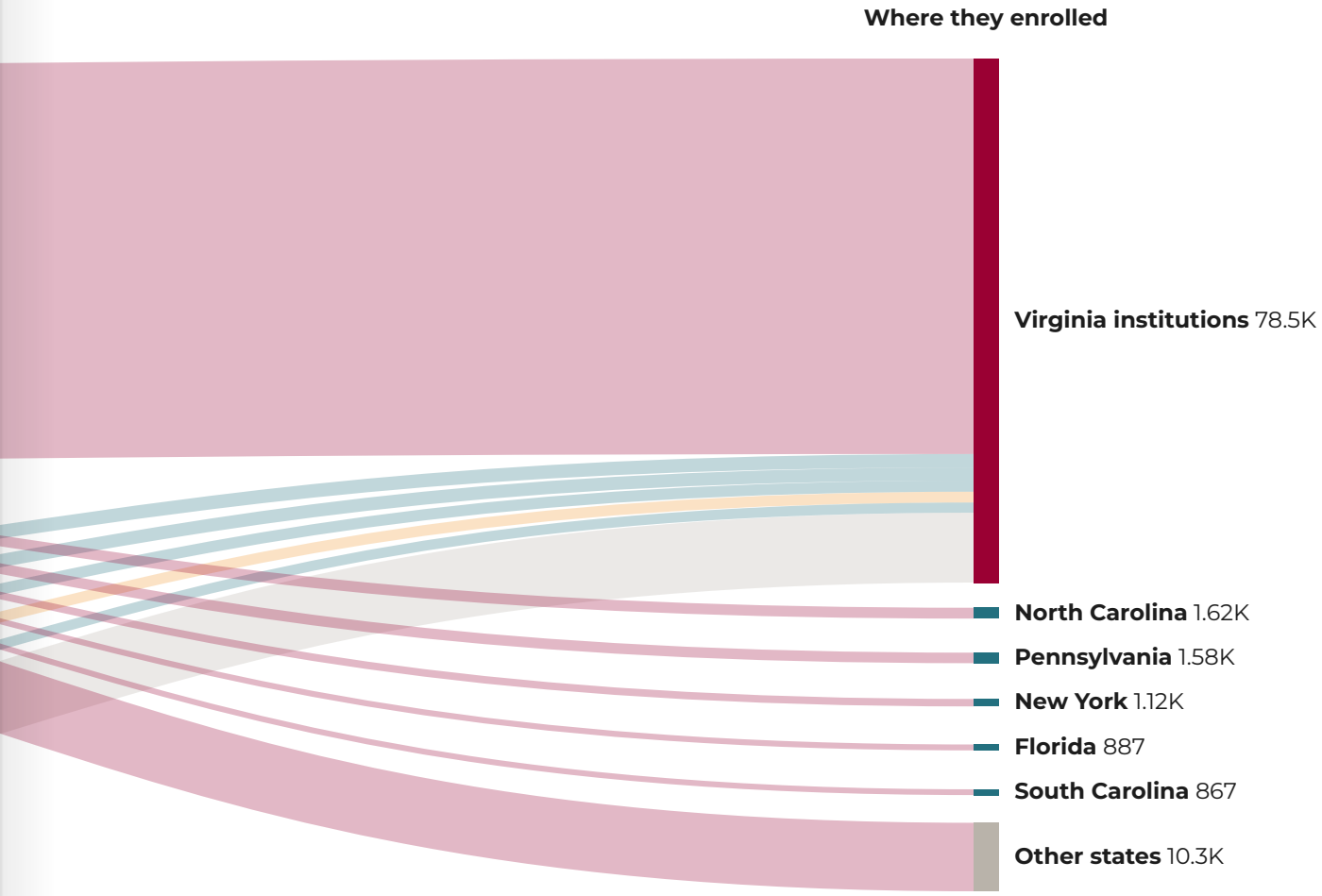
Figure 10 First-Time Students by Home and Destination, Fall 2024



Home of first-time degree-seeking undergraduates (left) and the state of the institution entered (right); the five largest s
Data from NCES, IPEDS Fall Enrollment.

59.2K residents stayed in Virginia for college in fall 2024; 17.6K students arrived from other states and 1.62K from abroad, while 16.3K residents left.

Each band runs from a student's home on the left to the state where they enrolled on the right. 75% of the state's first-time students are residents.



ources and destinations (more when states tie at the printed figure), the rest grouped; ties are ordered by exact count.

75%

First-time students from in state, 2024

20th of 51 · −1.5 pp since 2016

2.0%

First-time students from abroad, 2024

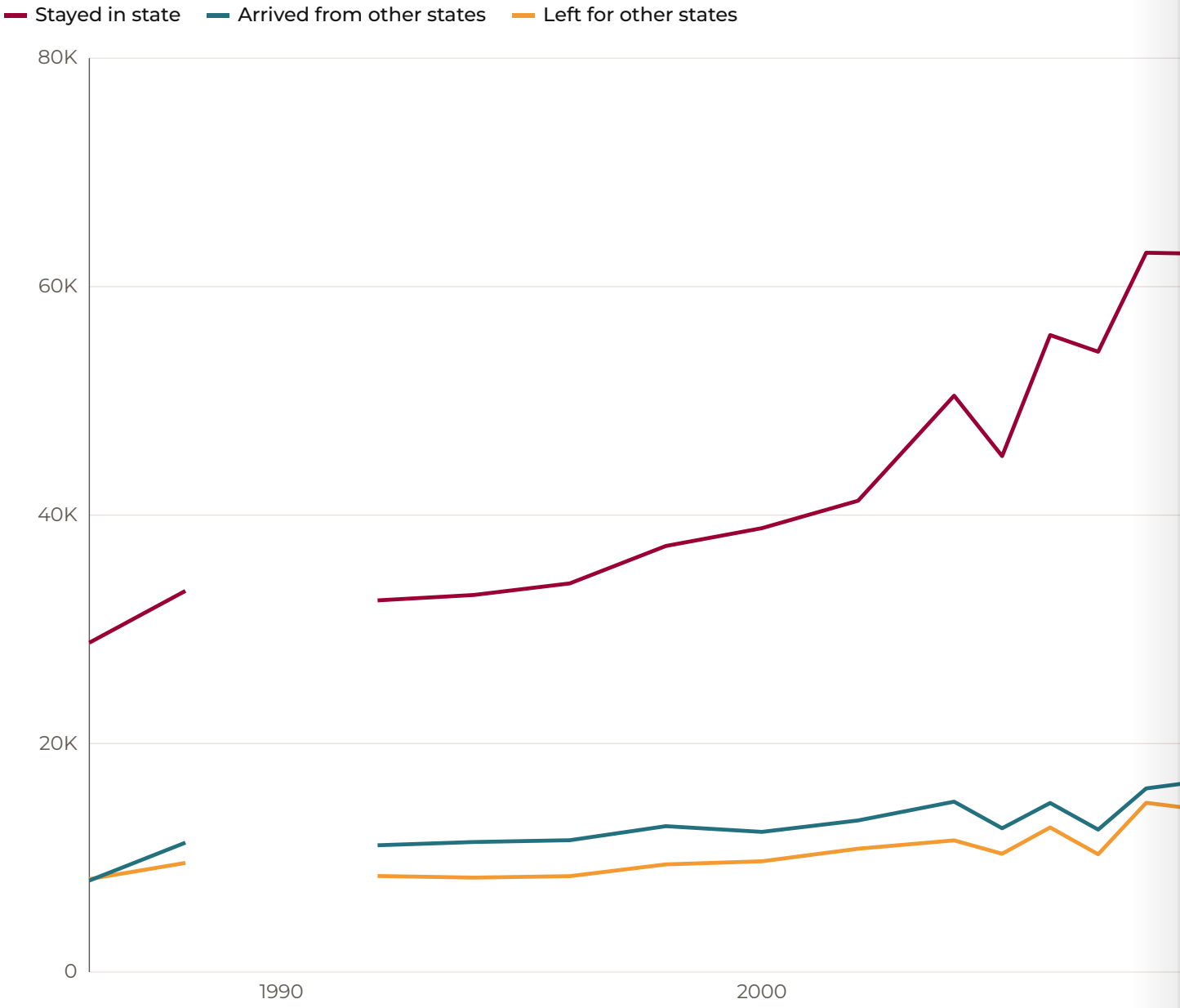
U.S. **3.1%** · **35th** of 51 · −0.2 pp since 2016

+2

Net migration per 100 residents, 2024

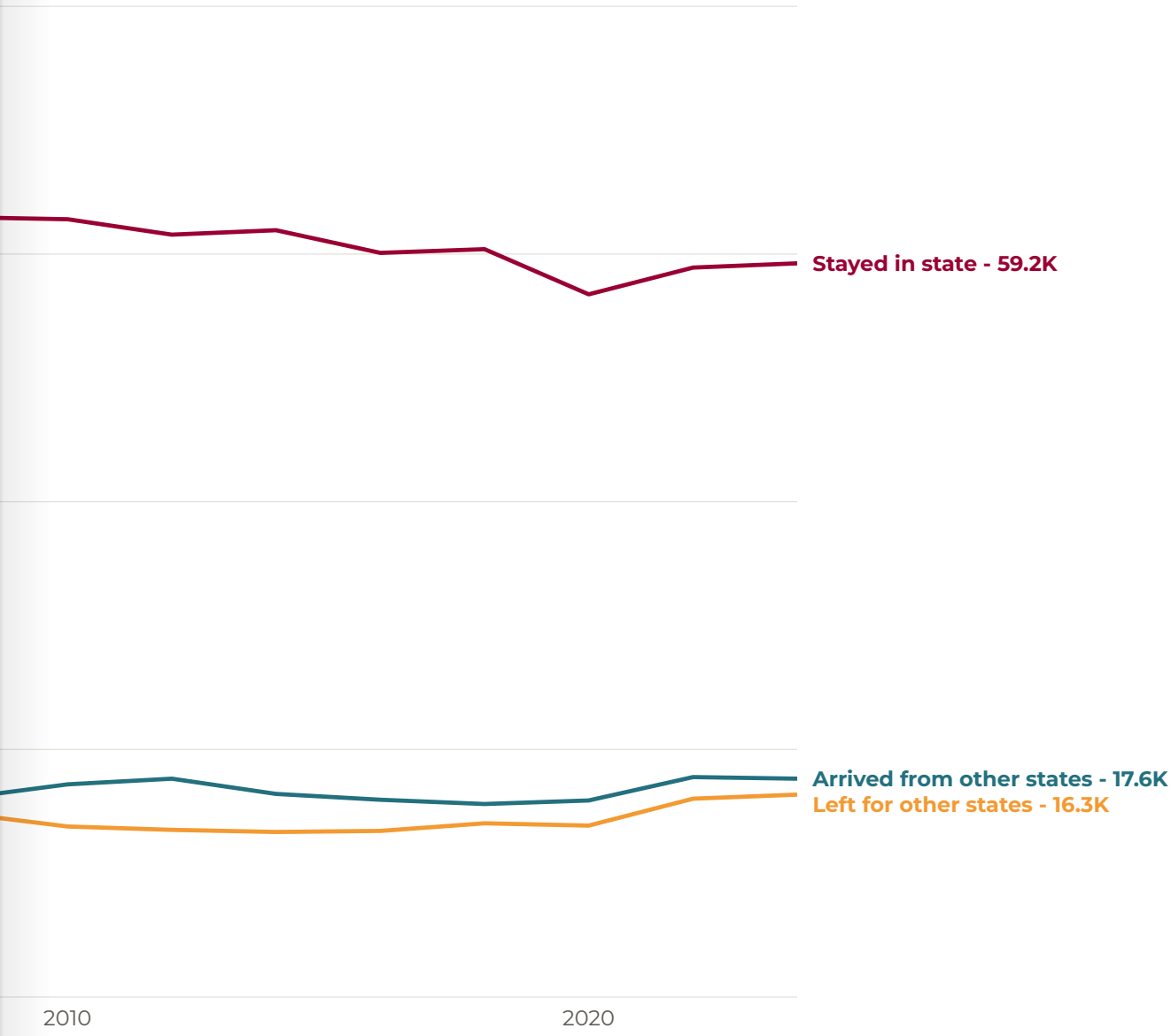
32nd of 51 · −1.7 since 2016

Figure 11 How Have the Flows Changed?



First-time degree-seeking undergraduates by flow; even years before 2001. Data from NCES, IPEDS Fall Enrollment.

Since 2015, the number of residents staying in state for college is roughly flat, arrivals from other states are up 11%, and residents leaving are up 22%.



Net migration was +2 per 100 resident first-time students (32nd of 51). The top destination for residents who left was North Carolina (1.62K).

Figure 12

Where Do Virginia's Newcomers Come From?

The bands show where first-time degree-seeking undergraduates who enrolled at Virginia's institutions in fall 2024 but did not live in the state came from: the 12 largest home states and students from abroad, with all other states grouped. Band widths are proportional to the number of students. Home is the state where the student lived when admitted.

19.3K

First-time students from other states and abroad, fall 2024

Abroad **1.62K**

2.0%

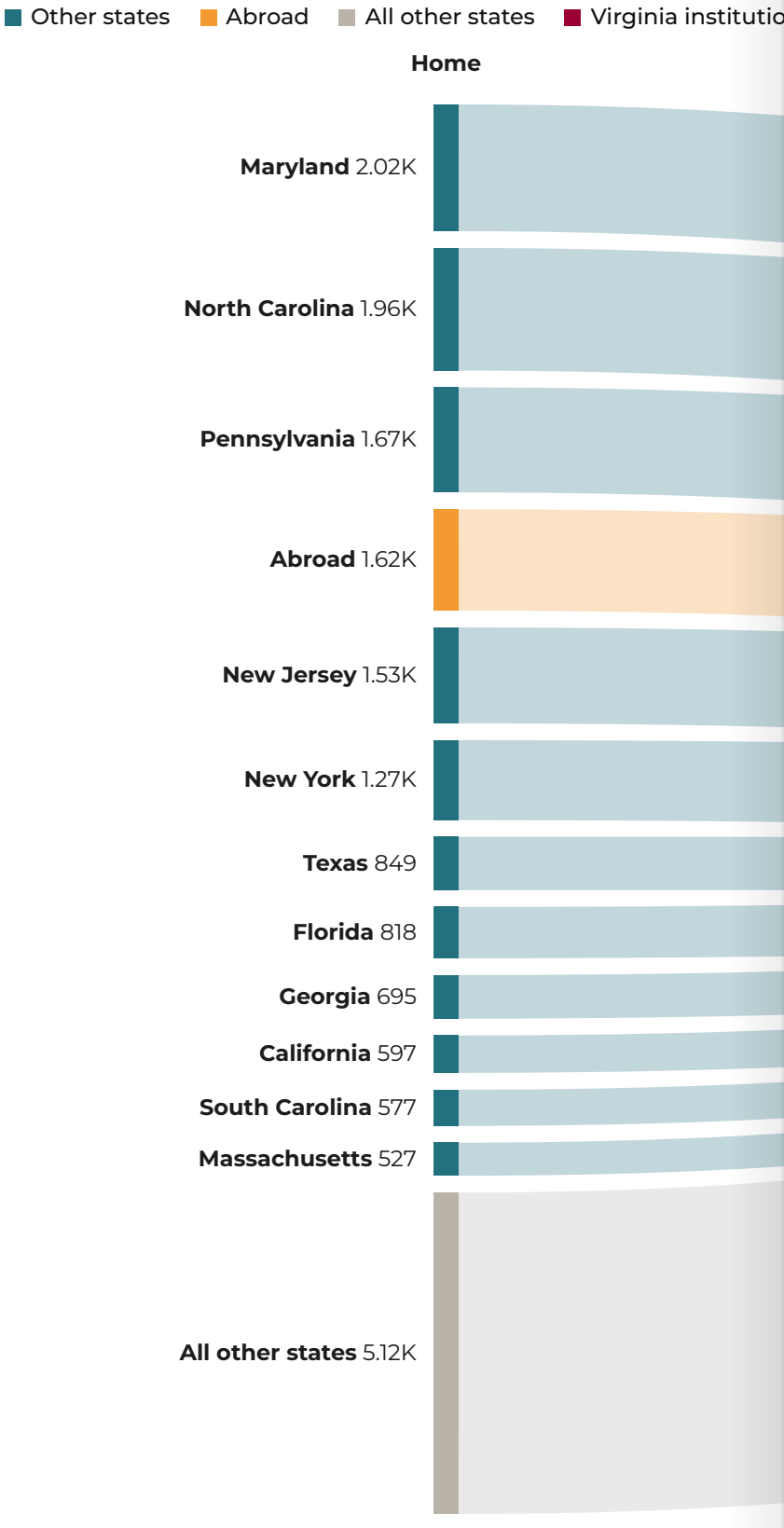
First-time students from abroad, 2024

U.S. **3.1%** · **35th** of 51 · −0.2 pp since 2016

75%

First-time students from in state, 2024

20th of 51 · −1.5 pp since 2016



First-time degree-seeking undergraduates, fall 2024; the 12 largest flo

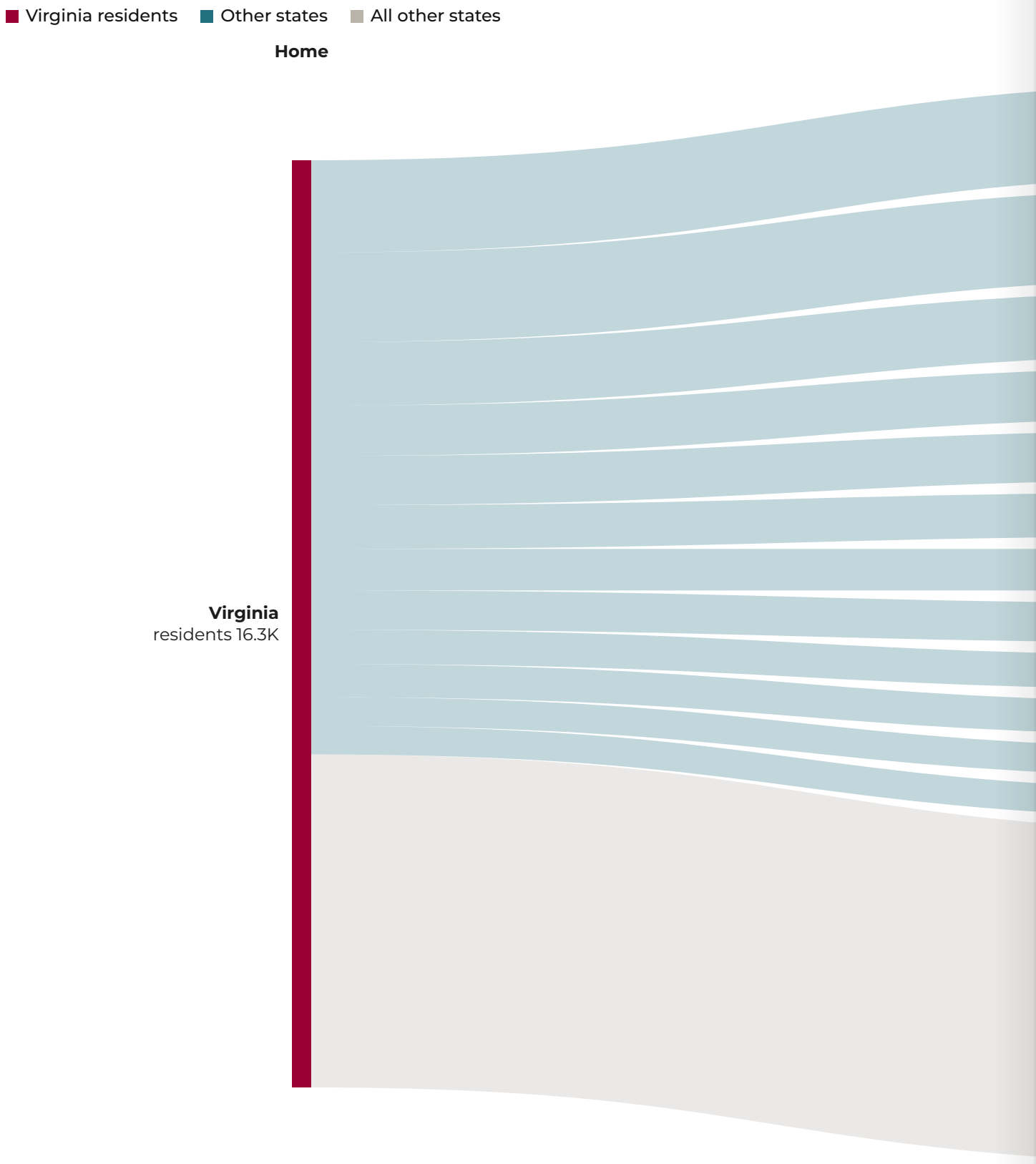
ons

Enrolled in



Virginia
institutions 19.3K

ws (more when states tie at the printed figure), ordered by exact count. Data from NCES, IPEDS Fall Enrollment.



First-time degree-seeking undergraduates, fall 2024; the 12 largest flows (more when states tie at the printed figure), ord

Figure 13

Where Do Virginia's Residents Go When They Leave?

The bands show where Virginia residents who left the state to start college in fall 2024 enrolled: the 12 largest destination states, with all other states grouped. Band widths are proportional to the number of students. Residents who enrolled in the state are not shown.

Residents leaving for other states. The number was 16.3K in fall 2024, up 2.0% from 16K in fall 2022 and up 22% since fall 2016, when it was 13.4K.

16.3K

Residents who started college in another state, fall 2024

Arrived from other states **17.6K** · **19.3K** with abroad

+2

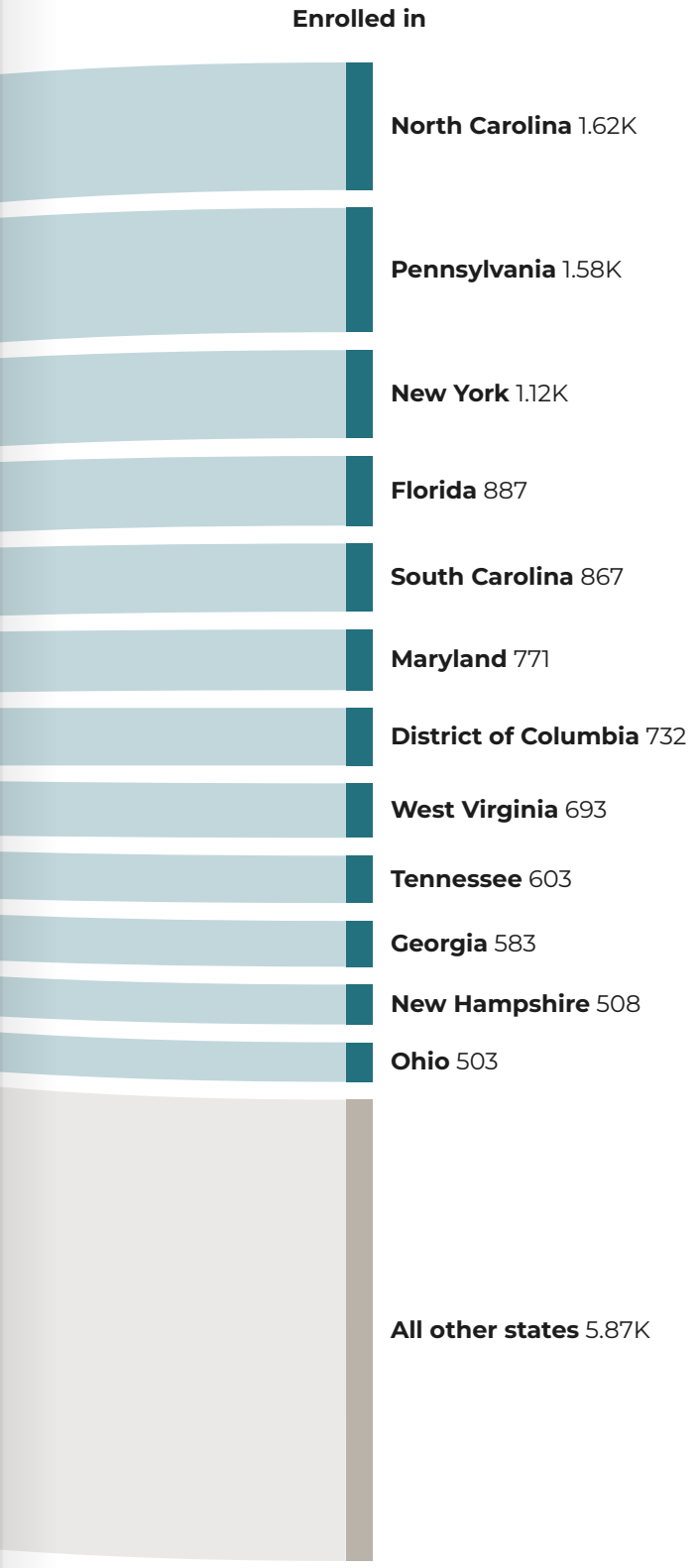
Net migration per 100 residents, 2024

32nd of 51 · −1.7 since 2016

1.62K

Went to North Carolina

The largest destination



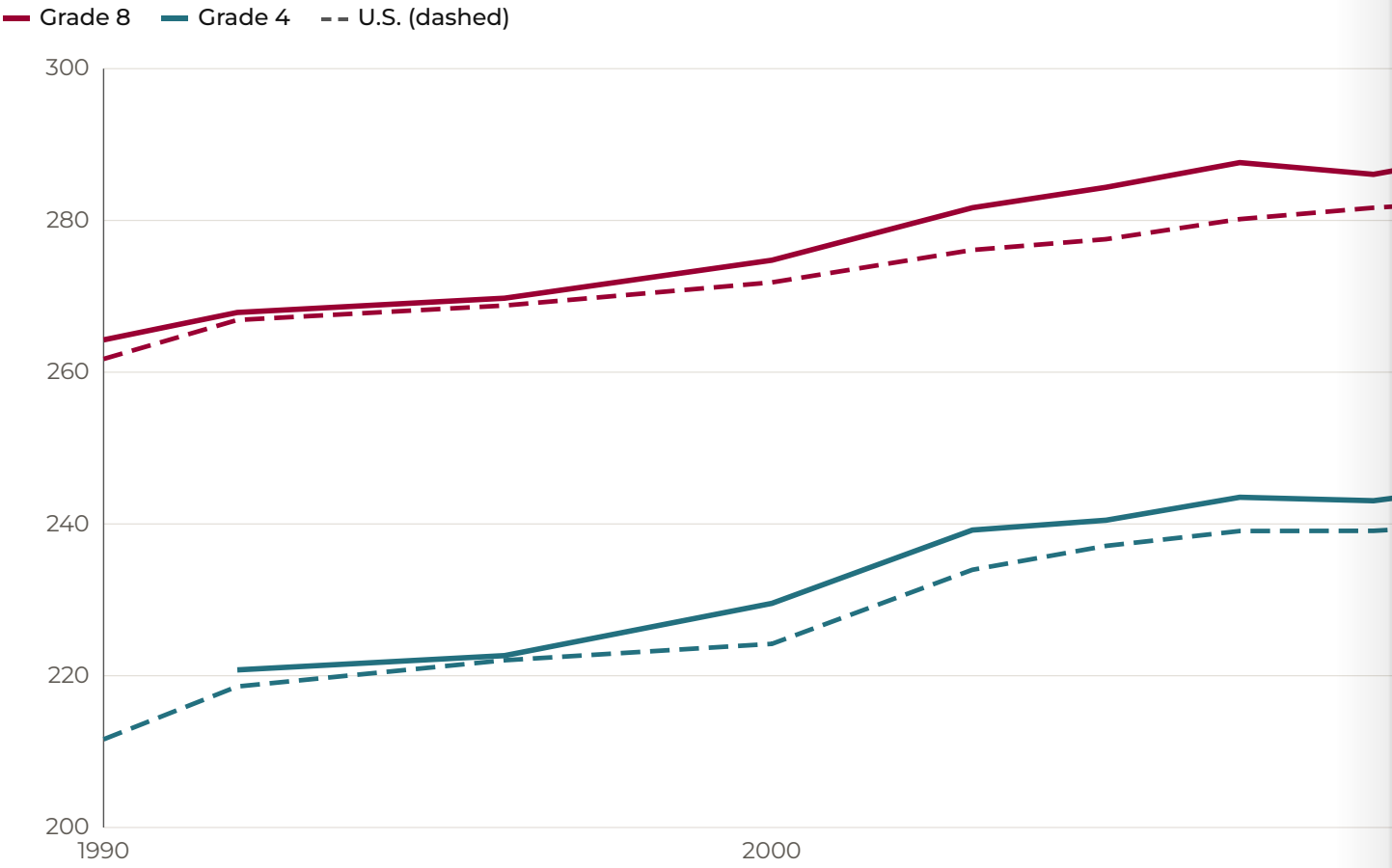
ered by exact count. Data from NCES, IPEDS Fall Enrollment.

SECTION 4

K-12 Preparation

Virginia ranks 22nd of 51 on the average grade 8 mathematics score, at 275 in 2024 against 272 for the U.S.

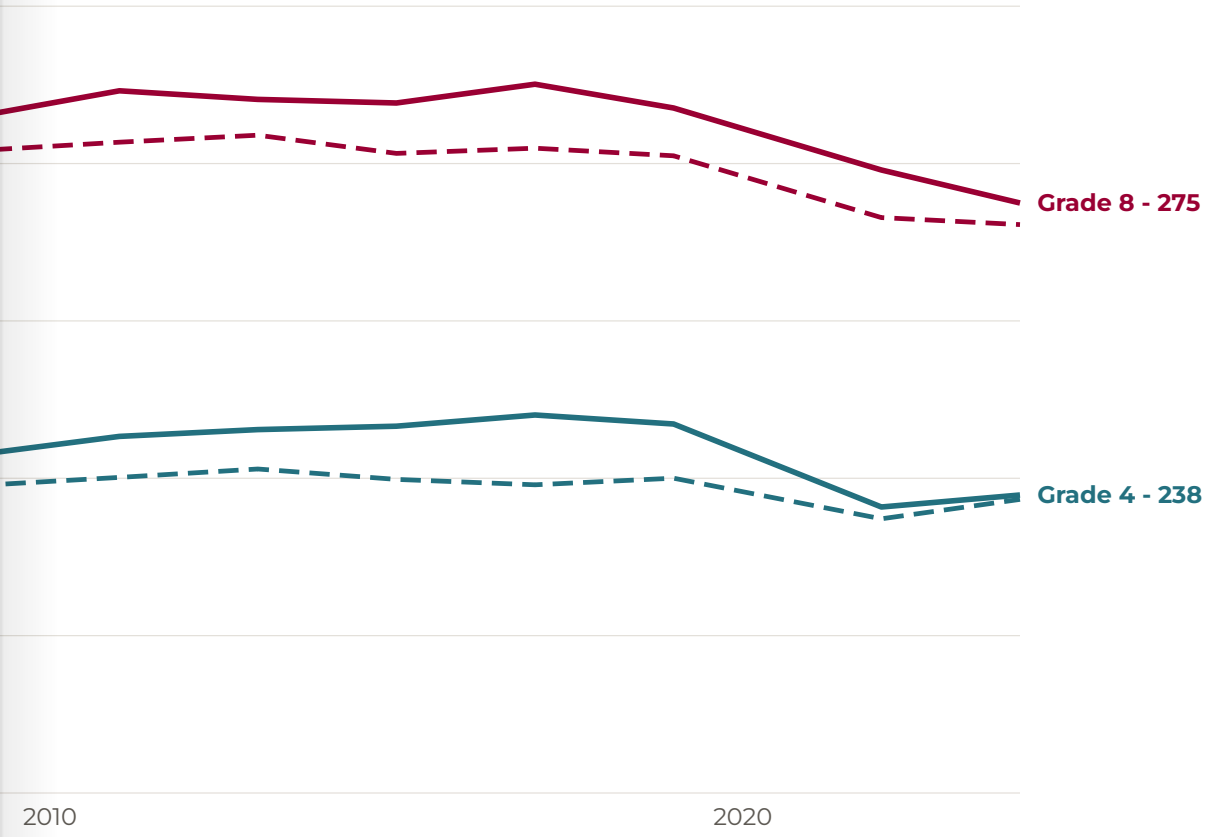
Figure 14 How Well Prepared in Mathematics Are Virginia's Students?



Average NAEP mathematics score, public schools, 0–500 scale; state solid, U.S. public schools dashed. Data from NCES, N

The talent pipeline starts in school. Virginia's eighth graders averaged 275 in mathematics on the 2024 NAEP, against 272 for U.S. public schools, 22nd of 51 among the states.

29% scored at or above the Proficient level (U.S. 27%). Differences of a point or two are often within the margin of error.



lational Assessment of Educational Progress.

275

Grade 8 math score (NAEP), 2024

U.S. **272** · **22nd** of 51 · −12.7 since 2015

29%

Grade 8 math, at or above Proficient, 2024

U.S. **27%** · **22nd** of 51 · −9.1 pp since 2015

256

Grade 8 reading score (NAEP), 2024

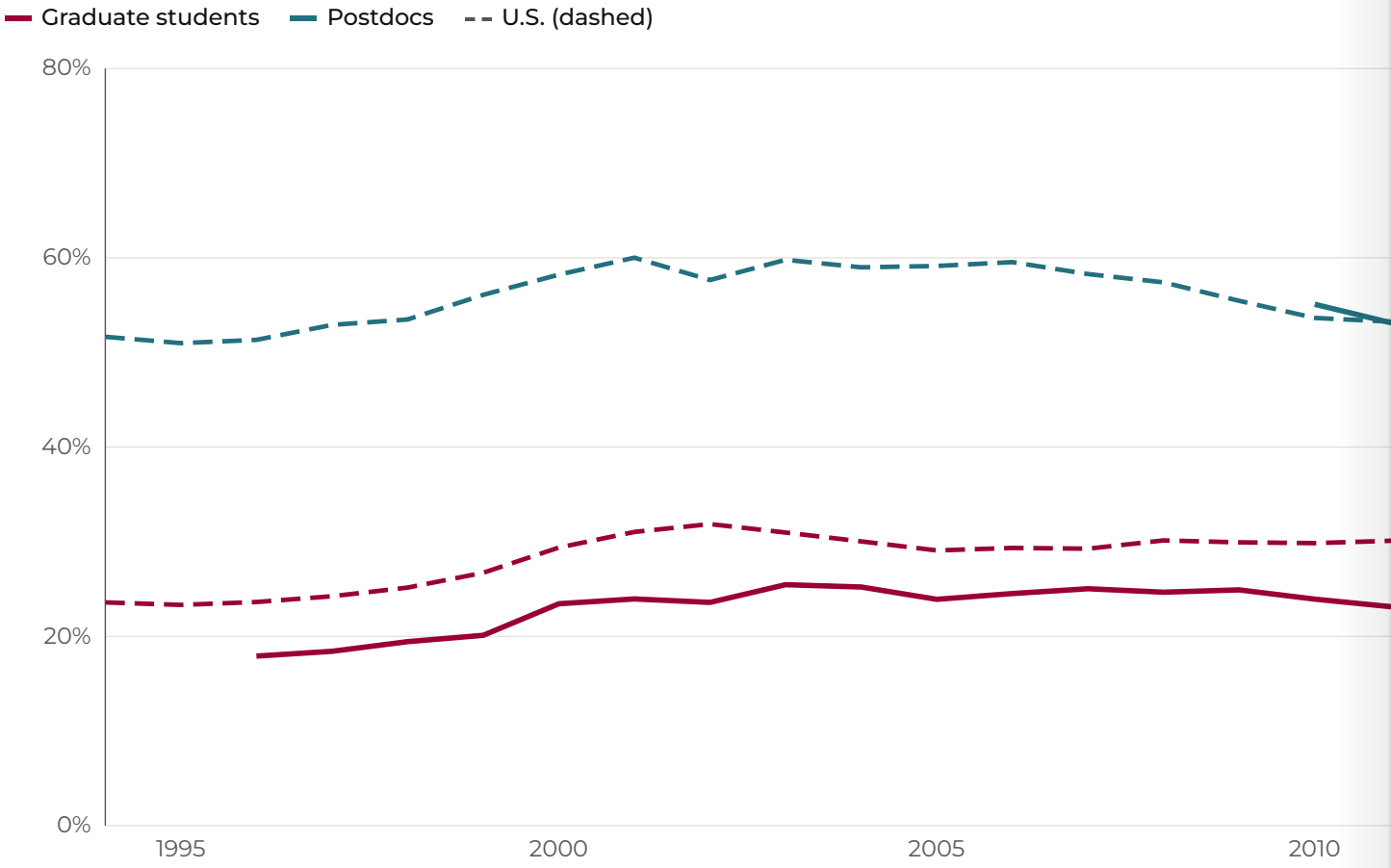
U.S. **257** · **29th** of 51 · −10.6 since 2015

SECTION 5

Graduate Research

Virginia ranks 11th of 51 on research doctorates awarded, at 1.76K in 2025 against 59K for the U.S.

Figure 15 How Many of Virginia's Science and Engineering Graduate Students Hold Ter



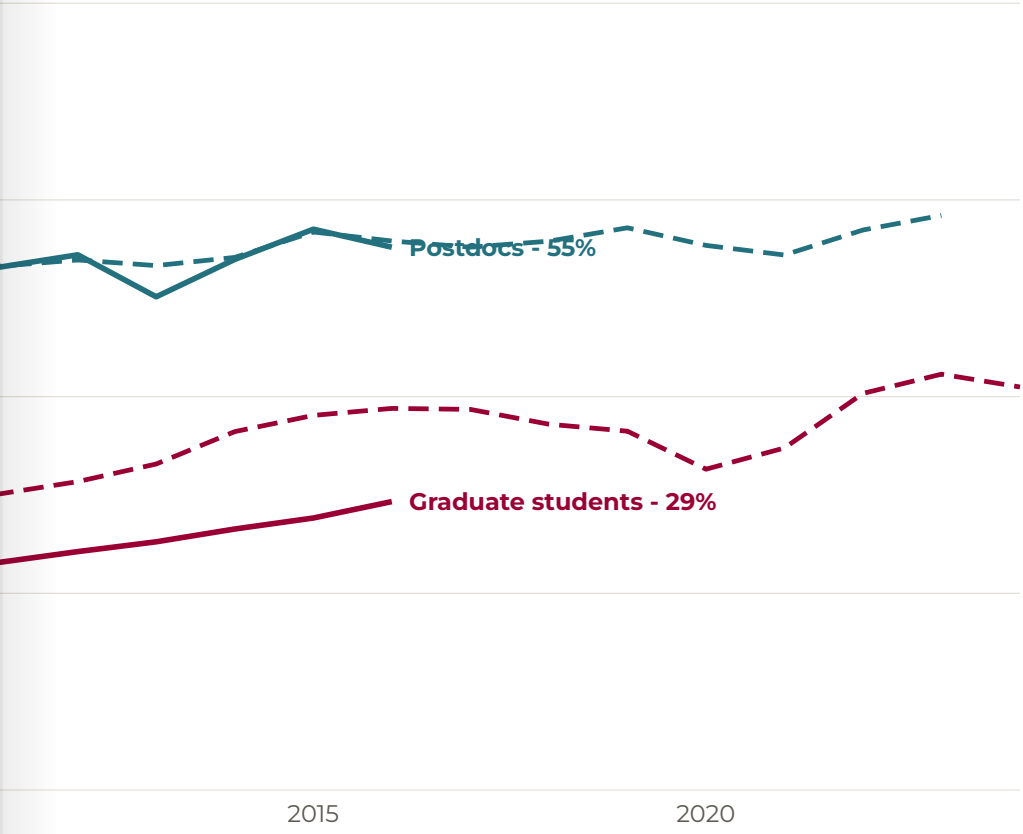
Temporary-visa share of science and engineering graduate students and postdoctoral appointees, by the institution's sta
Students and Postdoctorates.

In 2016, the last year NSF published state tables, 29% of Virginia's science and engineering graduate students held temporary visas (U.S. 39%; 34th of 51).

Among postdoctoral researchers, the share was 55% (U.S. 56%). Postdocs are the stage of research training most dependent on international talent, and IPEDS does not count them.

The state's institutions awarded 1.76K research doctorates in 2025, 3.0% of the U.S. total (Data from NSF NCSES, Survey of Earned Doctorates; NSF publishes doctorates by citizenship only for the nation).

mporary Visas?



te; state solid, U.S. dashed. NSF stopped publishing state tables after 2016. Data from NSF NCSES, Survey of Graduate

29%

S&E grad students on temporary visas, 2016

U.S. **39%** · **34th** of 51 · +1.7 pp since 2015

55%

S&E postdocs on temporary visas, 2016

U.S. **56%** · **26th** of 51 · −1.8 pp since 2015

1.76K

Research doctorates awarded (NSF SED), 2025

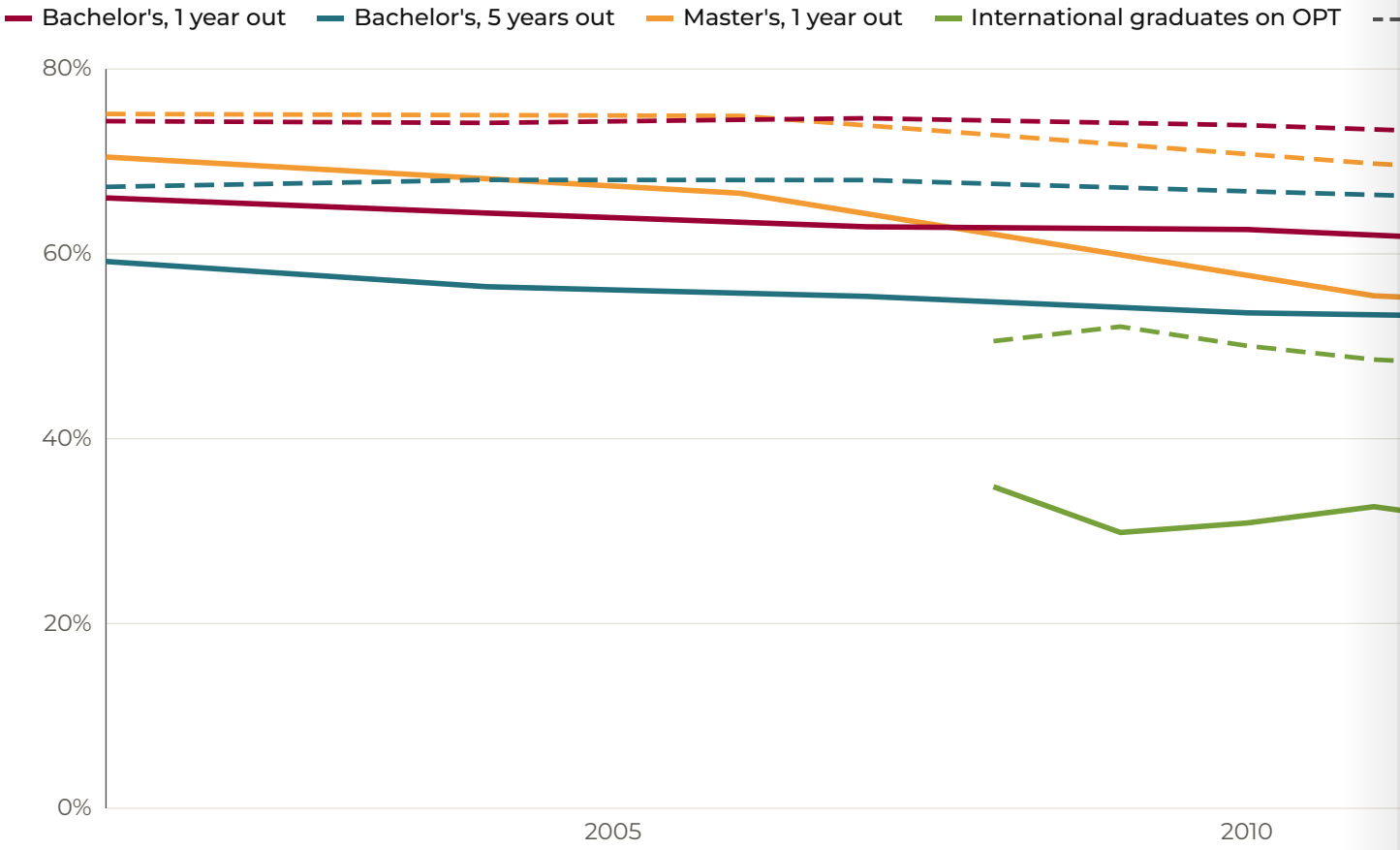
U.S. **59K** · **11th** of 51 · +15% since 2015

SECTION 6

Do Graduates Stay?

Virginia ranks 12th of 51 on net domestic migration of bachelor's-plus adults, per 1,000 residents, at +5.1 in 2025.

Figure 16 How Many of Virginia's Graduates Go On to Work in the State?

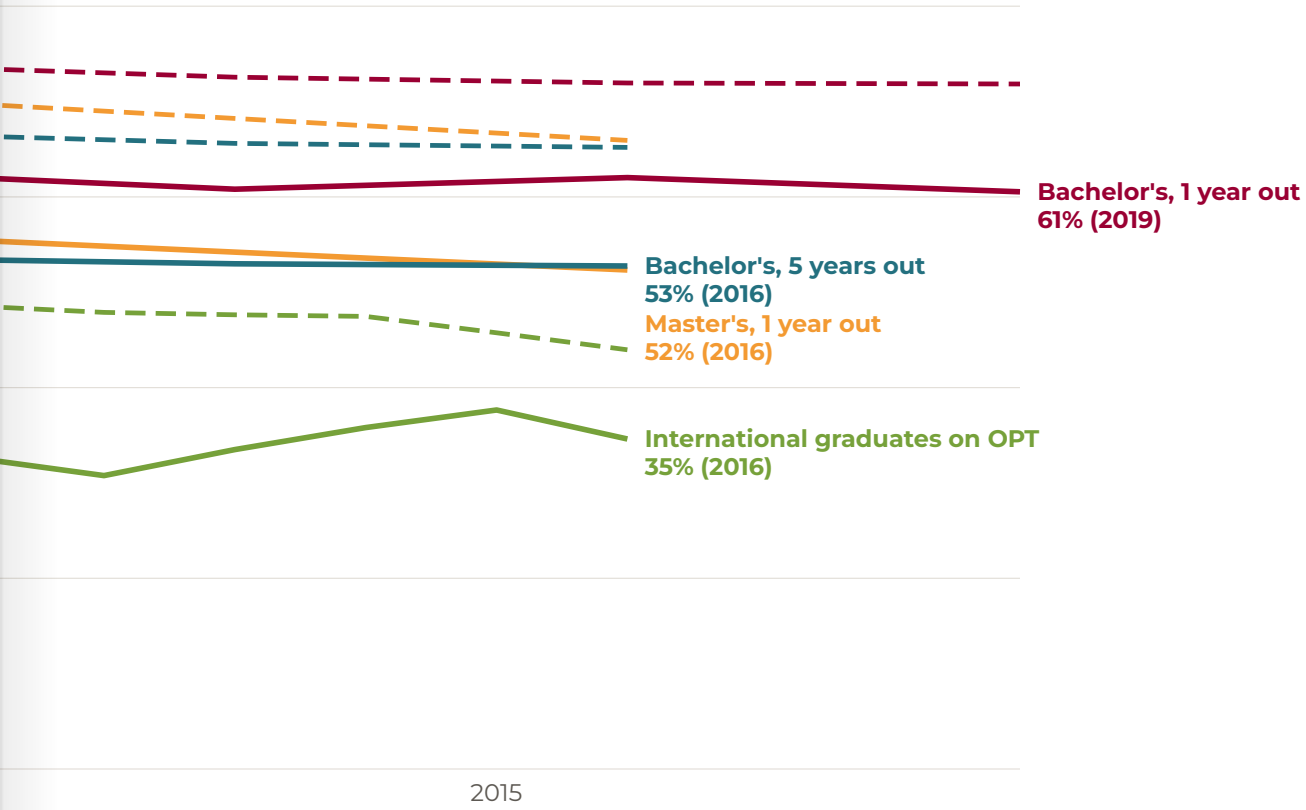


Share of employed graduates working in the state of their institution, by graduating cohort (Data from U.S. Census Bureau state where they studied (Data from ICE, SEVIS FOIA releases; shares only, the file is truncated). State solid, U.S. or particip

Of Virginia's bachelor's graduates who were working a year after graduating, 61% worked in the state (participating states 72%). Five years out the share was 53%.

Among international graduates on Optional Practical Training in 2016, 35% worked in the state where they studied (U.S. 44%). That is the last year the released records cover, before most of the growth in international STEM master's degrees since 2015. Through migration between states, the state gained 5.1 bachelor's-plus adults per 1,000 such residents in 2025 (Data from U.S. Census Bureau, American Community Survey).

U.S. (dashed)



au, Post-Secondary Employment Outcomes; participating institutions only); international graduates' OPT jobs in the
pating states dashed.

61%

Bachelor's grads working in-state, 1 year out, 2019

U.S. **72%** · **21st** of 33 · −1.5 pp since 2016

35%

OPT graduates working in the state, 2016

U.S. **44%** · **26th** of 51 · −3.0 pp since 2015

+5.1

Net migration, bachelor's-plus, per 1,000, 2025

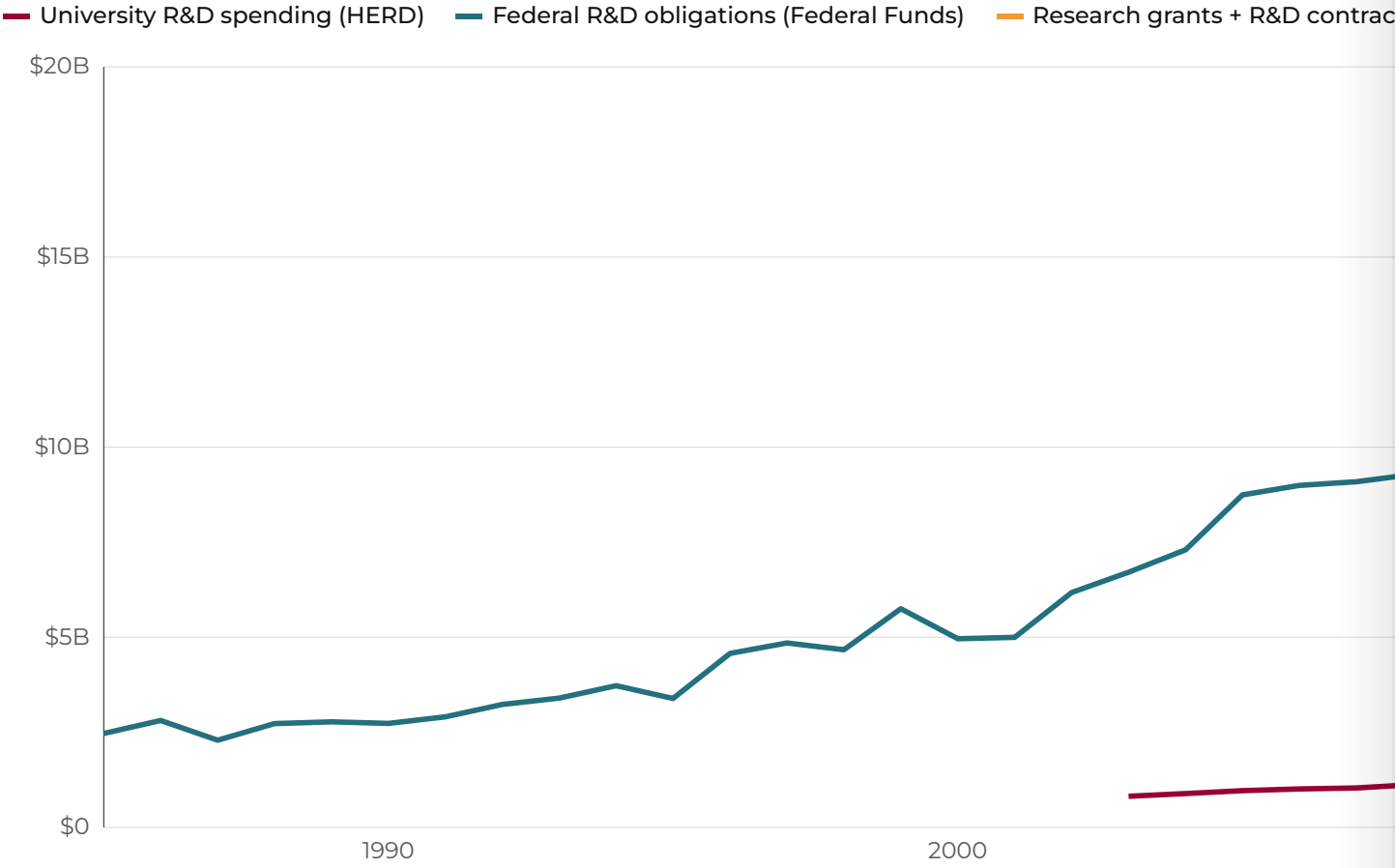
12th of 51 · +6.2 since 2015

SECTION 7

Research Funding

Virginia ranks 3rd of 51 on federal R&D obligations to the state, at $15.1B in 2024 against $193B for the U.S.

Figure 17 How Much Research Money Flows Into Virginia?

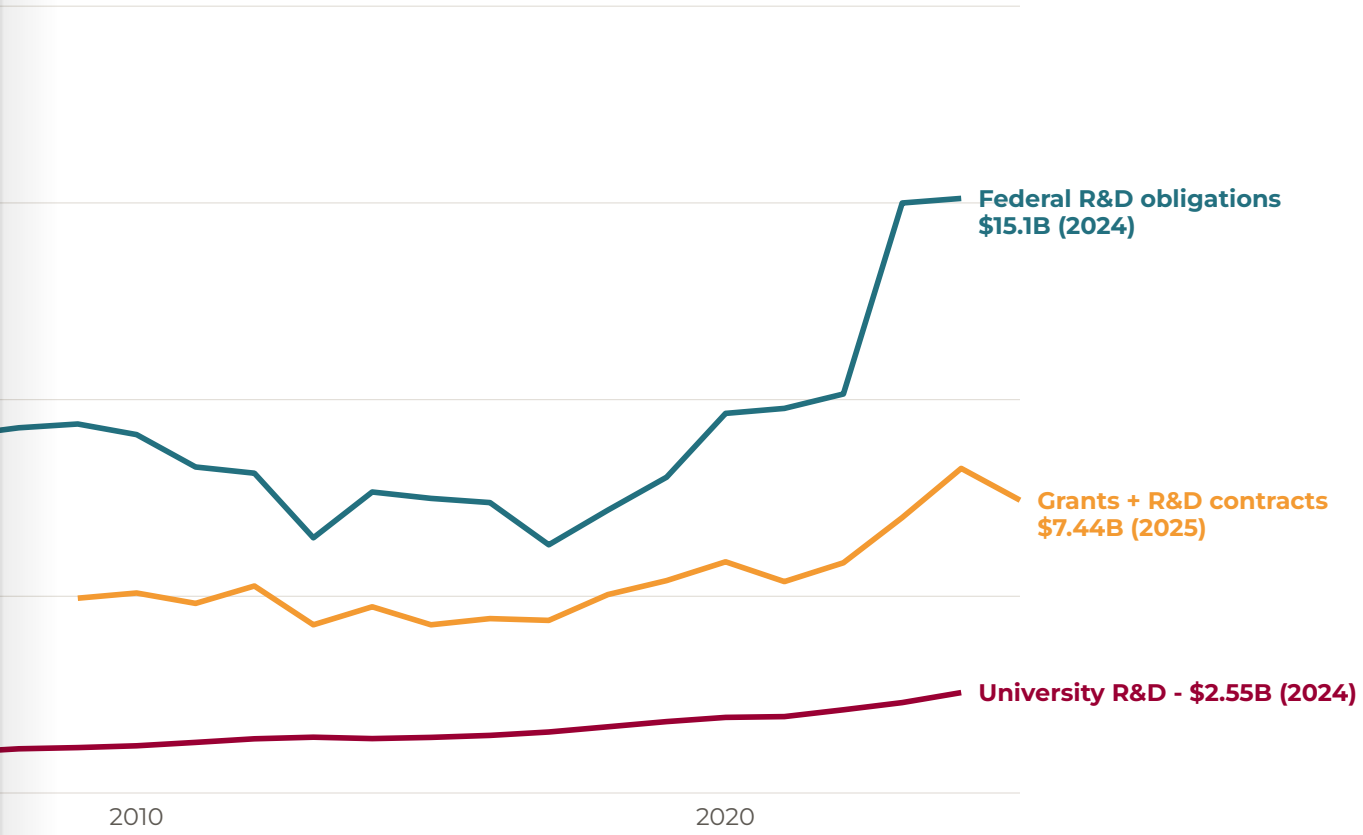


Nominal dollars (not adjusted for inflation), fiscal years; a line ending before the others is labeled with its last year. Univers
Federal R&D obligations by state of performance: Data from NSF NCSES, Survey of Federal Funds for R&D (coverage wide

Virginia's universities spent $2.55B on research and development in FY2024, 13th of 51 among the states; 47% of it was federally funded (U.S. 55%).

Federal agencies obligated $15.1B for R&D performed in the state in FY2024. Research funding drives demand for graduate students and postdocs, the stage of training most dependent on international talent.

ts (USAspending)



sity R&D: Data from NSF NCSES, Higher Education Research and Development Survey (FY2003-09 predecessor survey). ens in FY2021). Grants from NSF, NIH, NASA and DOE Science plus R&D contracts: Data from USAspending.gov.

$2.55B

University R&D spending, 2024

U.S. **$118B** · **13th** of 51 · +81% since 2015

47%

Federal share of university R&D, 2024

U.S. **55%** · **41st** of 51 · −5.8 pp since 2015

$15.1B

Federal R&D obligations, 2024

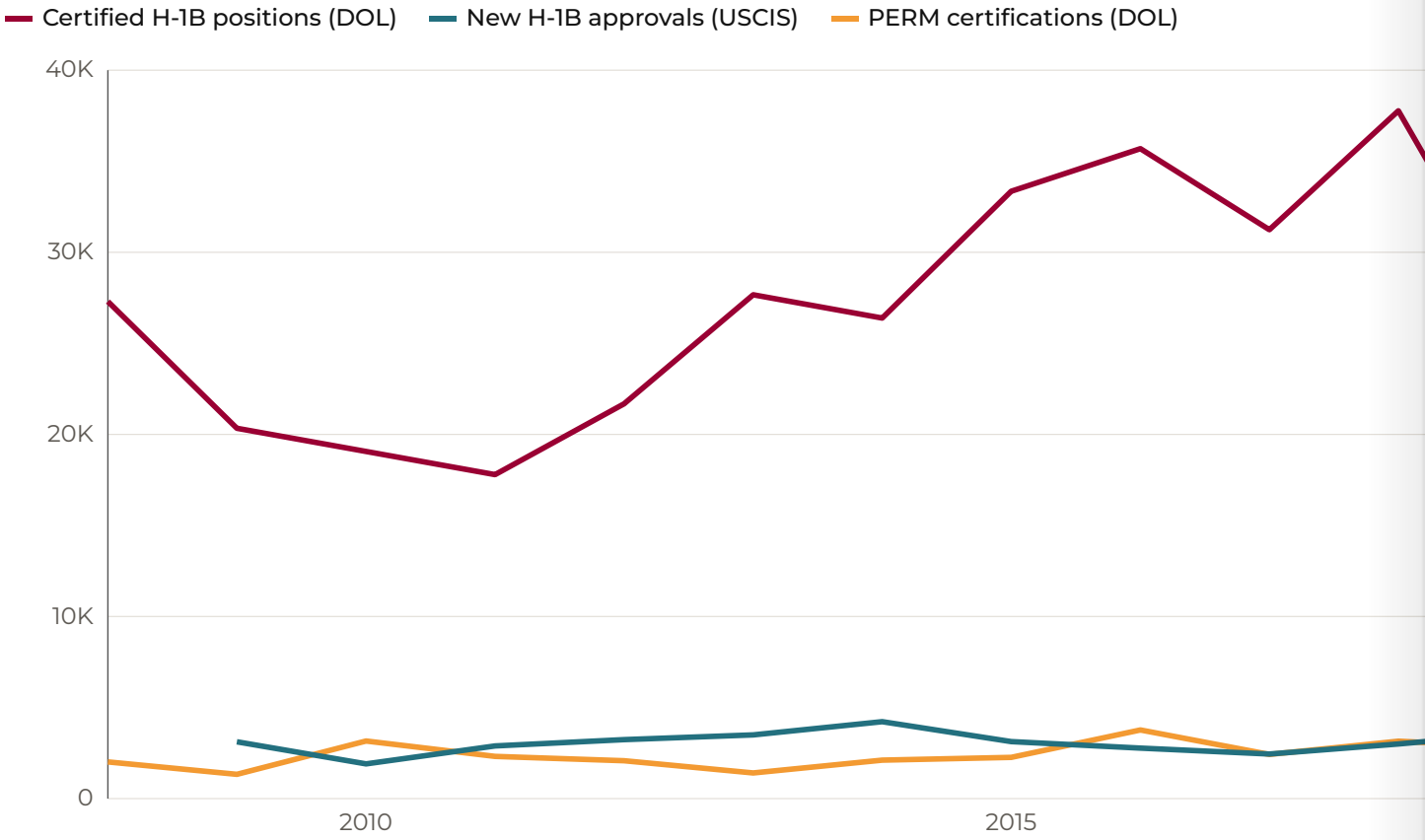
U.S. **$193B** · **3rd** of 51 · 2.02× since 2015

SECTION 8

H-1B Jobs and Green Cards

Virginia ranks 11th of 51 on certified H-1B positions, by worksite, at 22.4K in 2025 against 914K for the U.S.

Figure 18 How Many Skilled-Visa Jobs and Green-Card Certifications Does Virginia Gen

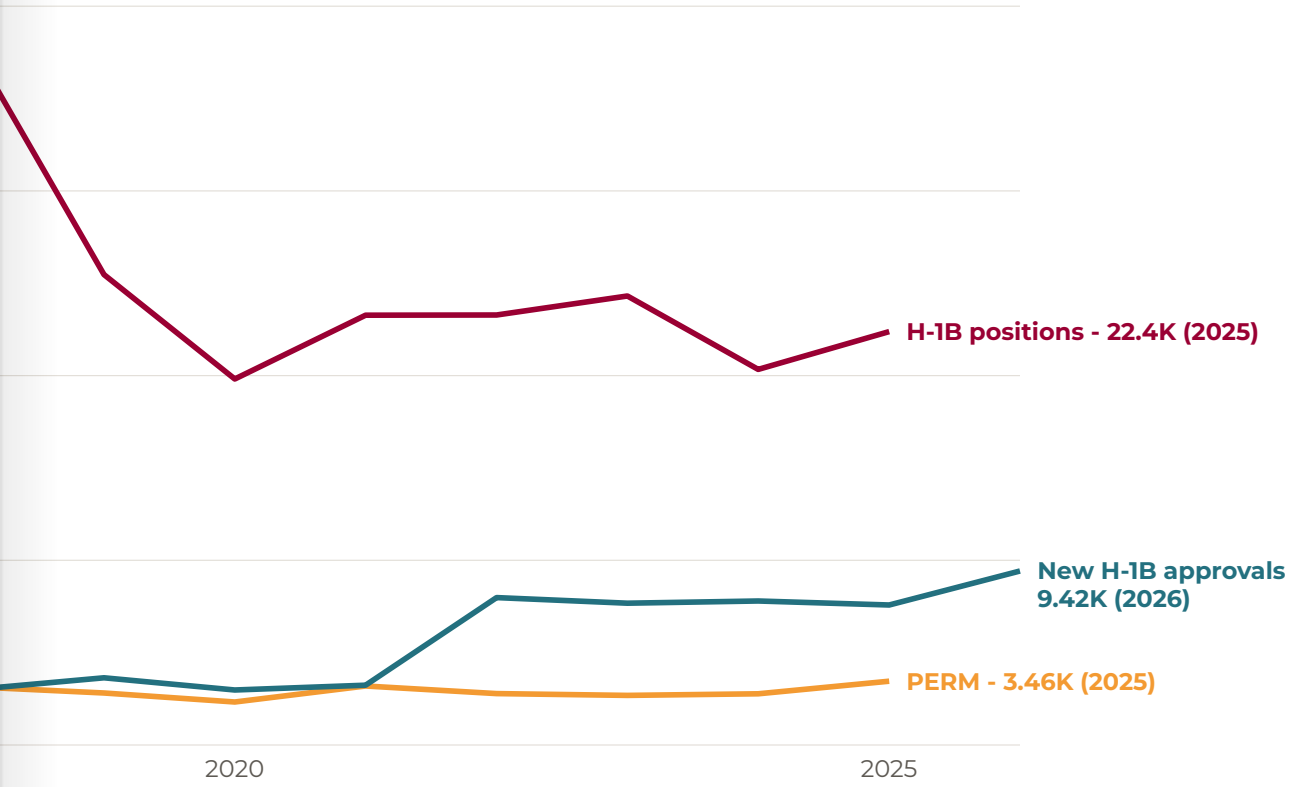


Certified H-1B positions on labor condition applications by worksite and certified PERM green-card applications, by fiscal from USCIS, H-1B Employer Data Hub).

Employers were certified for 22.4K H-1B positions at Virginia worksites in FY2025, 11th of 51 among the states. A labor condition application comes before the petition and covers renewals and job changes as well as new hires, by the worksite's state, so it runs well above approvals. USCIS approved 9.42K new H-1B petitions in FY2026 and 18.5K continuing ones, counted by the employer's address.

DOL certified 3.46K PERM applications, the employer's step toward an employment-based green card, for jobs in the state in FY2025. In FY2024, 10% were for workers still on F-1 student visas (U.S. 8.7%): the most direct link from classroom to permanent residence.

erate?



year (Data from DOL, Office of Foreign Labor Certification); new H-1B petition approvals by the employer's address (Data

22.4K

Certified H-1B positions (worksite), 2025

U.S. **914K** · **11th** of 51 · −33% since 2015

3.46K

PERM green-card certifications, 2025

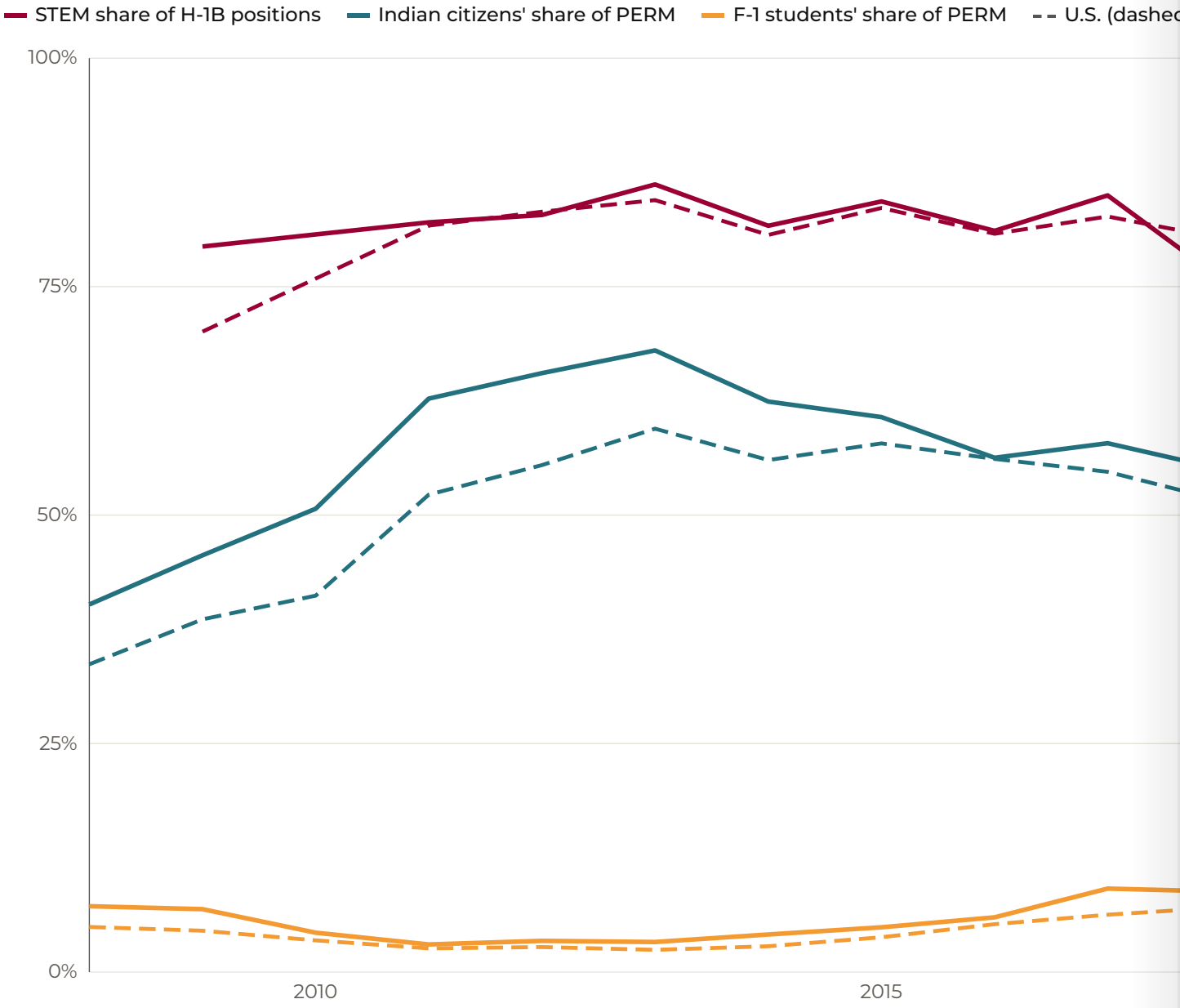
U.S. **138K** · **11th** of 51 · +53% since 2015

10%

PERM workers on F-1 student visas, 2024

U.S. **8.7%** · **23rd** of 51 · +5.4 pp since 2015

Figure 19 Who Holds Virginia's Skilled-Visa Jobs?

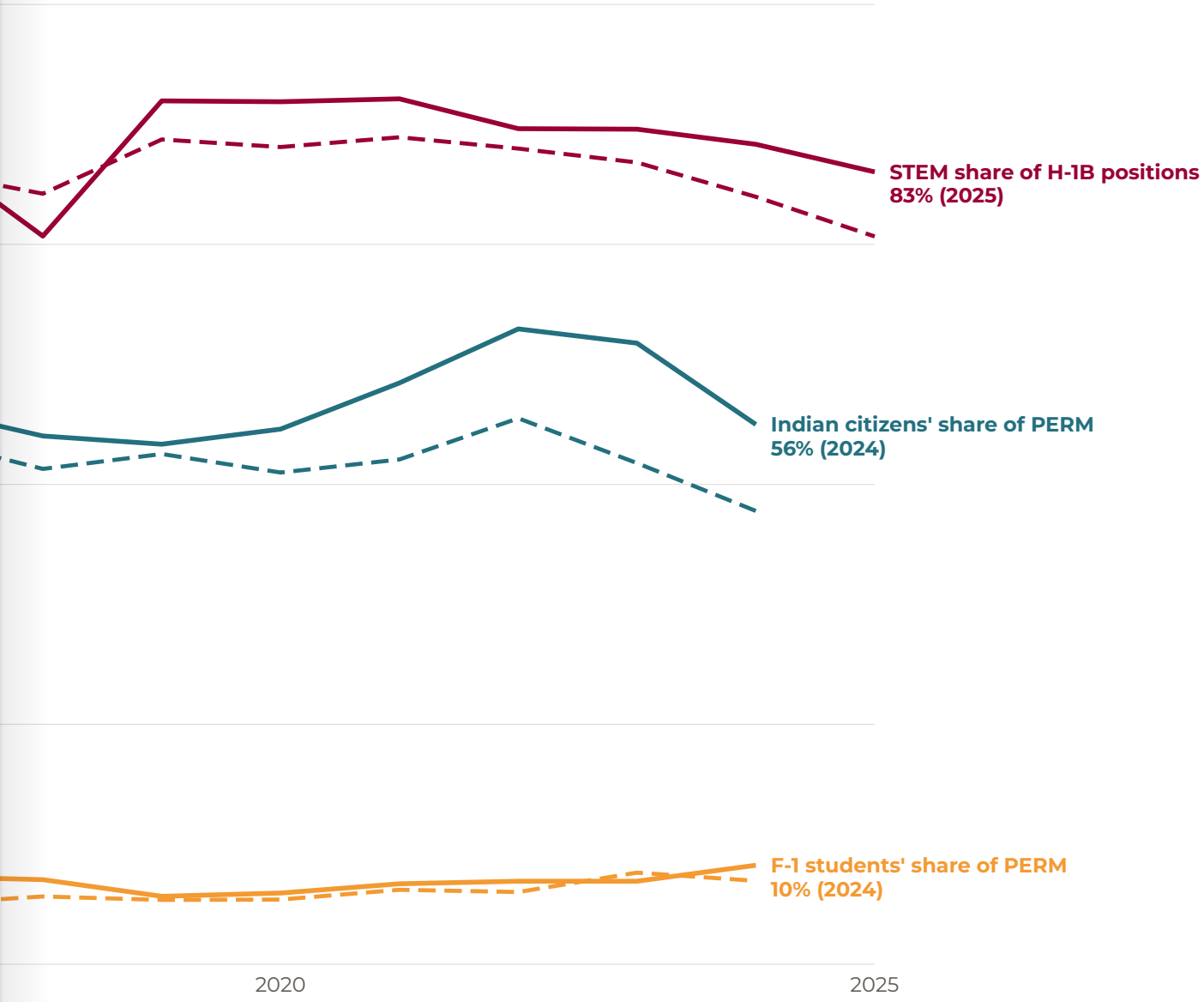


State solid, U.S. dashed. PERM citizenship and class of admission are not on the form DOL uses from FY2025. Data from D

83% of the state's certified H-1B positions were in STEM occupations in FY2025 (U.S. 76%).

Citizens of India held 56% of the state's PERM certifications in FY2024 (U.S. 47%). Under per-country caps they face the longest waits for a green card, so this share measures the state's exposure to the backlog.

d)



OL, Office of Foreign Labor Certification.

83%

STEM share of H-1B positions, 2025

U.S. **76%** · **8th** of 51 · −1.8 pp since 2015

56%

PERM certifications for Indian citizens, 2024

U.S. **47%** · **8th** of 51 · −4.5 pp since 2015

$115K

Median offered H-1B wage (nominal), 2025

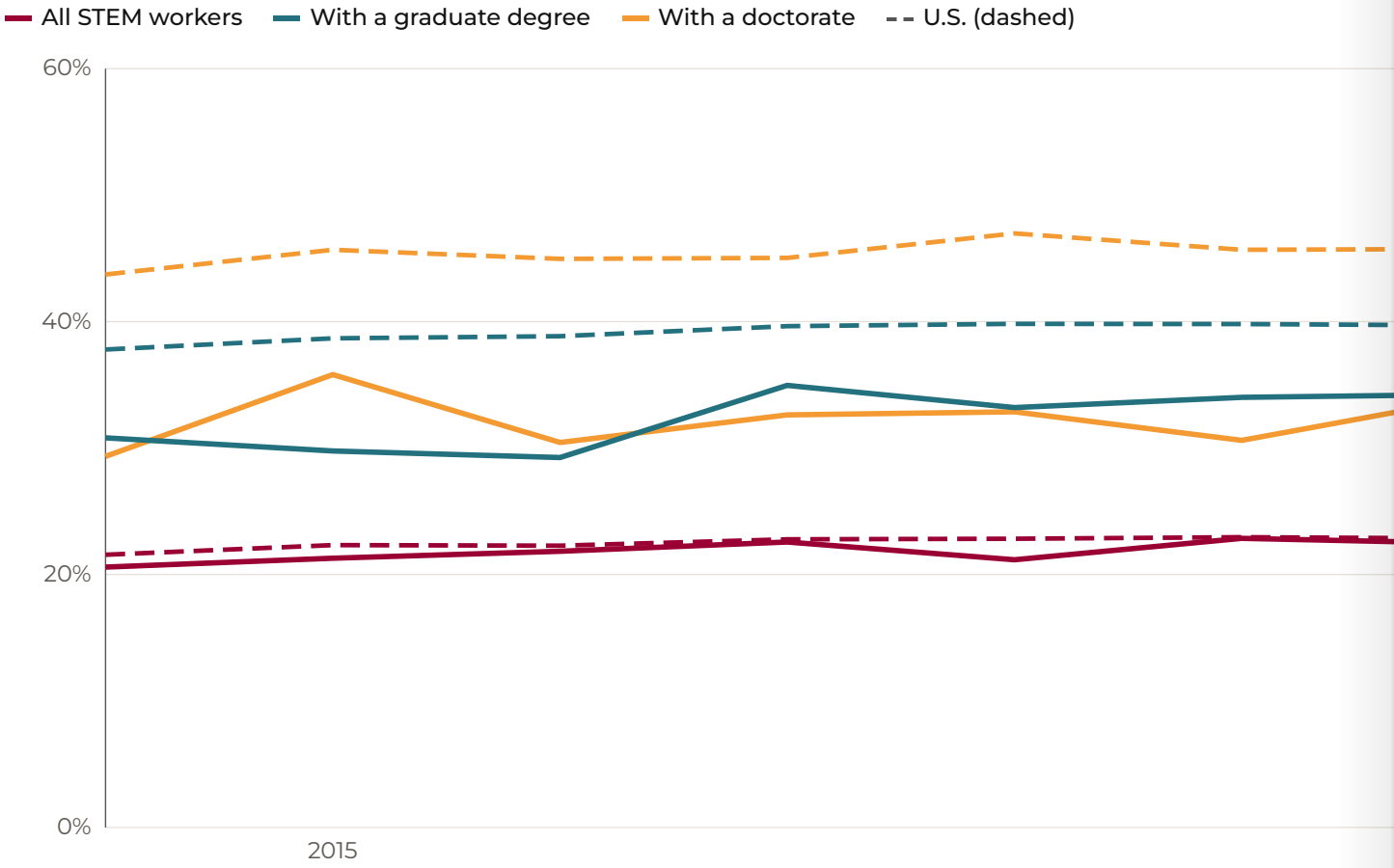
U.S. **$119K** · **10th** of 51 · +64% since 2015

SECTION 9

STEM Workforce

Virginia ranks 13th of 51 on the foreign-born share of STEM workers, at 22% in 2024 against 24% for the U.S.

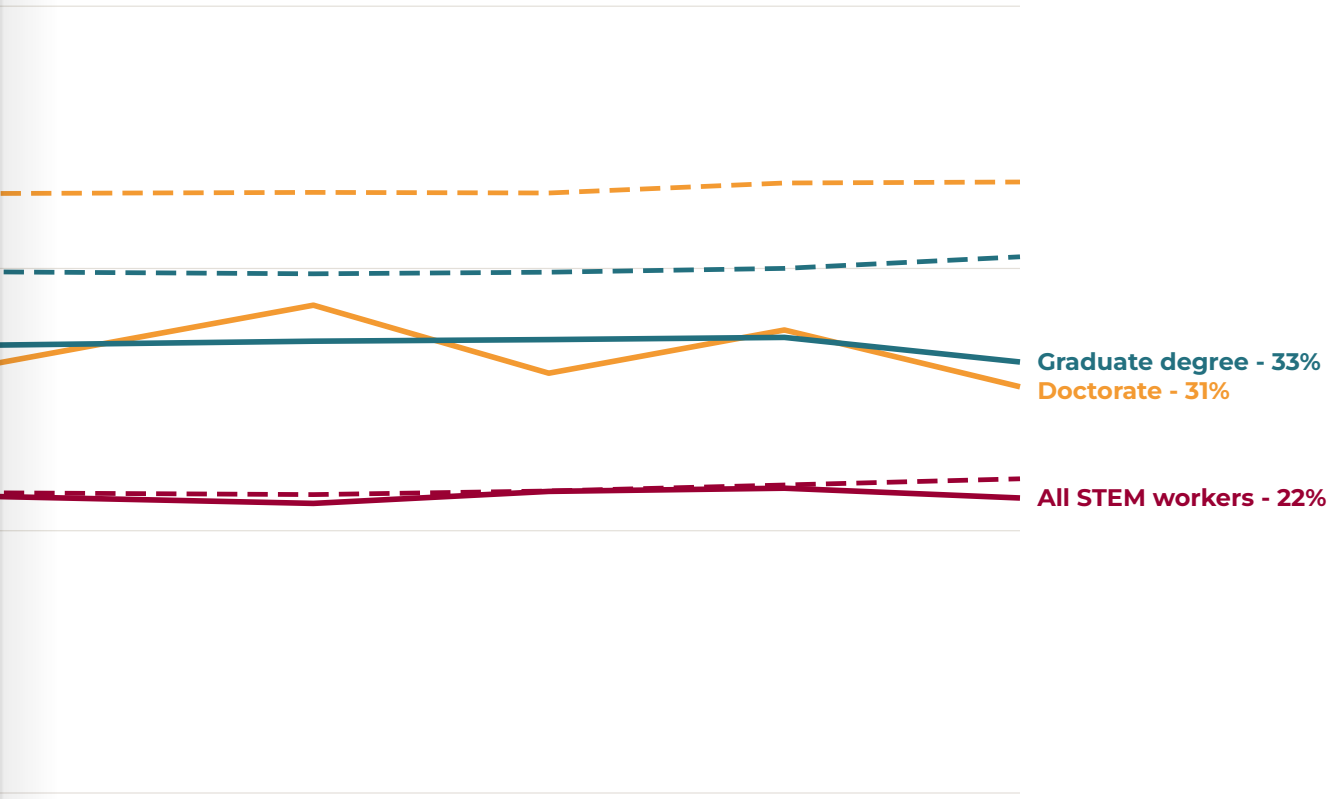
Figure 20 How Much of Virginia's STEM Workforce Was Born Abroad?



Foreign-born share of employed workers in STEM occupations, by state of residence; state solid, U.S. dashed. Shares resti
Community Survey.

In 2024, 22% of the people working in STEM occupations in Virginia were born abroad (U.S. 24%; 13th of 51).

The share rises with education: 33% of STEM workers with a graduate degree were foreign-born (U.S. 41%), and 31% of those with a doctorate (U.S. 47%; 18th of 20). 7.5% of all STEM workers were not U.S. citizens.



ng on fewer than 100 survey records are left out. No 2020 (experimental data). Data from U.S. Census Bureau, American

22%

Foreign-born share of STEM workers, 2024

U.S. **24%** · **13th** of 51 · +1.2 pp since 2015

33%

Foreign-born, STEM workers with grad degrees, 2024

U.S. **41%** · **19th** of 44 · +3.1 pp since 2015

7.5%

Noncitizen share of STEM workers, 2024

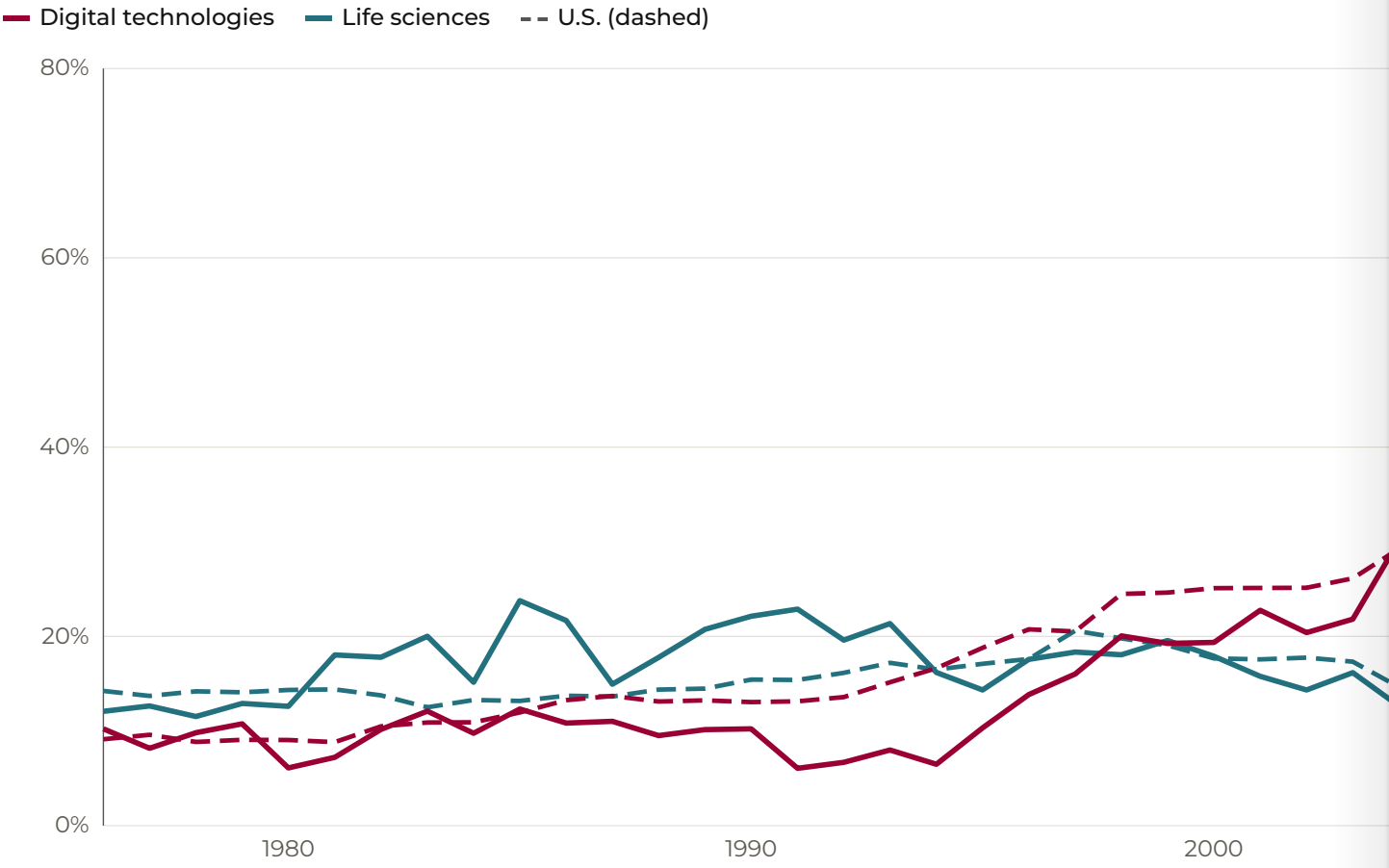
U.S. **11%** · **23rd** of 51 · −0.2 pp since 2015

SECTION 10

Patents

Virginia ranks 50th of 51 on the share of the state's patents in life sciences, at 6.0% in 2025 against 17% for the U.S.

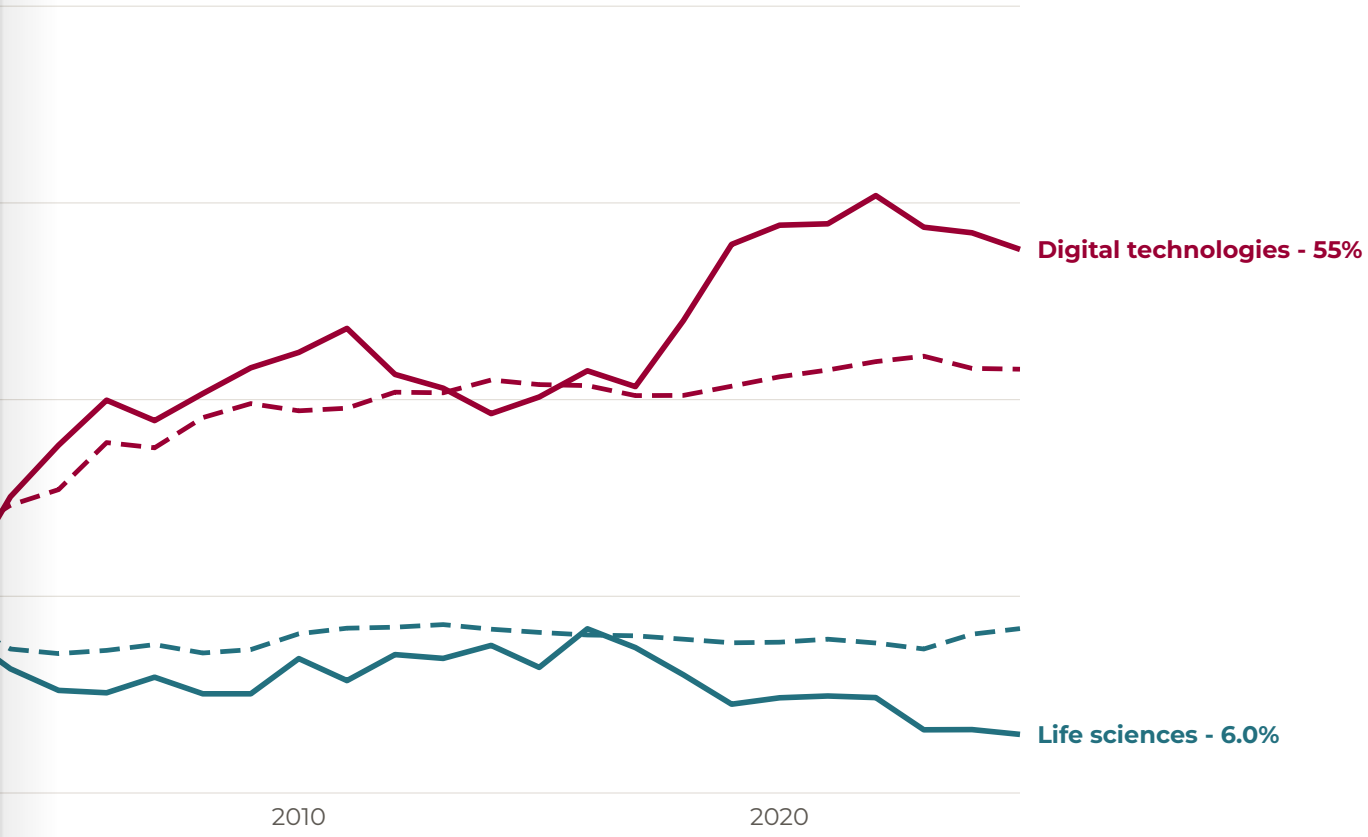
Figure 21 What Kinds of Technology Does Virginia Patent?



Share of patents granted to owners in the state, by WIPO technology field of the first classification; state solid, U.S. dashed

Organizations and people in Virginia were granted 3.96K U.S. patents in 2025, 2.9% of all patents with a U.S. owner (9th of 51).

55% were in digital technologies (U.S. 43%) and 6.0% in the life sciences (U.S. 17%).



d. Fractional counts by the owner's (assignee's) location, not where inventors live. Data from USPTO, PatentsView.

3.96K

Patents granted to owners in the state, 2025

U.S. **135K** · **9th** of 51 · 2.94× since 2015

55%

Patents in digital technologies, 2025

U.S. **43%** · **6th** of 51 · +15.0 pp since 2015

6.0%

Patents in life sciences, 2025

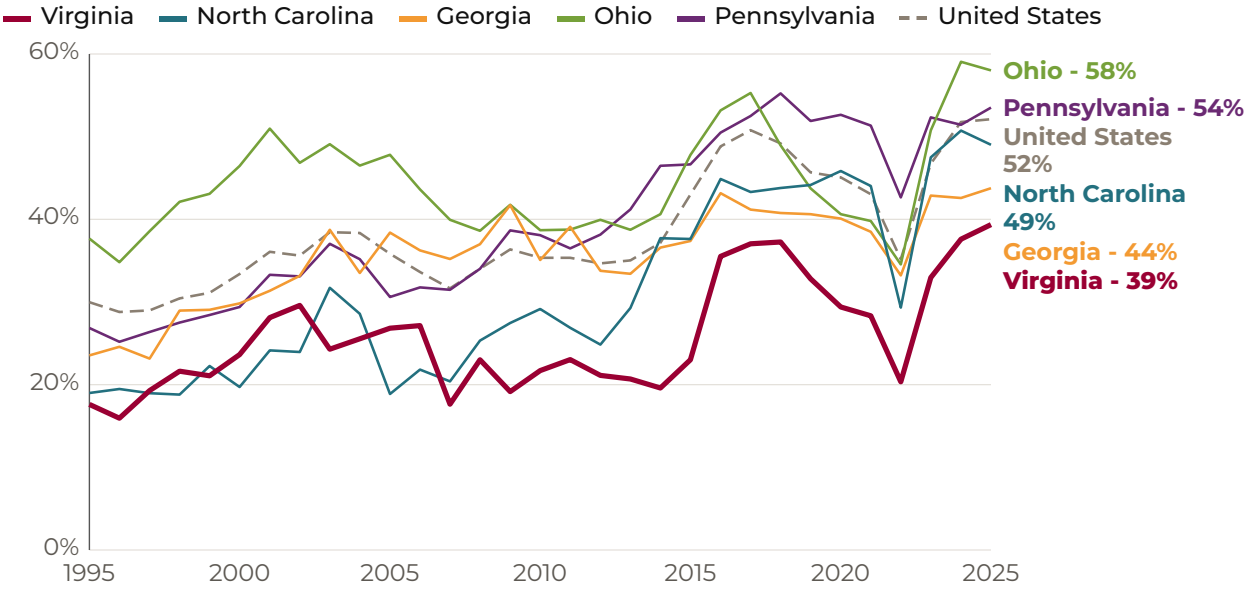
U.S. **17%** · **50th** of 51 · −6.8 pp since 2015

SECTION 11

Competitors

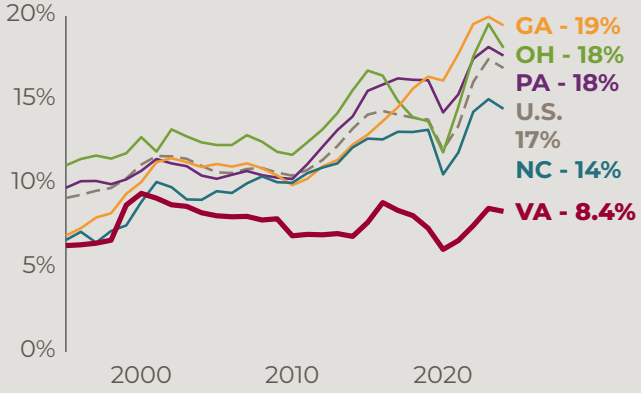
The four states closest in total fall enrollment, measured proportionally (a state 1.5 times as large counts as near as one two-thirds the size), are North Carolina, Georgia, Ohio, and Pennsylvania. Virginia ranks 5th of the five on the international share of STEM master's; Ohio is the most dependent at 58%.

Figure 22 How Dependent Are Virginia and Its Peers on International STEM Master's?



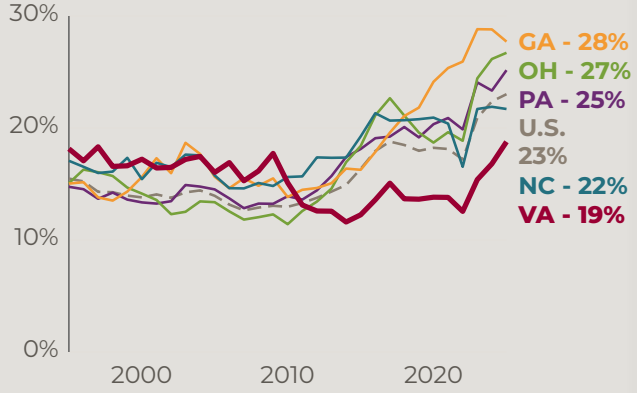
Data from NCES, IPEDS Completions.

Figure 23 Int. share of grad students



Data from NCES, IPEDS.

Figure 24 STEM share of master's



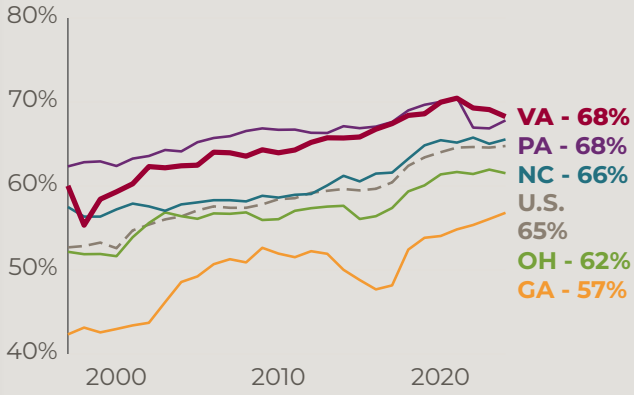
Data from NCES, IPEDS.

Table 4 How Does Virginia Compare With States of Similar Size?

Measure	VA	NC	GA	OH	PA	U.S.
Students enrolled, 2024	**575K**	582K	594K	611K	637K	19.7M
Graduate share of students, 2024	**22%**	14%	18%	14%	23%	17%
STEM share of bachelor's, 2025	**24%**	26%	27%	24%	27%	24%
STEM share of master's, 2025	**19%**	22%	28%	27%	25%	23%
International STEM master's, 2025	**2.07K**	2.42K	3.78K	3.84K	5.51K	117K
Share of U.S. international STEM master's, 2025	**1.8%**	2.1%	3.2%	3.3%	4.7%	–
Int. share of STEM master's, 2025	**39%**	49%	44%	58%	54%	52%
Int. share of STEM research doctorates, 2025	**41%**	39%	44%	45%	45%	43%
Int. share of grad students, 2024	**8.4%**	14%	19%	18%	18%	17%
Int. share of undergraduate enrollment, 2024	**2.7%**	2.1%	2.6%	3.3%	3.9%	3.0%
Six-year graduation rate, 2024	**68%**	66%	57%	62%	68%	65%
In-state tuition, public 4-year, 2023	**$15.7K**	$7.54K	$7.14K	$11K	$17K	$9.71K
Appropriations per student, 2024	**$11.3K**	$15.6K	$10.8K	$7.76K	$7.14K	$11.4K
Average faculty salary, 2024	**$106K**	$95.1K	$95.3K	$93.3K	$109K	$106K
First-time students from in state, 2024	**75%**	79%	82%	76%	68%	–
Net migration per 100 residents, 2024	**+2**	+4	−6	+5	+8	–

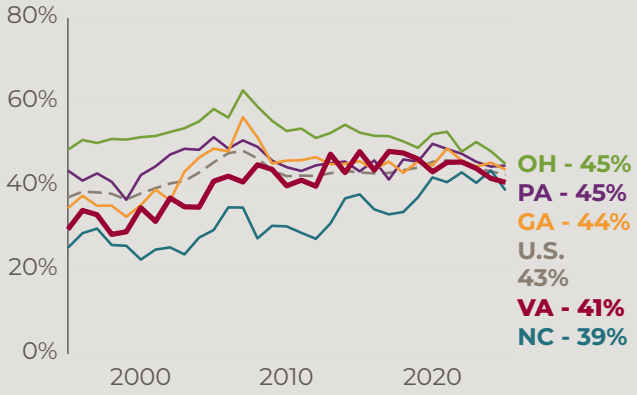
Latest year for each measure, given after its name. Dollars are 2024 dollars (CPI-U). Data from NCES, IPEDS.

Figure 25 Six-year graduation rate



Data from NCES, IPEDS.

Figure 26 Int. share of STEM doctorates

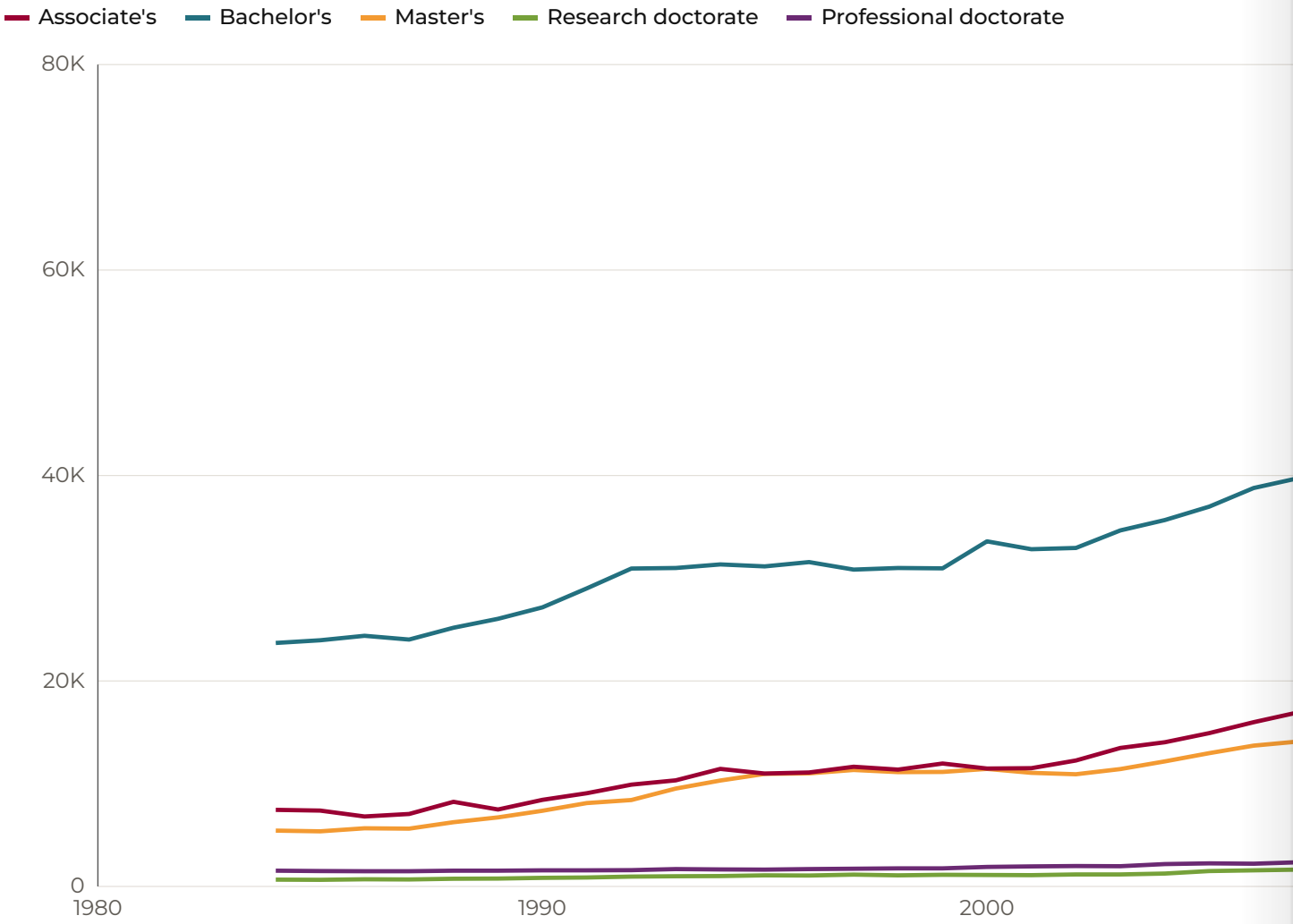


Data from NCES, IPEDS.

SECTION 12

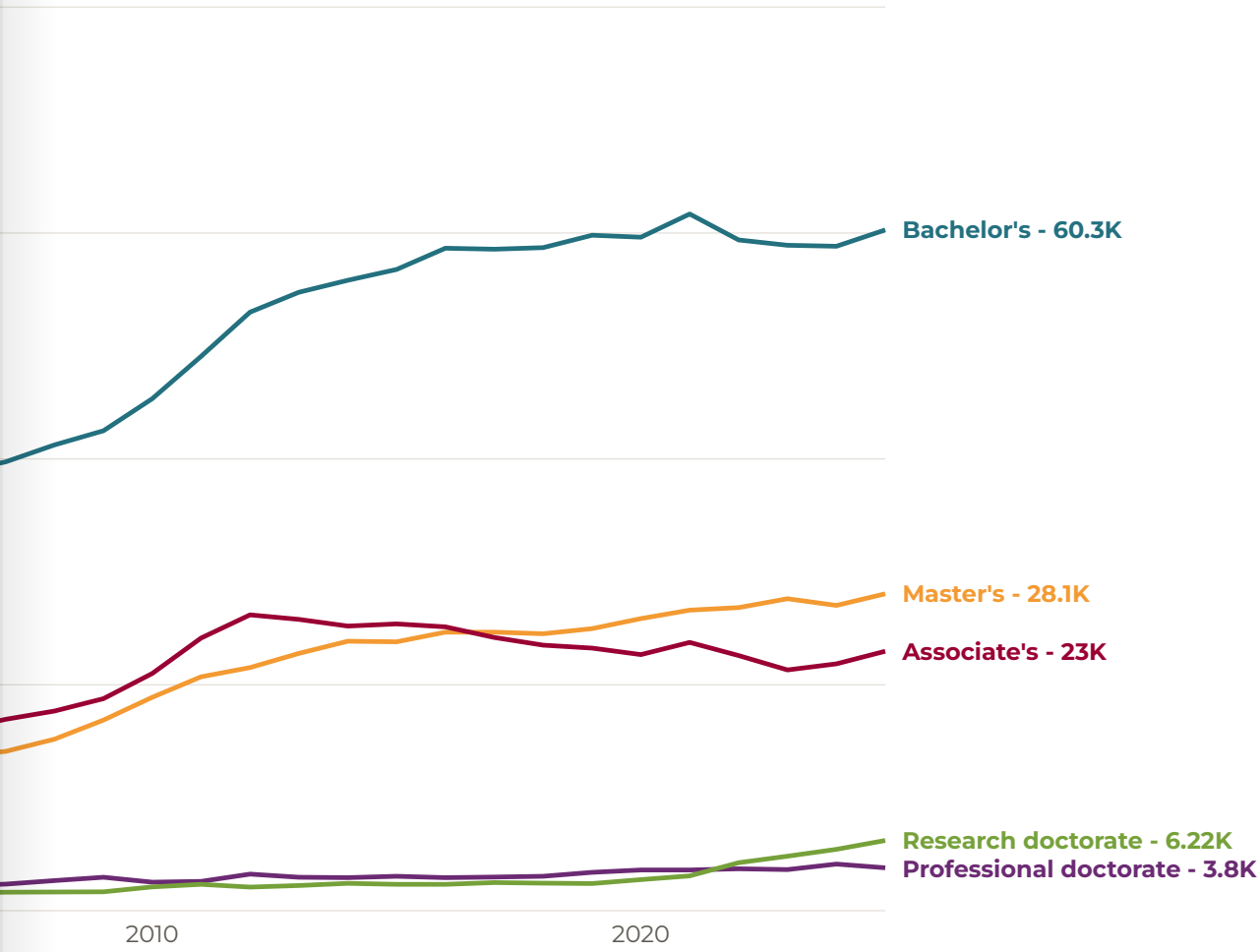
Fields of Study

Figure 27 How Many Degrees Does Virginia Award?



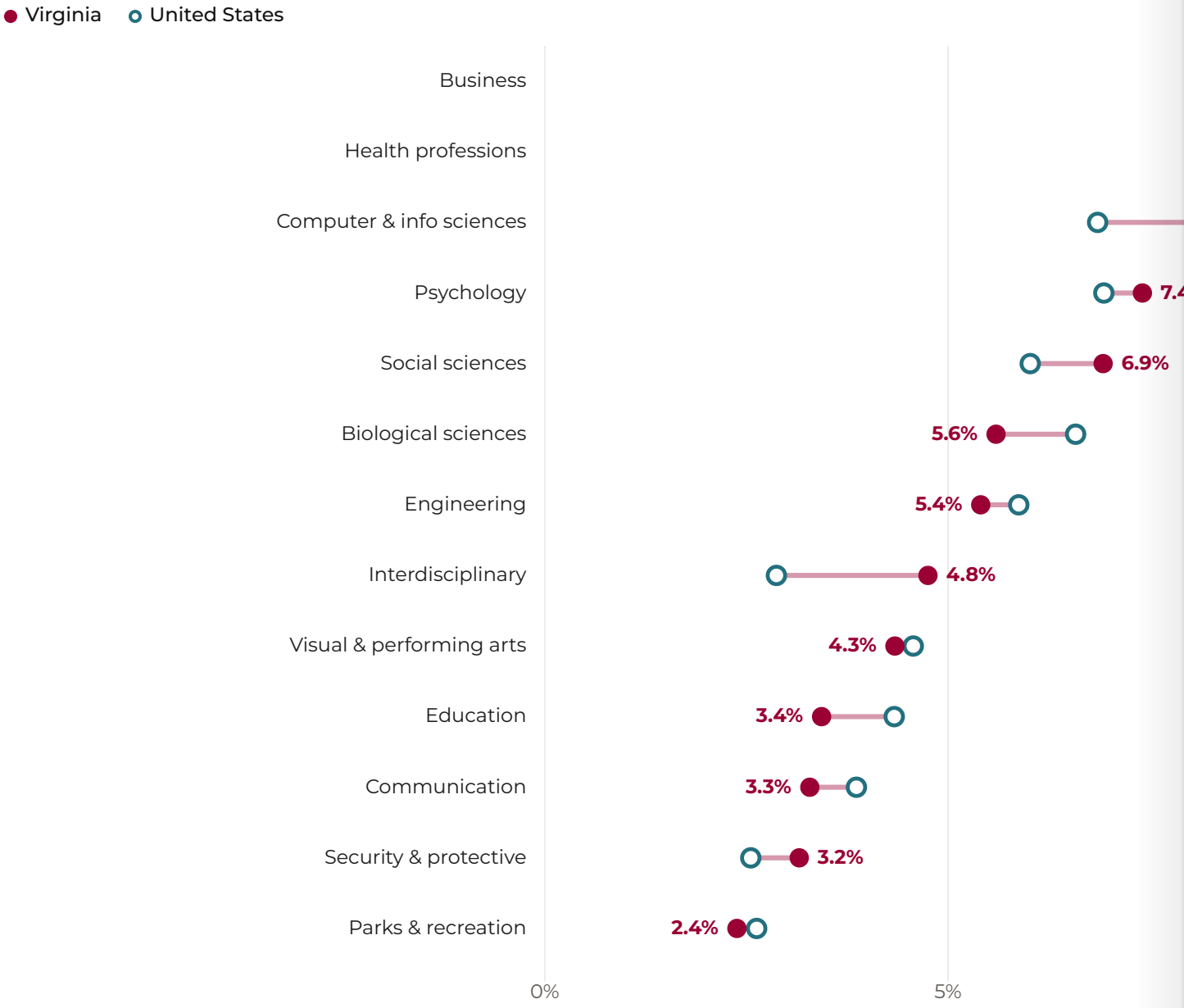
Degrees awarded by level, first majors. Data from NCES, IPEDS Completions.

In 2025, Virginia's institutions awarded 23K associate's degrees, 60.3K bachelor's degrees, 28.1K master's degrees, 6.22K research doctorate degrees, 3.8K professional doctorate degrees.



Since 2015: associate's down 10%; bachelor's up 6%; master's up 18%; research doctorate up 168%; professional doctorate up 25%. The state awards 13.4 bachelor's degrees for every 100 undergraduates (U.S. 12.3).

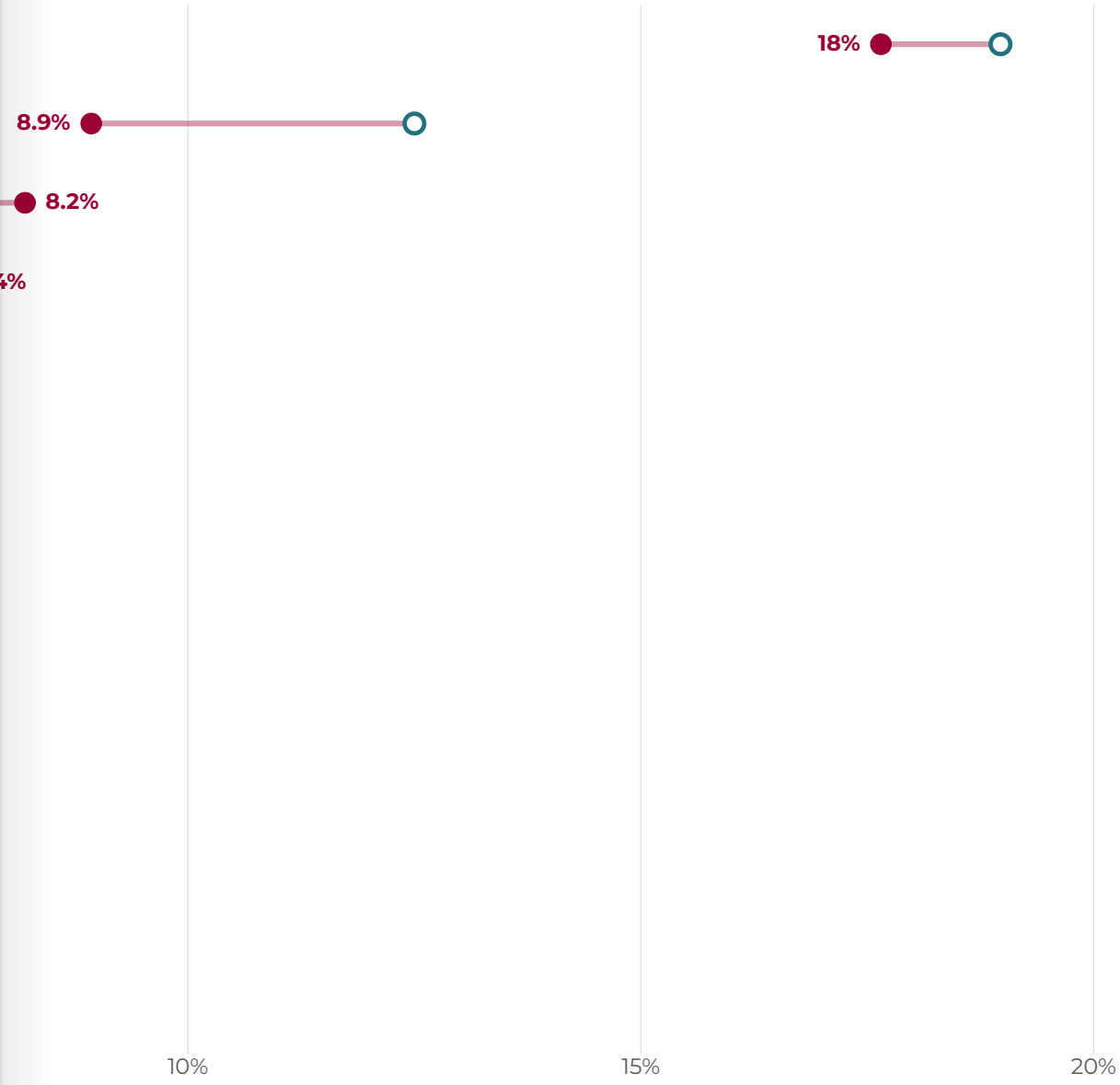
Figure 28 What Do Virginia's Bachelor's Graduates Study?



Share of bachelor's degrees by field, 2025. Data from NCES, IPEDS.

Virginia's largest bachelor's fields are business (18%; U.S. 19%), health professions (8.9%; U.S. 13%), computer & info sciences (8.2%; U.S. 6.9%).

Relative to the nation, Virginia's bachelor's degrees are most concentrated in transportation (2.46× the U.S. share), law (2.45× the U.S. share), military technologies (2.19× the U.S. share).



8.2%

Bachelor's in computer science, 2025

U.S. **6.9%** · **10th** of 51 · +3.7 pp since 2015

6.4%

Bachelor's in engineering & eng. technology, 2025

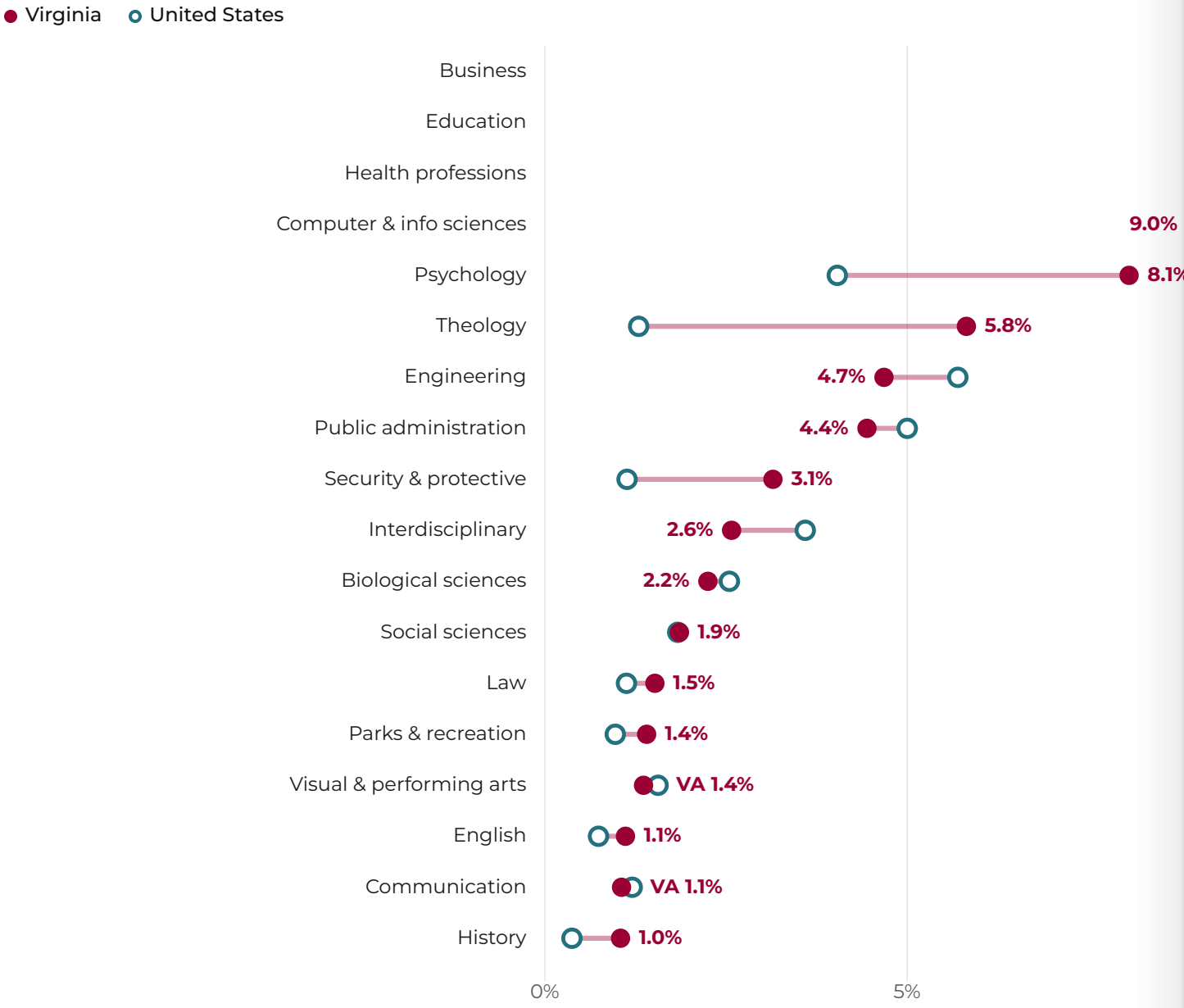
U.S. **6.8%** · **26th** of 51 · +0.4 pp since 2015

18%

Bachelor's in business, 2025

U.S. **19%** · **34th** of 51 · −0.4 pp since 2015

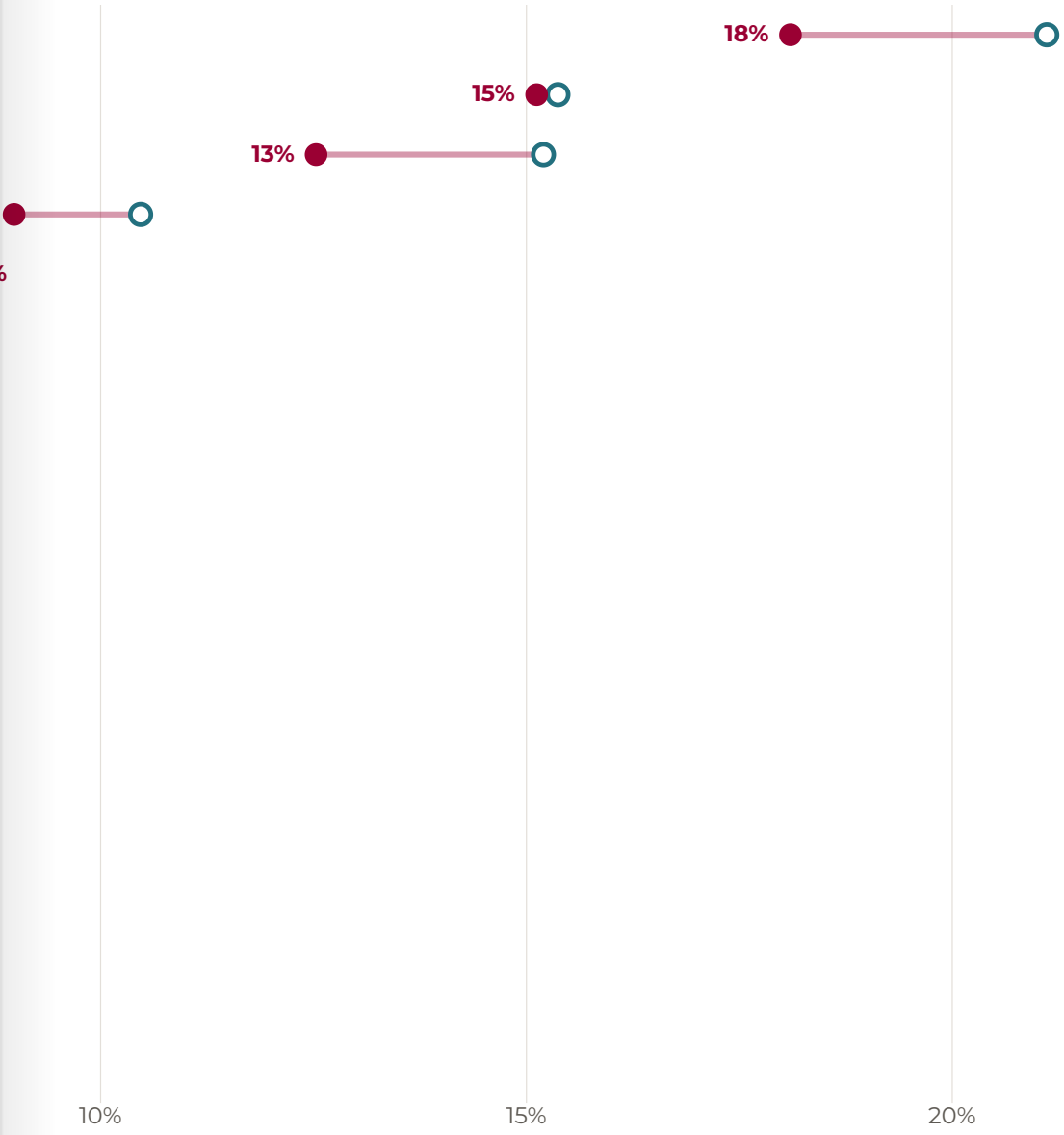
Figure 29 What Do Virginia's Master's Graduates Study?



Share of master's degrees by field, 2025; largest fields. Data from NCES, IPEDS.

Virginia's largest master's fields are business (18%; U.S. 21%), education (15%; U.S. 15%), health professions (13%; U.S. 15%).

International students earn 12% of all master's degrees in the state (U.S. 22%), and STEM fields account for 19% of them.



19%

STEM share of master's, 2025

U.S. **23%** · **29th** of 51 · +6.5 pp since 2015

12%

Int. share of all master's, 2025

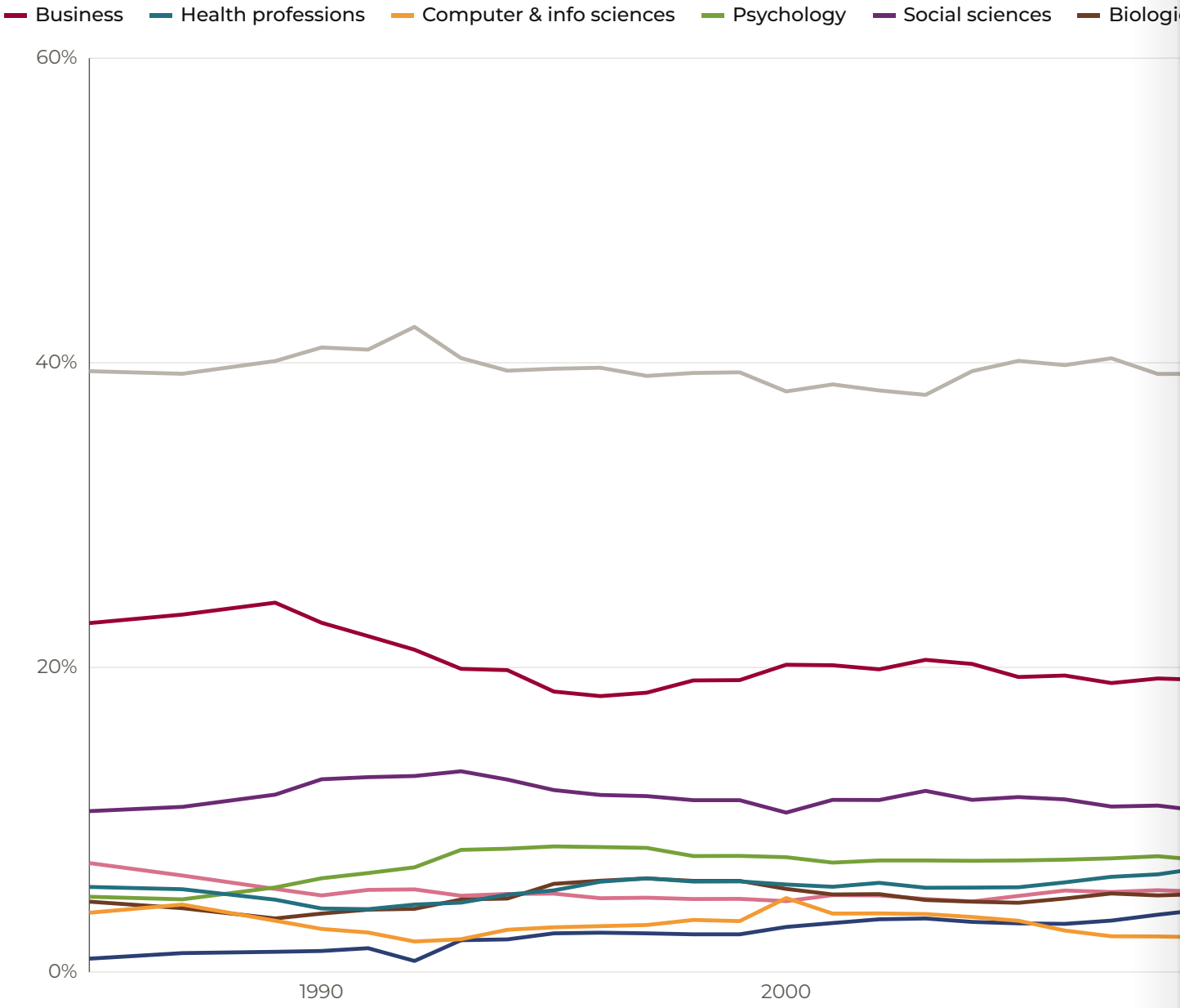
U.S. **22%** · **32nd** of 51 · +5.7 pp since 2015

59%

Int. share of CS master's, 2025

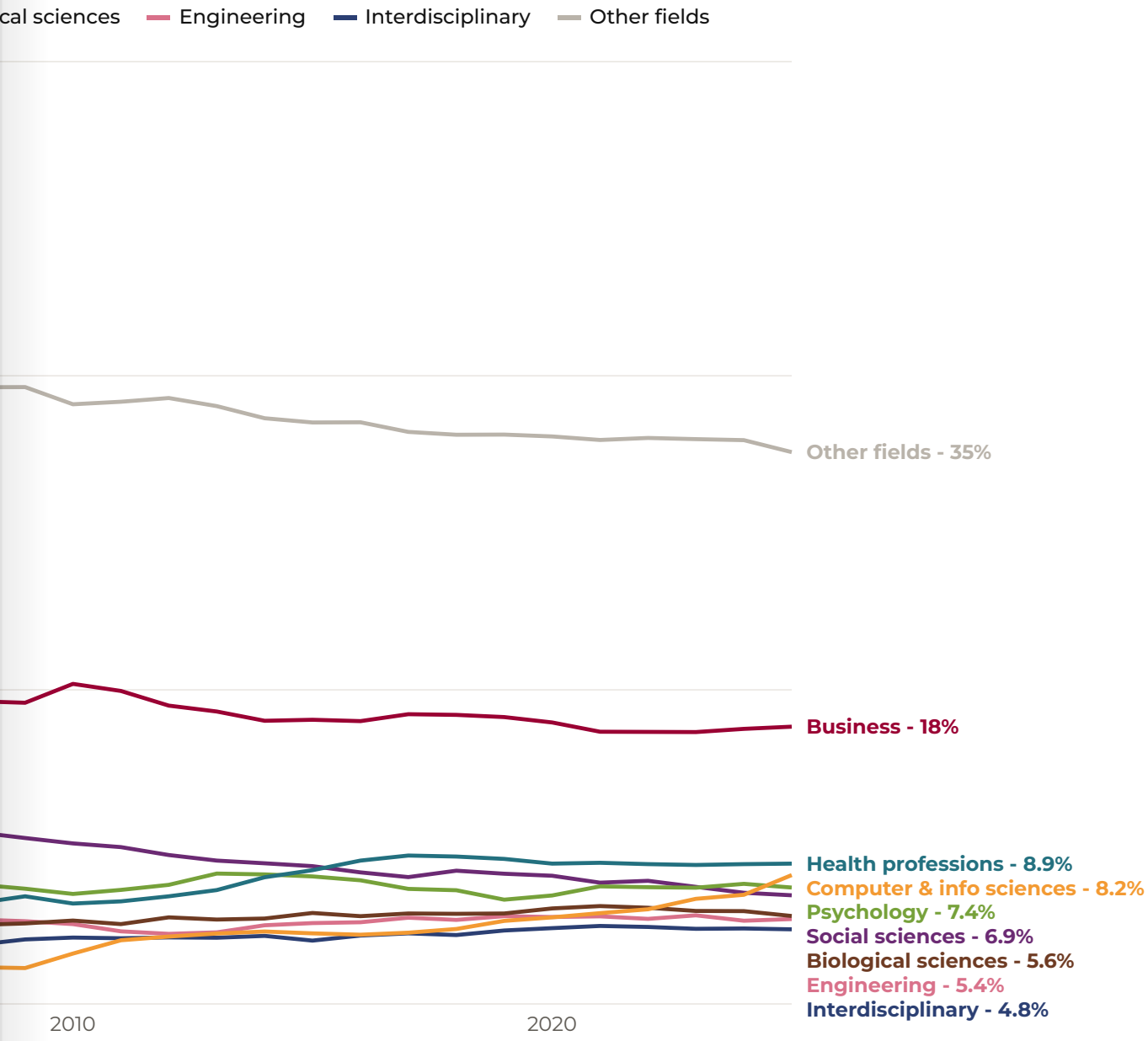
U.S. **65%** · **29th** of 47 · +30.9 pp since 2015

Figure 30 How Has Virginia's Bachelor's Field Mix Changed?



Share of bachelor's degrees in the largest fields. Data from NCES, IPEDS Completions.

Since 2015, computer & info sciences has gained the most ground, 3.7 points of the state's bachelor's degrees, while social sciences lost the most (1.8 points).



12

The chart follows the share of bachelor's degrees in each of the state's largest fields; smaller fields are grouped as other fields.

SECTION 12 · FIELDS OF STUDY

Where Is Virginia Specialized, and What Do Its Doctoral Graduates Study?

The location quotient divides a field's share of Virginia's bachelor's degrees by its share of all U.S. bachelor's degrees in 2025. A quotient of 1.00 means the same share as the nation; 2.00 means twice the national share. The table lists the 10 fields with the highest quotients among fields with at least 50 degrees.

Highest location quotients. Transportation (2.46, with 0.9% of the state's bachelor's degrees against 0.4% for the U.S.), law (2.45, with 0.6% of the state's bachelor's degrees against 0.2% for the U.S.), and military technologies (2.19, with 0.2% of the state's bachelor's degrees against 0.1% for the U.S.) had the highest quotients.

Table 5 Where Is Virginia Most Specialized?

Field	Location quotient	State share	U.S. share	Degrees
Transportation	**2.46**	0.9%	0.4%	539
Law	**2.45**	0.6%	0.2%	347
Military technologies	**2.19**	0.2%	0.1%	116
Theology	**2.16**	0.6%	0.3%	371
Philosophy & religion	**2.13**	1.2%	0.6%	718
History	**1.70**	1.7%	1.0%	1.05K
Interdisciplinary	**1.65**	4.8%	2.9%	2.86K
English	**1.35**	2.0%	1.5%	1.22K
Family & consumer sci.	**1.30**	1.1%	0.8%	644
Physical sciences	**1.29**	1.7%	1.3%	1K

Bachelor's degrees; location quotient = state share ÷ U.S. share; fields with 50+ degrees. Data from NCES, IPEDS.

24%
STEM share of bachelor's, 2025
U.S. **24%** · **21st** of 51

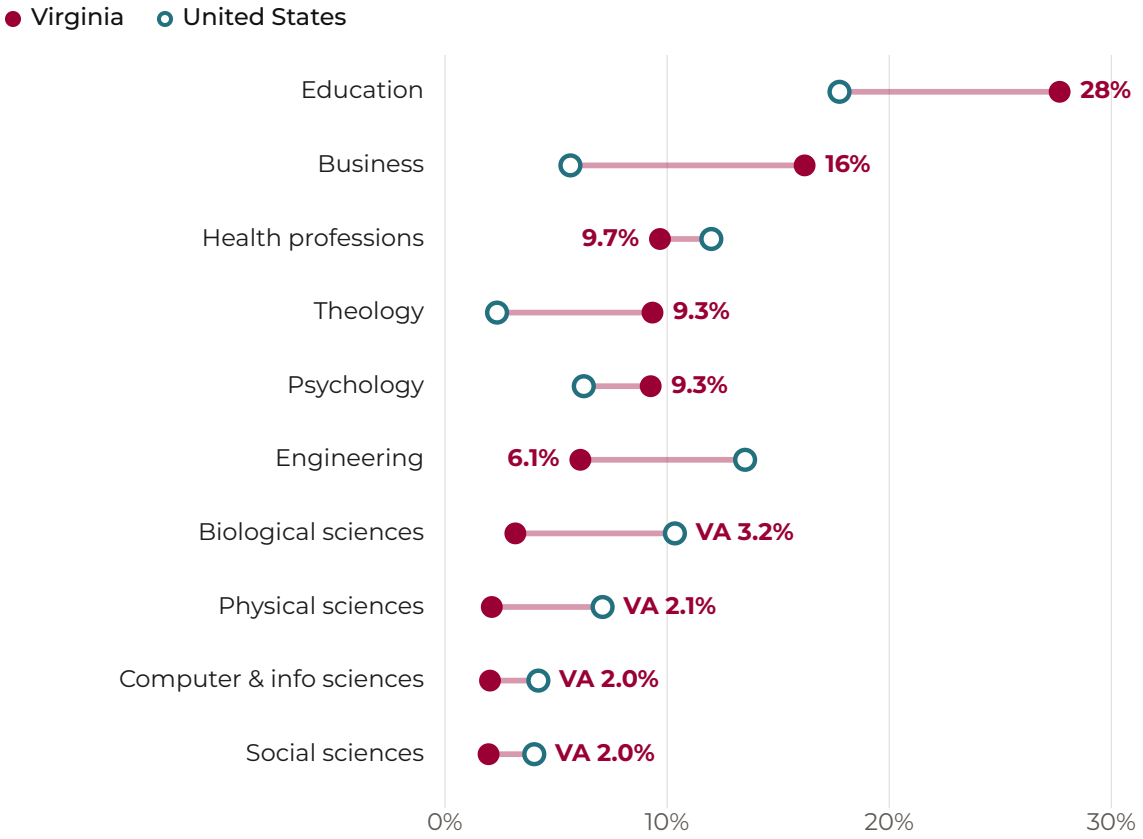
8.2%
Bachelor's in computer science, 2025
U.S. **6.9%** · **10th** of 51

6.4%
Bachelor's in engineering & eng. technology, 2025
U.S. **6.8%** · **26th** of 51

Each row is a field's share of the research doctorates awarded in Virginia in 2025 (filled dot), against its share nationally (open dot), for the ten largest fields with at least 20 doctorates. Research doctorates exclude professional doctorates such as the M.D. and J.D.

Largest doctoral fields. The state's largest doctoral fields were education (28% of research doctorates, against 18% for the U.S.), business (16% of research doctorates, against 5.6% for the U.S.), and health professions (9.7% of research doctorates, against 12% for the U.S.).

Figure 31 What Do Virginia's Doctoral Graduates Study?



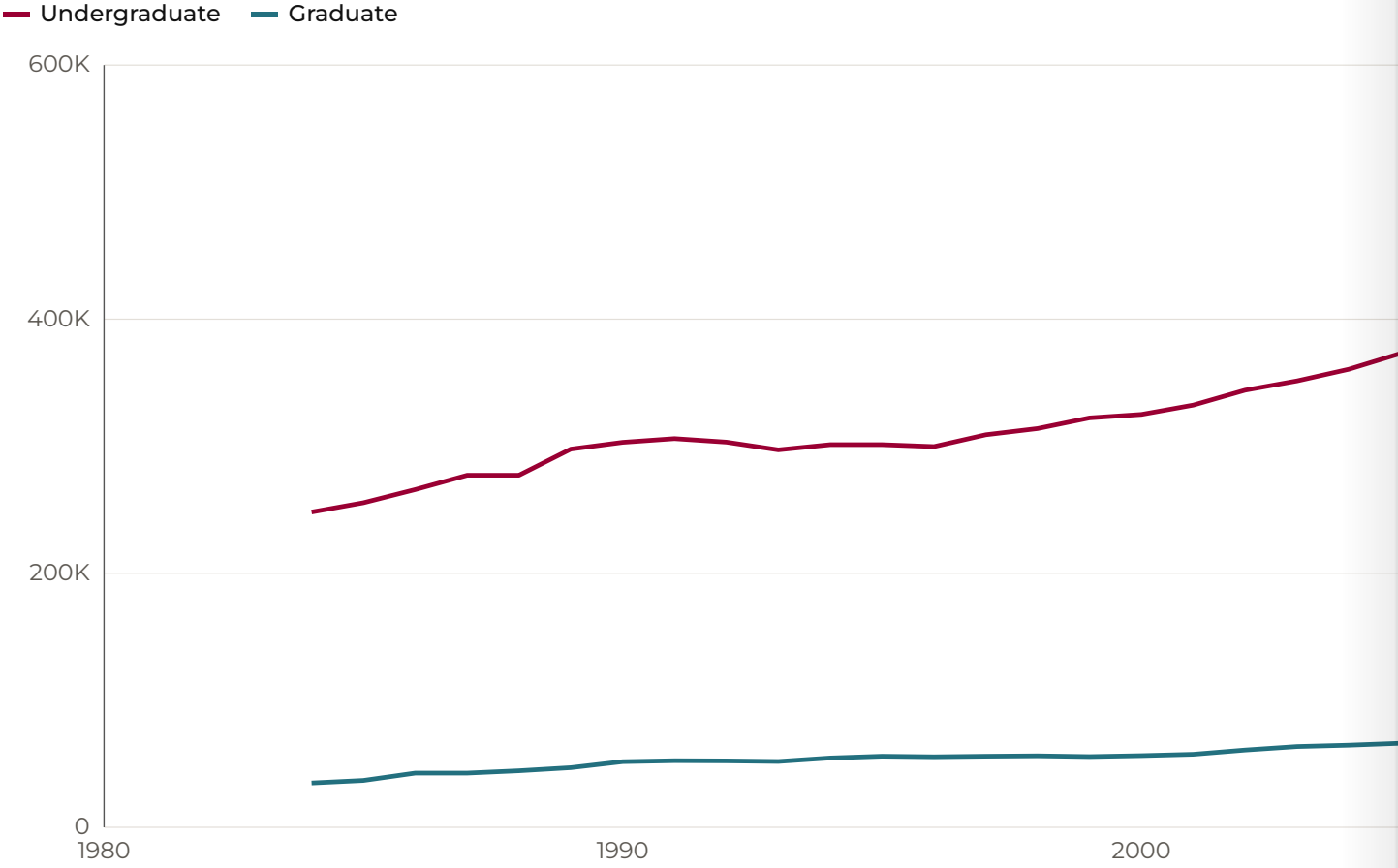
Share of research doctorates by field, 2025; fields with 20+ doctorates. Data from NCES, IPEDS.

SECTION 13

Students and Money

Virginia ranks 9th of 51 on the share of students who are graduate students, at 22% in 2024 against 17% for the U.S.

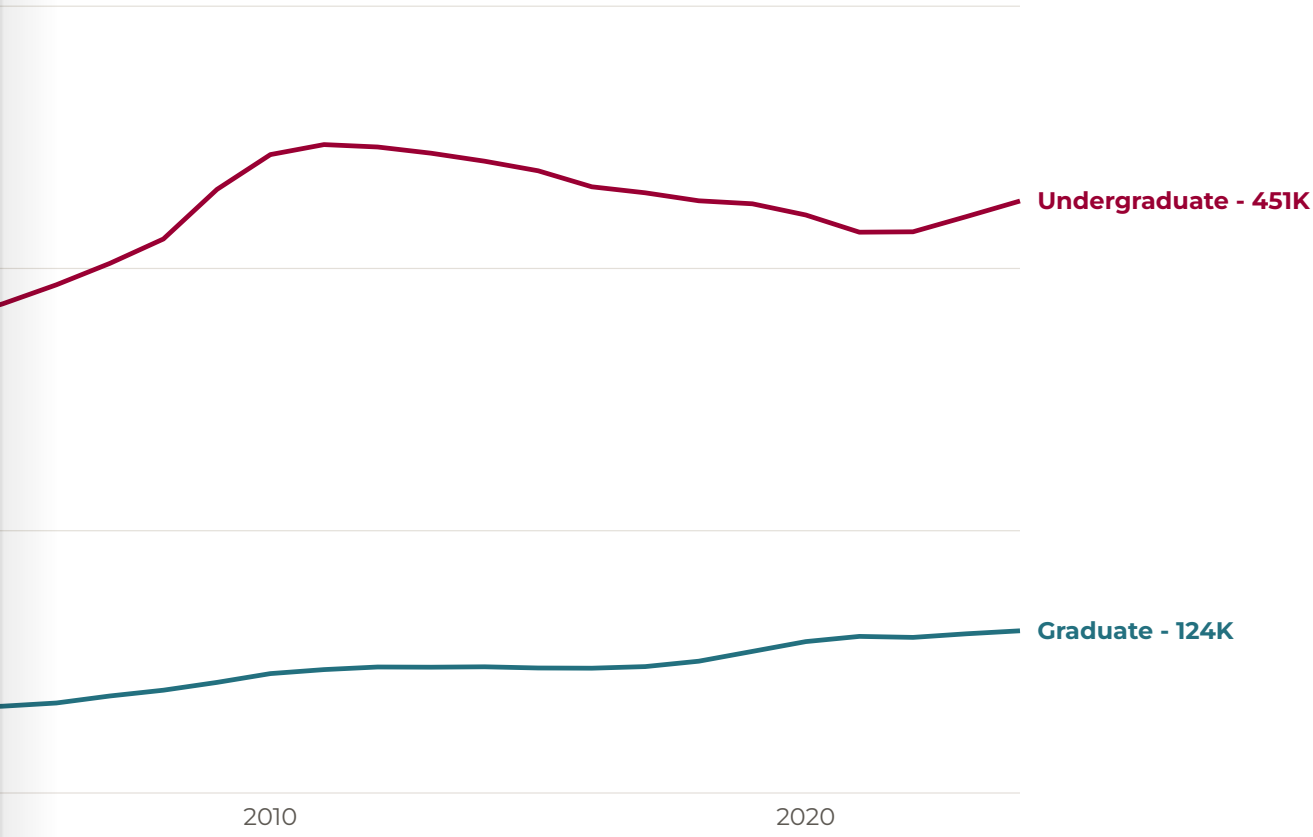
Figure 32 How Has Enrollment Changed?



Fall headcount enrollment. Data from NCES, IPEDS Fall Enrollment.

In fall 2024, Virginia enrolled 451K undergraduates and 124K graduate students, 575K in all (11th of 51).

Undergraduate enrollment is down 5% since 2015 and graduate enrollment up 30% since 2015. Graduate students are 22% of all students (U.S. 17%).



575K

Students enrolled, 2024

U.S. **19.7M** · **11th** of 51 · +0.9% since 2015

22%

Graduate share of students, 2024

U.S. **17%** · **9th** of 51 · +4.8 pp since 2015

38%

Students attending part-time, 2024

U.S. **39%** · **31st** of 51 · −1.2 pp since 2015

SECTION 13 · STUDENTS AND MONEY

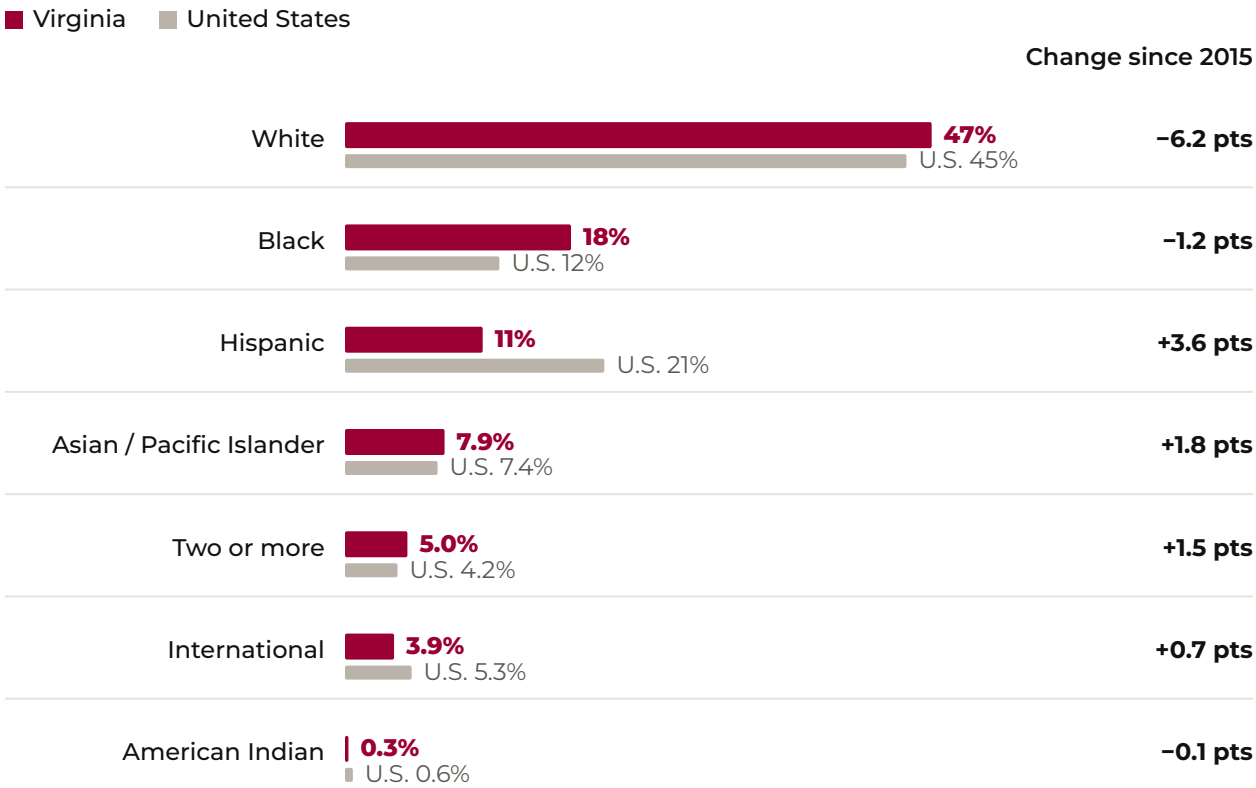
Who Enrolls in Virginia, and Where?

In fall 2024, White students were the largest group at Virginia's degree-granting institutions, **47%** of enrollment against 45% nationally, down 6.2 points since 2015.

Black students were **18%** (U.S. 12%), down 1.2 points since 2015, and Hispanic students were **11%** (U.S. 21%), up 3.6 points since 2015.

Asian and Pacific Islander students were **7.9%** (U.S. 7.4%), up 1.8 points since 2015, and students of two or more races were **5.0%** (U.S. 4.2%), up 1.5 points since 2015. "International" counts U.S. nonresidents on temporary visas, whatever their race.

Figure 33 Who Enrolls in Virginia?



Share of fall 2024 headcount enrollment by race and ethnicity; students of unknown race are left out of the chart. Data from NCES, IPEDS.

47%
White students
U.S. **45%** · −6.2 pp since 2015

18%
Black students
U.S. **12%** · −1.2 pp since 2015

39%
At public 4-year institutions
+1.7 pp since 2015

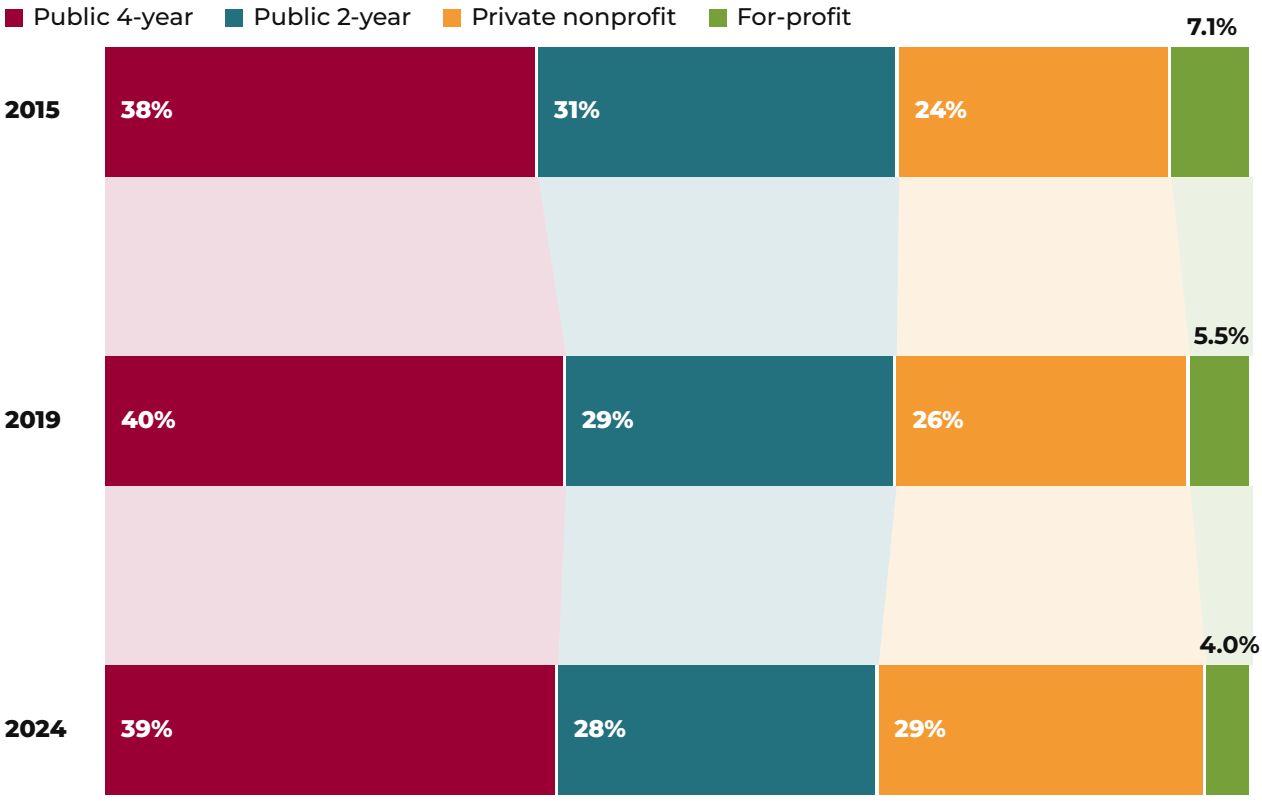
29%
At private nonprofit institutions
+4.9 pp since 2015

In fall 2024, public 4-year institutions enrolled **39%** of students, up 1.7 points from 38% in 2015. Private nonprofit institutions enrolled **29%** of students, up 4.9 points from 24% in 2015.

Public 2-year institutions enrolled **28%** of students, down 3.5 points from 31% in 2015, and for-profit institutions enrolled **4.0%** of students, down 3.1 points from 7.1% in 2015.

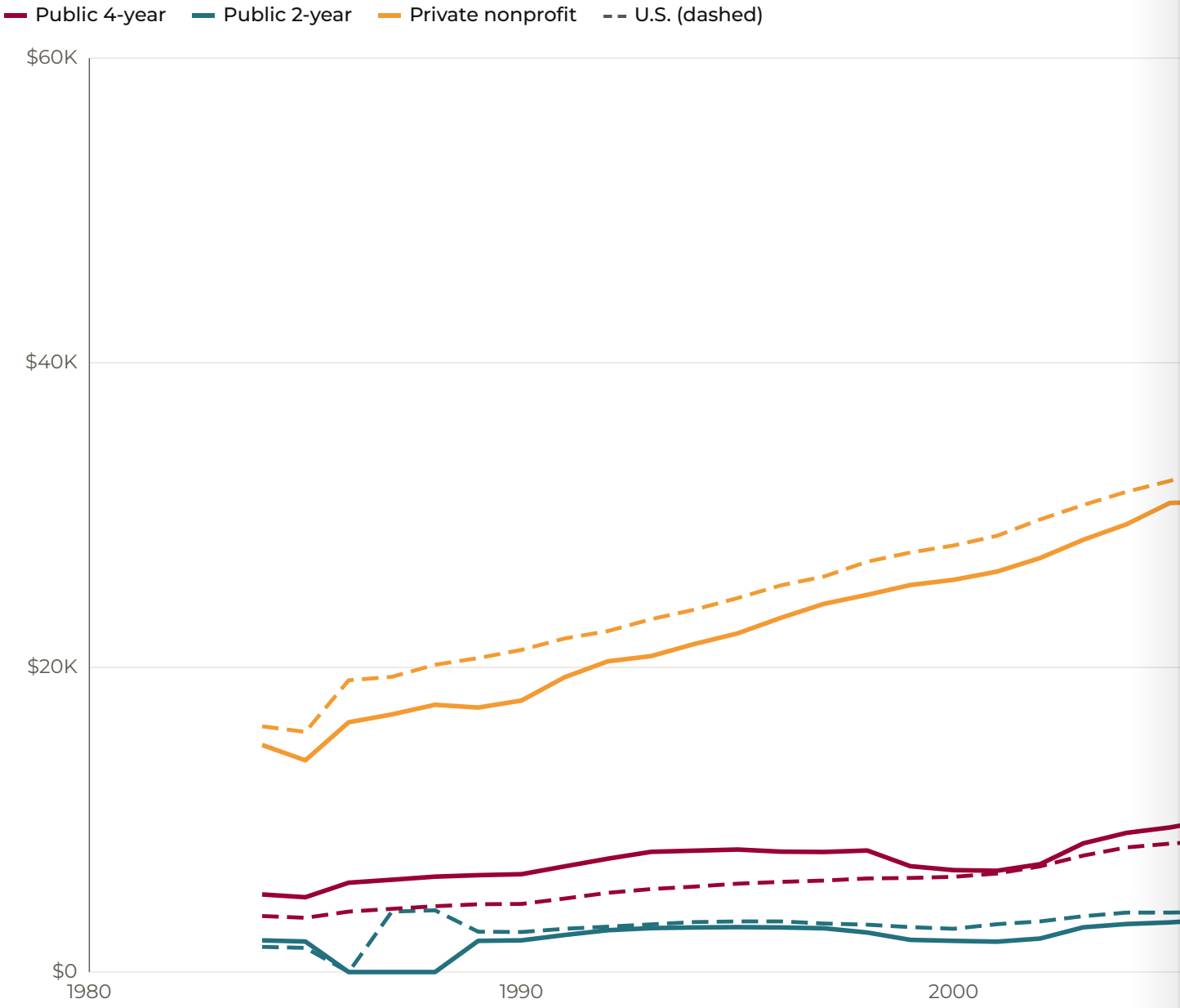
Public institutions together enrolled **67%** of students, against 73% nationally, 35th among the states.

Figure 34 Where Do Students Enroll?



Share of fall headcount enrollment by sector, 2015, 2019, 2024. Sectors as IPEDS classifies them each year. Data from NCES, IPEDS.

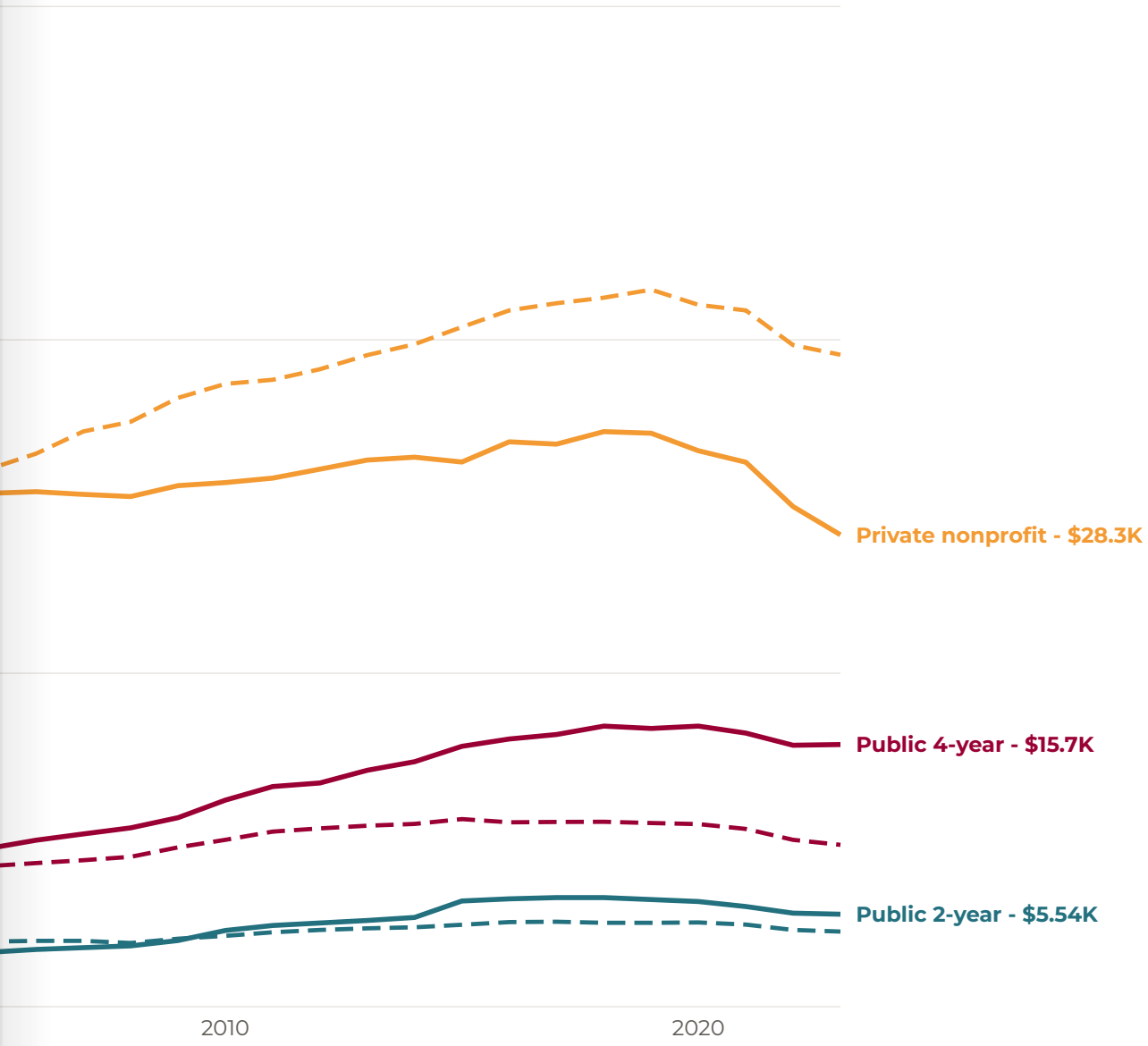
Figure 35 What Does In-State Tuition Cost?



Published in-state tuition and fees, 2024 dollars (CPI-U); state solid, U.S. dashed. Data from NCES, IPEDS.

In-state tuition and fees at Virginia's public four-year institutions are $15.7K (2023, 2024 dollars), 62% above the U.S. ($9.71K); the state ranks 6th of 51.

State and local appropriations come to $11.3K per full-time-equivalent student at public institutions (22nd of 51), and net tuition makes up 50% of net tuition plus state and local appropriations (U.S. 43%).



$15.7K

In-state tuition, public 4-year, 2023

U.S. **$9.71K** · **6th** of 51 · +0.7% since 2015

$11.3K

Appropriations per student, 2024

U.S. **$11.4K** · **22nd** of 51 · +51% since 2015

50%

Tuition share of tuition plus appropriations, 2024

U.S. **43%** · **19th** of 51 · –11.5 pp since 2015

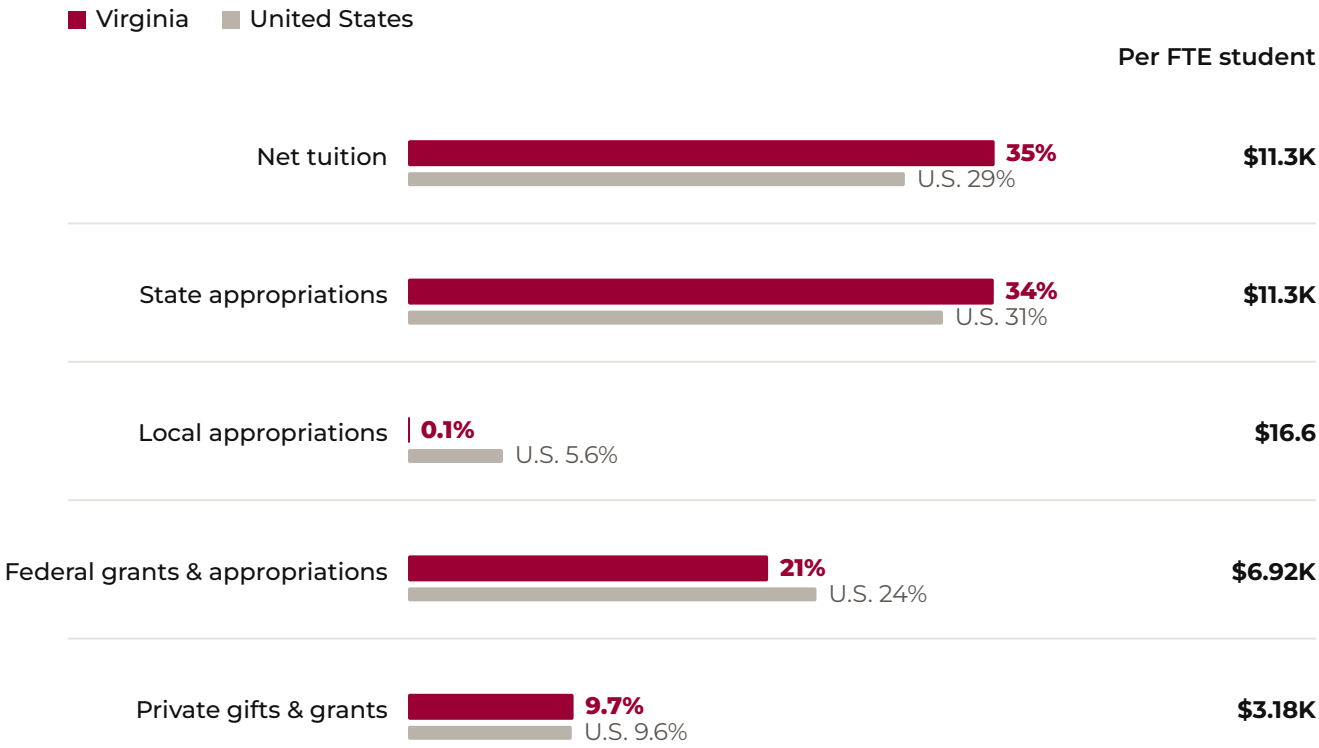
SECTION 13 · STUDENTS AND MONEY

How Are Virginia's Public Institutions Funded, and What Do They Produce?

The bars give each source's share of the core revenue of Virginia's public institutions, with the nation for comparison, and at right the dollars per full-time-equivalent (FTE) student it provides. Values are in 2024 dollars (CPI-U). Net tuition is tuition and fees after institutional discounts and allowances.

Net tuition per FTE student. The figure was $11.3K in FY2024, against $8.96K for the U.S., down 3.1% from $11.6K in FY2023 and down 5.6% since FY2015, when it was $11.9K.

Figure 36 How Are Public Institutions Funded?



Share of core revenue per FTE student by source, public institutions, FY2024; dollars per FTE at right, 2024 dollars (CPI-U). Shares are of all five sources; the tuition share elsewhere in this profile divides net tuition by net tuition plus state and local appropriations only. Data from NCES, IPEDS Finance.

$15.7K
In-state tuition, public 4-year, 2023
U.S. **$9.71K** · **6th** of 51

$11.3K
Appropriations per student, 2024
U.S. **$11.4K** · **22nd** of 51

50%
Tuition share of tuition plus appropriations, 2024
U.S. **43%** · **19th** of 51

The table brings together five measures of what public higher education costs and what it produces in Virginia: in-state tuition at public four-year institutions, state and local appropriations per full-time-equivalent student, tuition's share of public institutions' core revenue, the six-year bachelor's graduation rate, and the average full-time faculty salary.

In-state tuition, public 4-year. The figure was $15.7K in 2023, against $9.71K for the U.S., 6th among the states, up 0.3% from $15.7K in 2022 and up 0.7% since 2015, when it was $15.6K.

Table 6 Money and Results

Measure	Year	Virginia	U.S.	Rank	Change since 2015
In-state tuition, public 4-year	2023	**$15.7K**	$9.71K	6th of 51	+0.7%
Appropriations per student	2024	**$11.3K**	$11.4K	22nd of 51	+51%
Tuition share of tuition plus appropriations	2024	**50%**	43%	19th of 51	−11.5 pp
Six-year graduation rate	2024	**68%**	65%	11th of 51	+2.4 pp
Average faculty salary	2024	**$106K**	$106K	16th of 51	+1.3%

Rank among the 50 states and DC (1 = highest). Dollars in 2024 dollars (CPI-U). Data from NCES, IPEDS.

SECTION 14

Student Outcomes

Virginia ranks 11th of 51 on the six-year bachelor's graduation rate, 4-year institutions, at 68% in 2024 against 65% for the U.S.

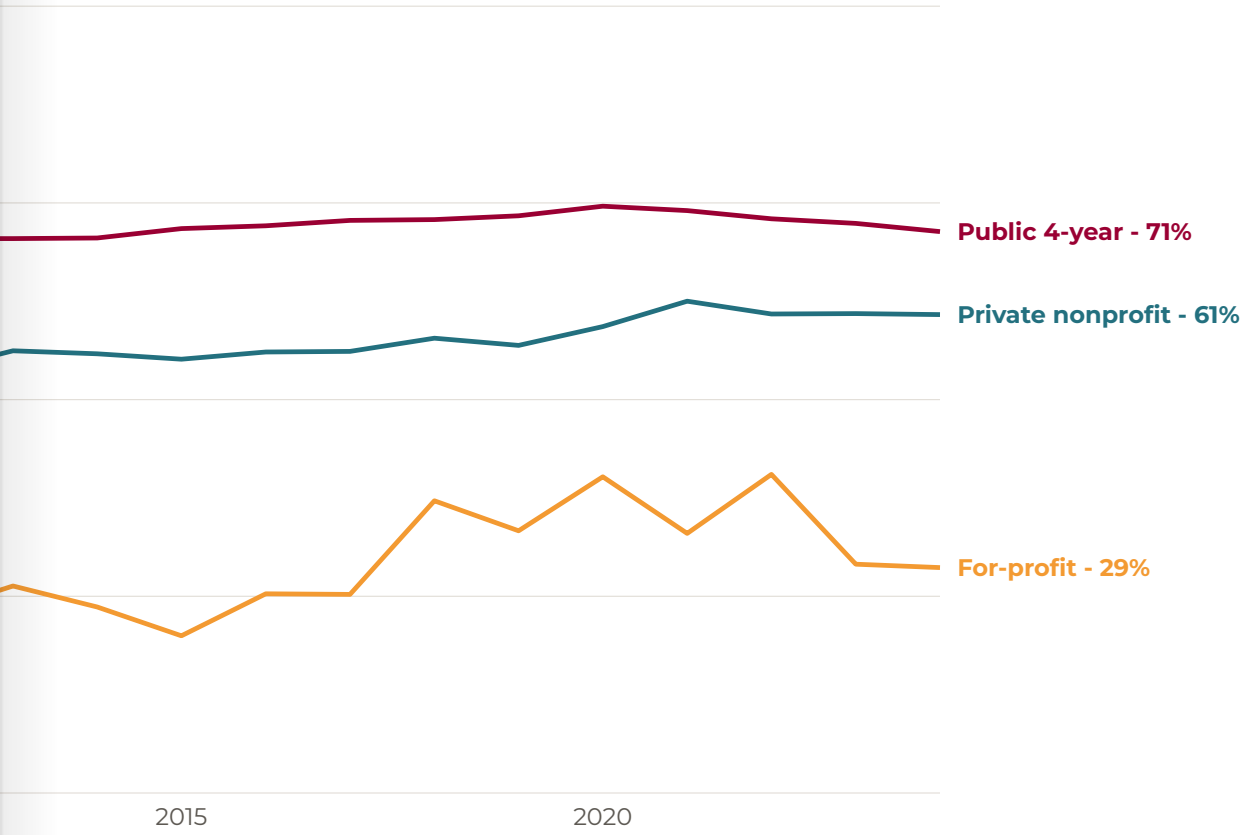
Figure 37 How Do Graduation Rates Differ by Sector?



Six-year bachelor's graduation rate by sector of institution. Data from NCES, IPEDS Graduation Rates.

68% of students who started a bachelor's degree at a four-year institution in Virginia graduated within six years (2024), 3.5 points above the U.S. (65%); the state ranks 11th of 51.

By sector: public 4-year 71%, private nonprofit 61%, for-profit 29%. At two-year institutions, 37% complete within 150% of normal time (U.S. 41%).



68%

Six-year graduation rate, 2024

U.S. **65%** · **11th** of 51 · +2.4 pp since 2015

37%

Two-year completion rate, 2024

U.S. **41%** · **29th** of 50 · +7.0 pp since 2015

13.4

Bachelor's per 100 undergraduates, 2025

U.S. **12.3** · **17th** of 51 · +1.6 since 2015

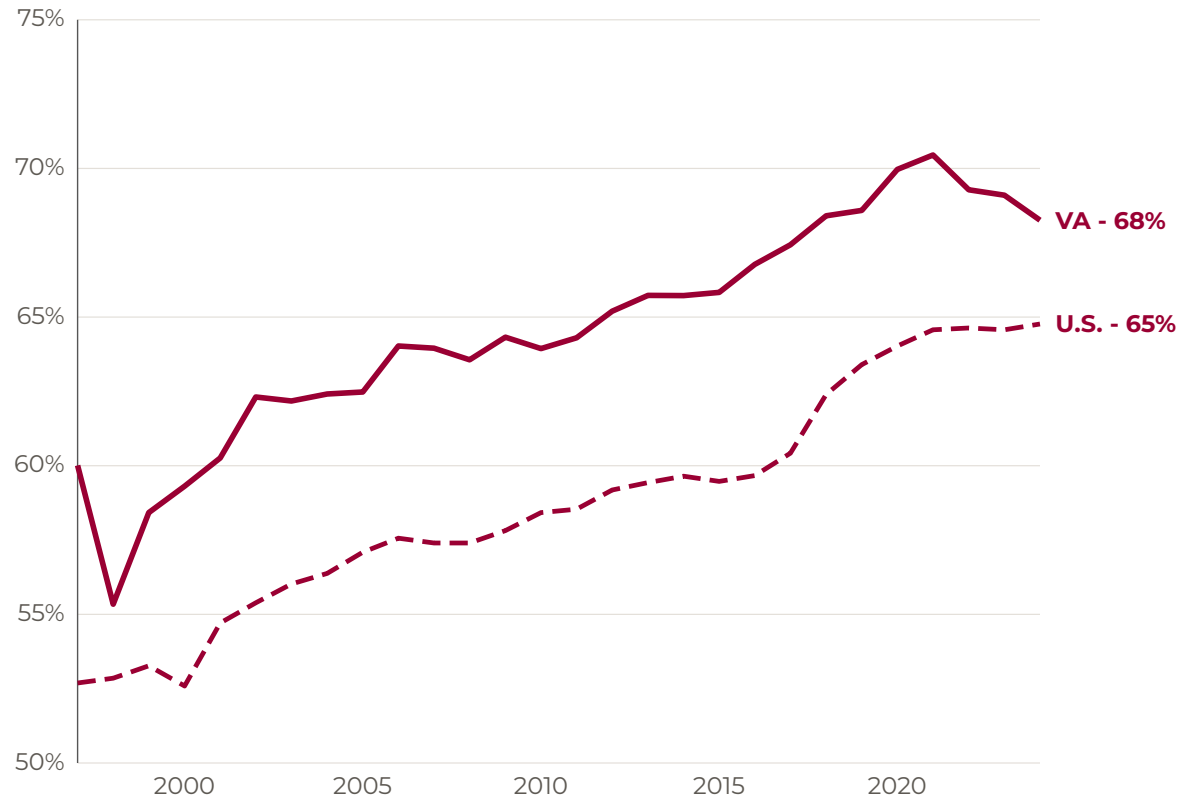
SECTION 14 · STUDENT OUTCOMES

How Many of Virginia's Students Graduate?

The line follows the share of first-time, full-time bachelor's-degree students at Virginia's four-year institutions who graduated from the institution where they started within six years (150% of normal time). Each year is the cohort that entered six years earlier. The rate leaves out students who start part-time (38% of the state's students attend part-time, against 39% nationally) and students who transfer and finish elsewhere. The state is solid and the United States dashed.

Six-year graduation rate. The rate was 68% in 2024, against 65% for the U.S., down 0.8 points from 69% in 2023 and up 2.4 points since 2015, when it was 66%.

Figure 38 Six-Year Bachelor's Graduation Rate



Four-year institutions; state solid, U.S. dashed. Data from NCES, IPEDS Graduation Rates.

68%
Six-year graduation rate, 2024
U.S. **65%** · **11th** of 51

37%
Two-year completion rate, 2024
U.S. **41%** · **29th** of 50

13.4
Bachelor's per 100 undergraduates, 2025
U.S. **12.3** · **17th** of 51

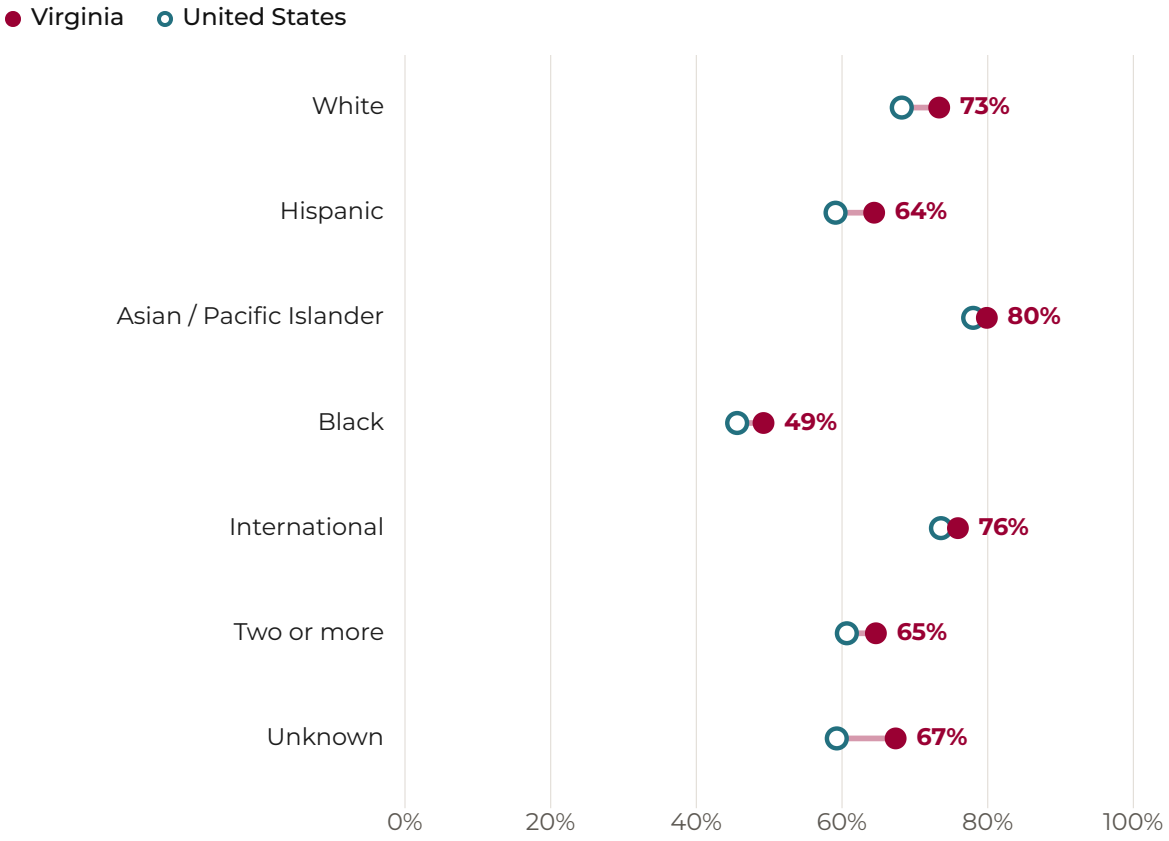
Each row is the six-year bachelor's graduation rate for one racial or ethnic group of students at Virginia's four-year institutions (filled dot), against the national rate for the same group (open dot). Groups with fewer than 100 students in the cohort are left out.

White. The six-year rate was 73% in 2024, against 68% for the U.S.

Hispanic. The six-year rate was 64% in 2024, against 59% for the U.S.

Asian / Pacific Islander. The six-year rate was 80% in 2024, against 78% for the U.S.

Figure 39 How Do Graduation Rates Differ by Race and Ethnicity?



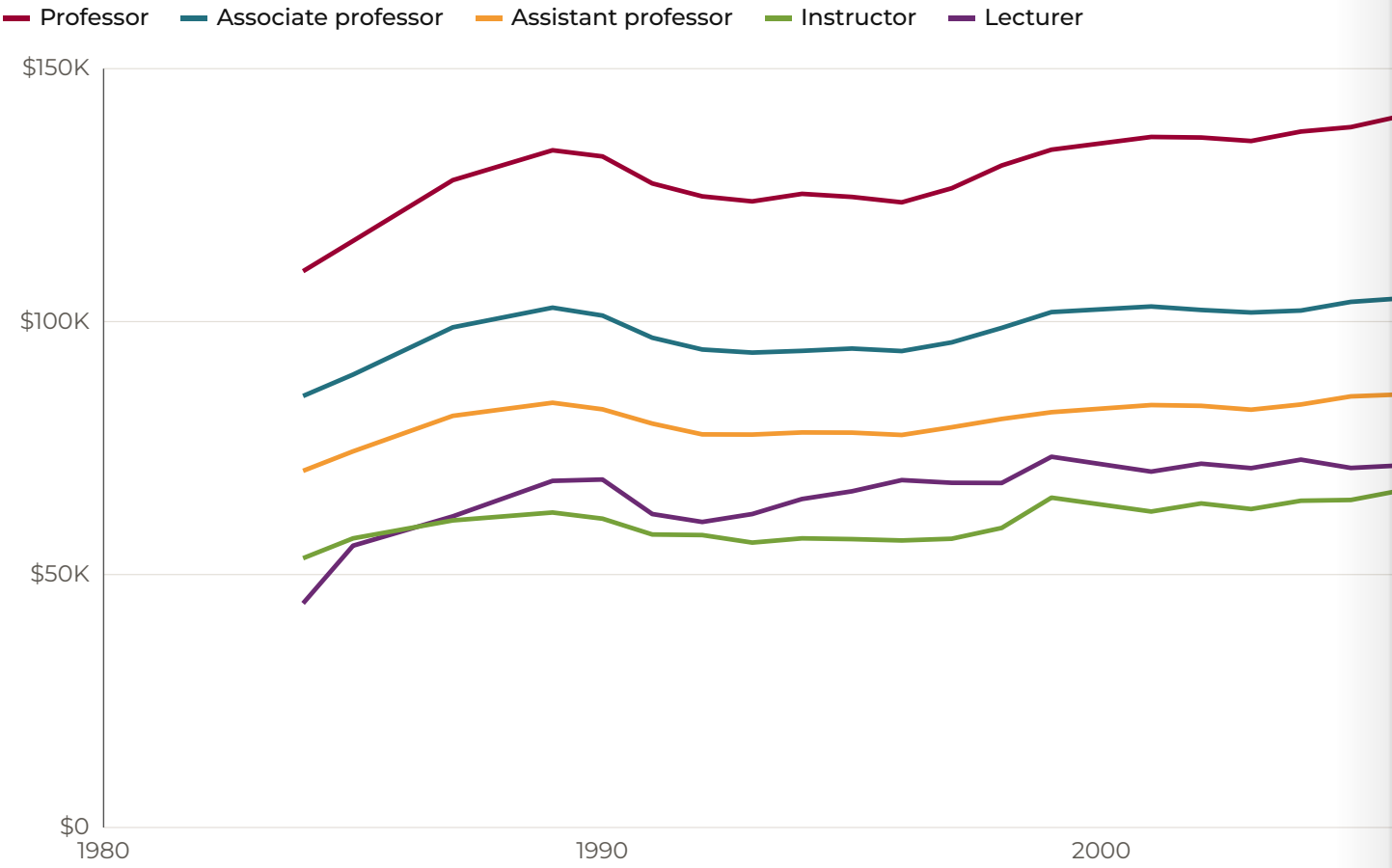
Six-year rate, 2024; groups with 100+ students. Data from NCES, IPEDS.

SECTION 15

Faculty

Virginia ranks 16th of 51 on the average full-time faculty salary, at $106K in 2024 against $106K for the U.S.

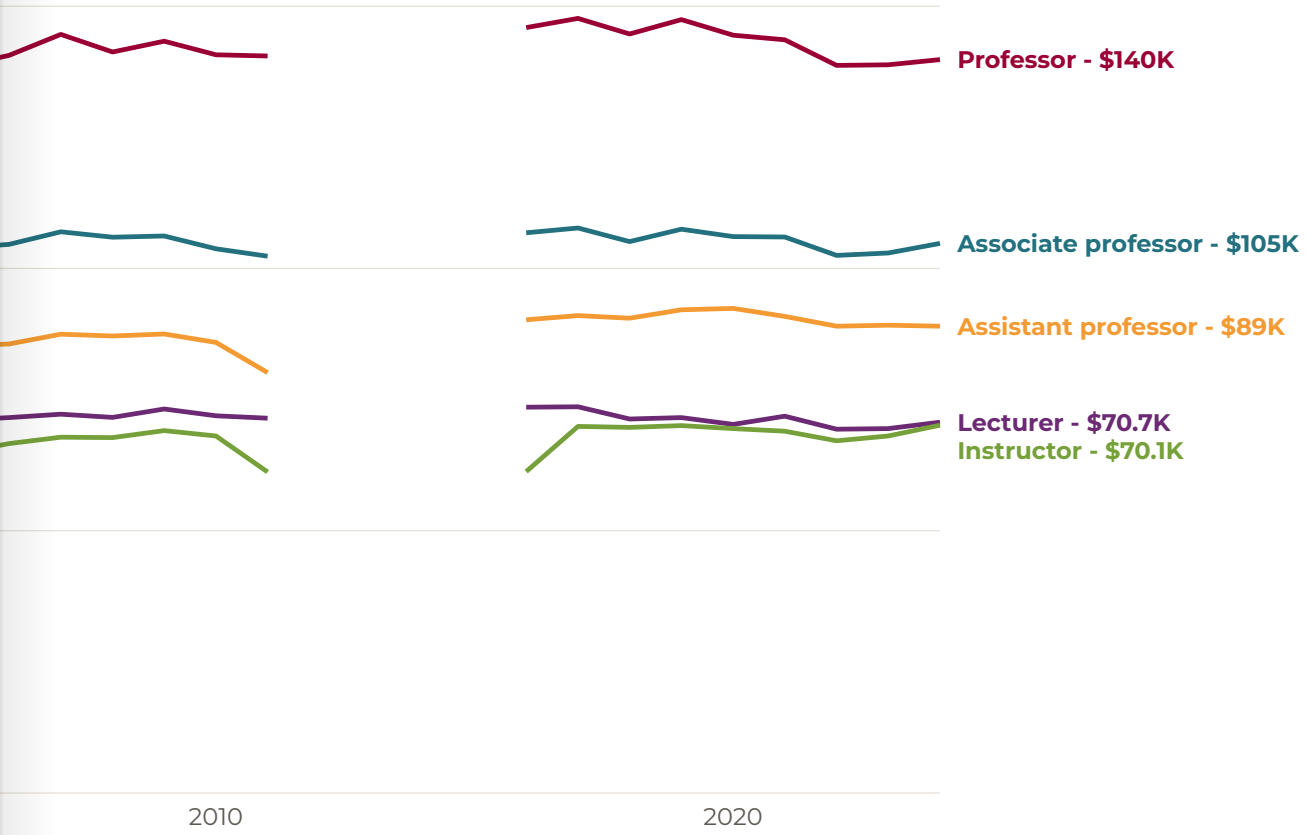
Figure 40 What Do Virginia's Faculty Earn, by Rank?



Average salary of full-time instructional faculty on 9/10-month contracts, by academic rank, 2024 dollars (CPI-U). Data fro

Full-time instructional faculty on 9/10-month contracts at Virginia's institutions earned an average of $106K in 2024 (2024 dollars), close to the U.S. figure ($106K); the state ranks 16th of 51. By rank (2024): professors $140K, associate professors $105K, assistant professors $89K, instructors $70.1K, lecturers $70.7K.

Virginia's institutions had 20.3K full-time instructional staff in 2024, roughly flat since 2015. Average salary across all ranks is roughly flat since 2016.



m NCES, IPEDS Human Resources.

$106K

Average faculty salary, 2024

U.S. **$106K** · **16th** of 51 · +1.3% since 2016

SECTION 15 · FACULTY

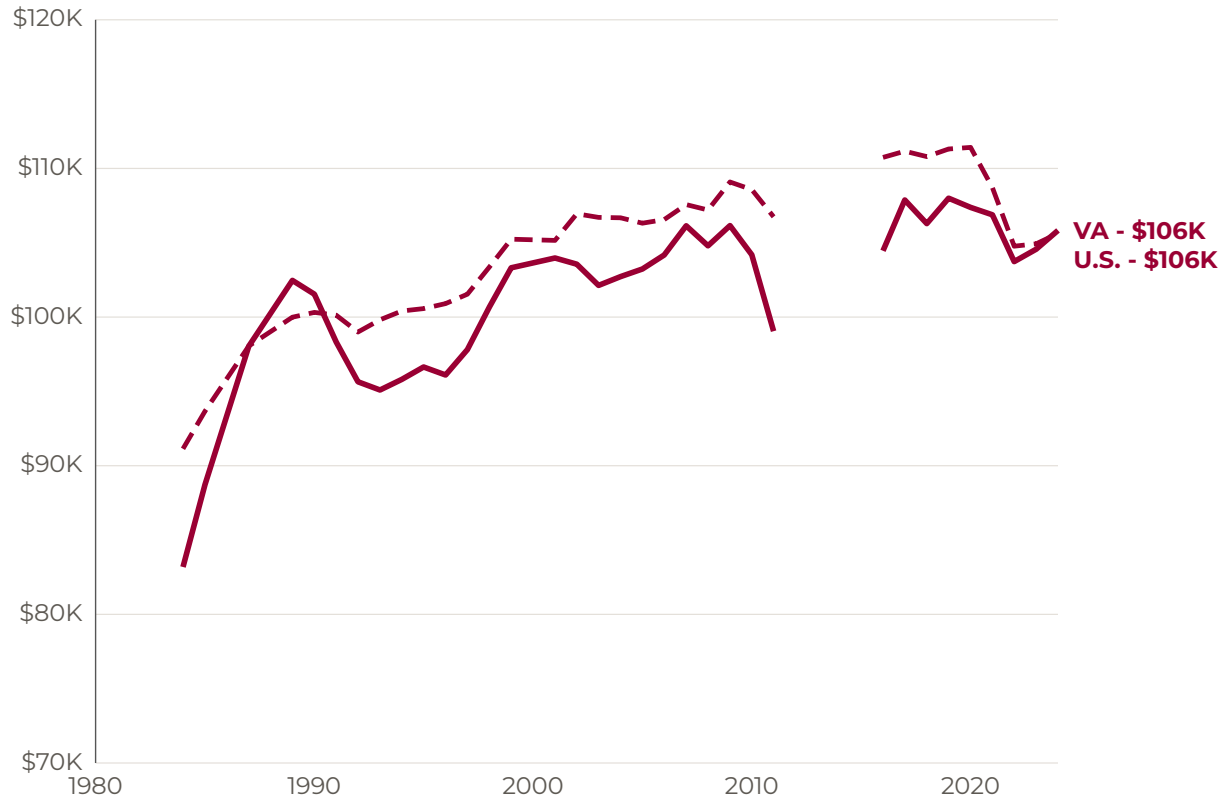
What Do Virginia's Faculty Earn, and How Many Hold Tenure?

The line follows the average salary of full-time instructional faculty on 9- or 10-month contracts at Virginia's degree-granting institutions, all ranks combined, in 2024 dollars (CPI-U). The state is solid and the United States dashed.

Average salary. The figure was $106K in 2024, against $106K for the U.S., 16th among the states, up 1.2% from $105K in 2023 and up 1.3% since 2016, when it was $104K.

Full-time instructional staff. The number was 20.3K in 2024, up 2.1% from 19.9K in 2023 and down 0.5% since 2015, when it was 20.4K.

Figure 41 Average Faculty Salary



Full-time instructional faculty, 9/10-month, 2024 dollars. Data from NCES, IPEDS.

$106K
Average faculty salary, 2024
U.S. **$106K** · **16th** of 51

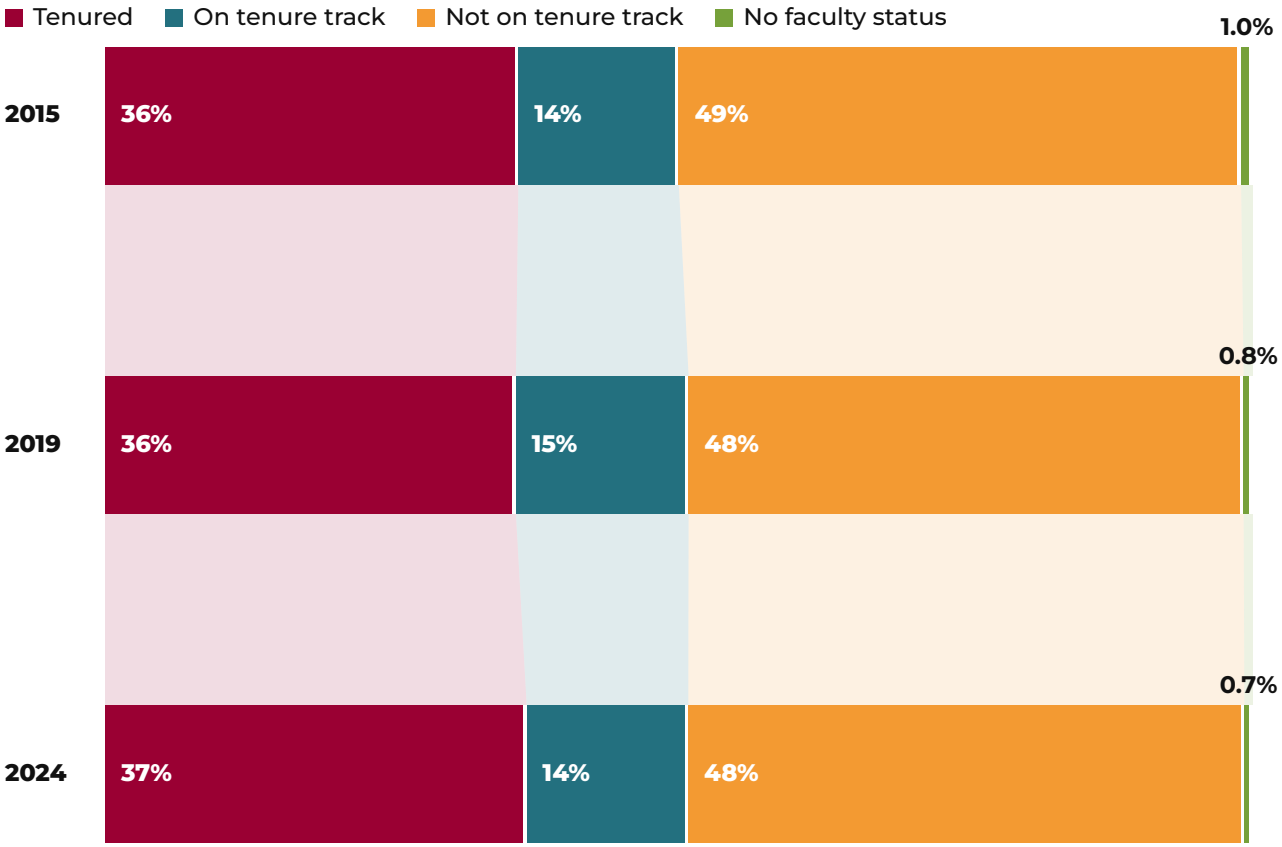
20.3K
Full-time instructional staff, 2024

37%
Tenured, 2024

The bars divide Virginia's full-time instructional staff by tenure status in 2015, 2019 and 2024: tenured, on the tenure track, not on the tenure track (for example, lecturers on fixed-term contracts), and staff at institutions with no tenure system.

Tenured, share of full-time instructional staff. The share was 37% in 2024, down 0.9 points from 38% in 2023 and up 0.7 points since 2015, when it was 36%.

Figure 42 Tenure Status



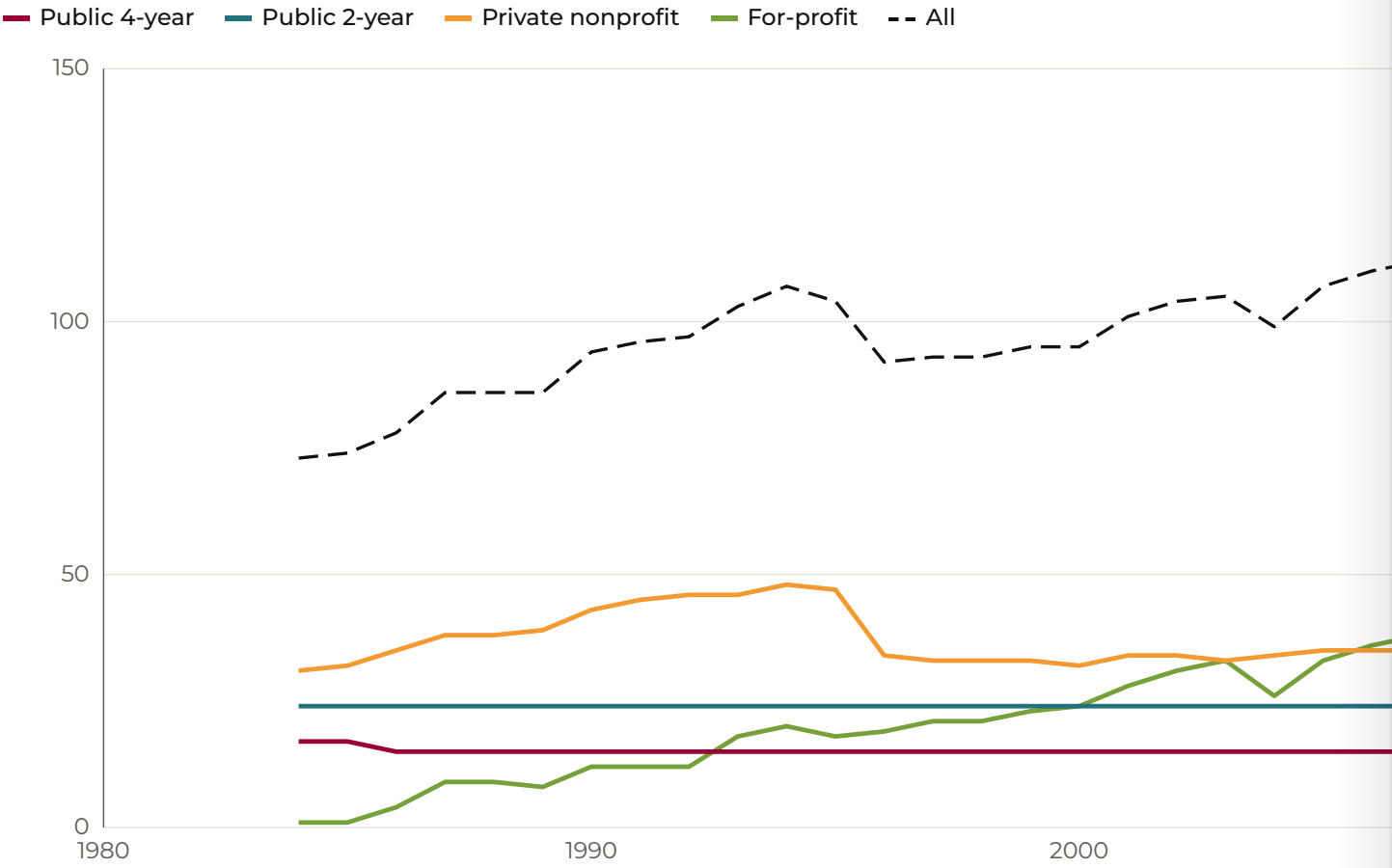
Full-time instructional staff by tenure status, 2015, 2019, 2024. Data from NCES, IPEDS.

SECTION 16

Institutions

Virginia ranks 9th of 51 on the share of students enrolled exclusively online, at 32% in 2024 against 27% for the U.S.

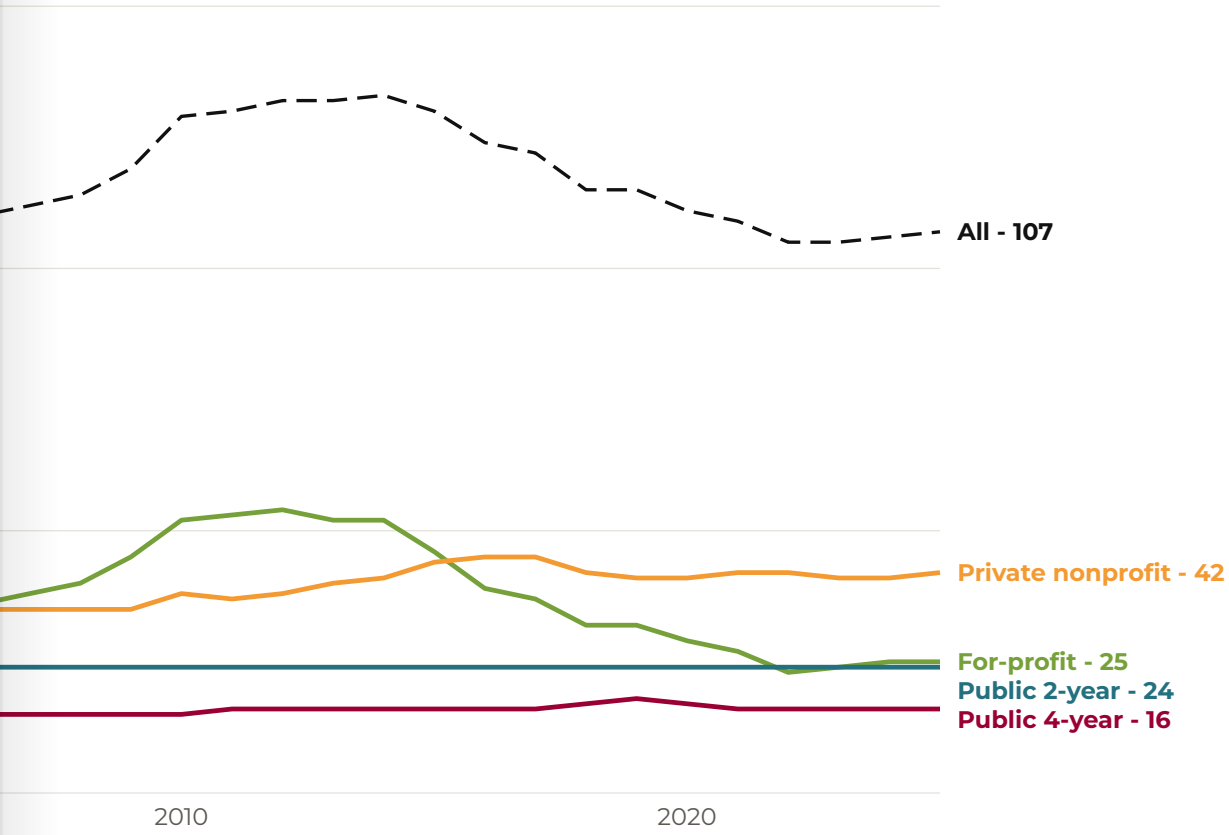
Figure 43 How Many Institutions Does Virginia Have?



Degree-granting institutions by sector. Data from NCES, IPEDS.

Virginia has 107 degree-granting institutions:16 public 4-year, 24 public 2-year, 42 private nonprofit, 25 for-profit (2025).

The number of institutions is down 18% since 2015. 67% of students attend public institutions and 4.0% attend for-profit ones.



67%

Students at public institutions, 2024

U.S. **73%** · **35th** of 51 · −1.8 pp since 2015

4.0%

Students at for-profit institutions, 2024

U.S. **5.5%** · **15th** of 51 · −3.1 pp since 2015

32%

Students fully online, 2024

U.S. **27%** · **9th** of 51 · +11.3 pp since 2015

SECTION 16 · INSTITUTIONS

Which Institutions Lead in Virginia?

The table ranks institutions in Virginia by the STEM master's degrees and research doctorates they awarded to international students in 2025, the two counted together. The share columns give international students' share of each institution's STEM degrees at that level.

Largest. George Mason University (897 international STEM master's degrees and 50 doctorates), Virginia Polytechnic Institute and State University (539 international STEM master's degrees and 190 doctorates), and Washington University of Science and Technology (268 international STEM master's degrees and – doctorates) awarded the most.

Table 7 Which Institutions Train the Most International STEM Graduates?

Institution	Intl STEM total	Intl STEM master's	Int. share	Intl STEM doctorates	Int. share
George Mason University Fairfax · Public 4-year	**947**	897	64%	50	46%
Virginia Polytechnic Institute and State University Blacksburg · Public 4-year	**729**	539	37%	190	48%
Washington University of Science and Technology Alexandria · For-profit	**268**	268	94%	–	–
University of Virginia-Main Campus Charlottesville · Public 4-year	**234**	161	29%	73	34%
Old Dominion University Norfolk · Public 4-year	**111**	82	19%	29	41%
Virginia Commonwealth University Richmond · Public 4-year	**56**	33	20%	23	31%
Marymount University Arlington · Private nonprofit	**37**	34	45%	3	7.9%

Ranked by the total (shaded) of international STEM master's degrees and research doctorates. Degrees awarded in 2025. Data from NCES, IPEDS.

107
Degree-granting institutions, 2025

67%
Students at public institutions, 2024
U.S. **73%** · **35th** of 51

4.0%
Students at for-profit institutions, 2024
U.S. **5.5%** · **15th** of 51

32%
Students fully online, 2024
U.S. **27%** · **9th** of 51

The table lists Virginia's largest degree-granting institutions by fall 2024 headcount enrollment, with the international share of graduate students, STEM bachelor's degrees awarded in 2025, the six-year graduation rate and the admission rate (a dash where the institution admits all applicants or does not report one).

Largest. The three largest were Liberty University (104K students), Northern Virginia Community College (56.5K students), and George Mason University (39.8K students).

Degree-granting institutions. The state had 107 in 2025, and 105 of them reported the enrollment and degree data in these tables.

Table 8 What Are Virginia's Largest Institutions?

Institution	Enrollment	Int. share, grad	STEM bachelor's	Six-year grad rate	Admit rate
Liberty University	**104K**	0.6%	733	67%	99%
Northern Virginia Community College	**56.5K**	–	–	–	–
George Mason University	**39.8K**	23%	1.74K	68%	87%
Virginia Polytechnic Institute and State University	**38.9K**	35%	3.31K	86%	55%
Virginia Commonwealth University	**28.5K**	9.5%	1.14K	63%	93%
University of Virginia-Main Campus	**26.4K**	19%	1.86K	96%	17%
Old Dominion University	**23.7K**	8.9%	1.03K	46%	90%
James Madison University	**22.7K**	6.0%	721	80%	72%
Tidewater Community College	**16.9K**	–	–	–	–
ECPI University	**12.6K**	0.2%	897	28%	75%

Fall 2024 enrollment; degrees 2025. Data from NCES, IPEDS.

NOTES

About These Data

International students

U.S. nonresidents on temporary visas, as IPEDS defines them. IPEDS does not record country of citizenship. Permanent residents count with U.S. citizens.

STEM

CIP families 03, 04, 11, 14, 15, 26, 27 and 40. Doctorates are research and scholarship doctorates; professional doctorates appear only in the degree-count figure. Degrees are first majors only.

Universe

Degree-granting institutions located in the state; Title IV participants from 1996. Geography is the institution's location, not the student's home. Enrollment is fall headcount.

Migration

First-time degree-seeking undergraduates by home state, reported in even years before 2001 and every year since; odd years with incomplete reporting are omitted. Net migration is students arriving from other states minus residents leaving, per 100 resident first-time students.

Money

Tuition, appropriations, revenue and faculty salaries are in 2024 dollars using the CPI-U. Research funding (HERD, Federal Funds, USAspending), wages and the NAFSA economic contribution are nominal dollars, and are labeled so. Appropriations and revenue are per full-time-equivalent student at public institutions; the tuition share divides net tuition by net tuition plus state and local appropriations.

Race and ethnicity

IPEDS did not collect race and ethnicity in every year before 2010, and categories changed in 2008–2010; “Two or more” exists only from then.

Countries of citizenship

From DHS SEVIS by the Numbers: active F-1 and M-1 student records in the month named, at every level from primary school to doctorate, by the state of the school. IPEDS records only whether a student is a U.S. nonresident, not where they come from. SEVIS counts are monthly snapshots and are not directly comparable with IPEDS fall enrollment or degree counts. Year-to-year comparisons use one fall snapshot per year (the month closest to November).

Peers

Comparison states are the four closest in total fall enrollment, measured proportionally: a state 1.5 times as large counts as near as one two-thirds the size.

Sources

IPEDS (National Center for Education Statistics) is the backbone: Institutional Characteristics, Fall Enrollment, Completions, Graduation Rates, Finance, Human Resources and Admissions, 1980–2025, harmonized into a state panel. The profile also draws on NAEP, NSF's Survey of Graduate Students and Postdoctorates, Survey of Earned Doctorates, HERD and Federal Funds surveys, USAspending.gov, DHS SEVIS by the Numbers and ICE SEVIS releases, the Department of Labor's Office of Foreign Labor Certification and BLS wage data, the USCIS H-1B Employer Data Hub, the Census Bureau's American Community Survey and Post-Secondary Employment Outcomes, USPTO PatentsView, and other federal sources; every figure names its source, and the Sources page lists them in full. Generated November 2026.

NOTES

How the Numbers Are Counted

Ranks

Ranks are among the 50 states and the District of Columbia (1 = highest value, whether or not a high value is good). Where fewer than 51 report, the count is given ("23rd of 33").

Sectors

Sectors are as IPEDS classifies institutions each year. IPEDS moves a whole institution to "public, 4-year or above" once it awards any bachelor's degree, so community colleges that began offering applied bachelor's degrees carry all of their students into the public 4-year sector. Sector series (enrollment shares, public 4-year tuition, graduation rates by sector) reflect these moves.

Research doctorates

Two counts appear. IPEDS counts degrees institutions report as "doctor's degree, research/scholarship", which includes some practice-oriented degrees such as many Ed.D.s; the NSF Survey of Earned Doctorates counts research doctorate recipients and is smaller. Each figure names which it uses. "STEM master's per STEM research doctorate" divides IPEDS STEM master's degrees by IPEDS STEM research doctorates.

Engineering

"Engineering" alone is CIP 14. Where engineering technology (CIP 15) is added, the label says "engineering & eng. technology".

Graduation rates

IPEDS graduation rates follow first-time, full-time students and count only those who finish at the institution where they started, so they leave out part-time starters and students who transfer and finish elsewhere.

International students in SEVIS

SEVIS counts active F-1 and M-1 records, which include graduates on Optional Practical Training (up to 36 months for STEM graduates, 12 for others), so STEM shares run high. STEM follows the DHS STEM Designated Degree Program List, which is broader than the eight CIP families above, and DHS draws its STEM pages on a different basis from its all-student pages, so a country's STEM count can exceed its all-level count.

Visas and green cards

DOL labor condition applications (H-1B positions) are counted by worksite and include renewals and job changes; USCIS approvals are counted by the employer's address and split into new and continuing. PERM certifications are the employer's step toward a green card.

Patents

Patents are counted by the owner's (assignee's) location, not where inventors live; a company headquartered in another state takes its patents there.

Years

Each measure uses its latest year, so years differ across a page; every figure, table and statement gives its year, and lines that stop early are labeled with their last year.

Virginia – By the Numbers

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
K-12 PREPARATION							
Grade 8 mathematics, average score	288	**275**[2024]	−12.7 ▼	272	−9.0 ▼	#22	[5]
Grade 8 mathematics: at or above Proficient	38%	**29%**[2024]	−9.1 pp ▼	27%	−5.3 pp ▼	#22	[5]
Grade 4 mathematics, average score	247	**238**[2024]	−8.7 ▼	237	−2.5 ▼	#24	[5]
Grade 4 mathematics: at or above Proficient	47%	**40%**[2024]	−7.3 pp ▼	39%	0.0 pp	#22	[5]
Grade 8 reading, average score	267	**256**[2024]	−10.6 ▼	257	−7.3 ▼	#29	[5]
Grade 8 reading: at or above Proficient	36%	**29%**[2024]	−6.9 pp ▼	29%	−4.0 pp ▼	#24	[5]
Grade 4 reading, average score	229	**214**[2024]	−14.9 ▼	214	−7.1 ▼	#30	[5]
Grade 8 science, average score, 2009 framework	–	**160**[2015]	–	153	–	#13	[5]
Current spending per public-school pupil	$14.7K	**$16.3K**[2021]	+11% ▲	$16.8K	+12% ▲	#22	[3]
Public high school 4-year graduation rate, ACGR	86%	**89%**[2022]	+3.0 pp ▲	87%	+4.0 pp ▲	#9	[3]
Public high school graduates	83.8K	**93.3K**[2024]	+11% ▲	3.49M	+8.8% ▲	#12	[3]
Average public-school teacher salary	$66.2K	**$65.8K**[2022]	−0.6% ▼	$72.9K	−3.0% ▼	#24	[3]
Pupils per teacher, public schools	14.2	**13.6**[2022]	−0.6 ▼	15.4	−0.6 ▼	#34	[3]
Public elementary and secondary enrollment	1.28M	**1.26M**[2024]	−2.0% ▼	49.4M	−2.1% ▼	#12	[3]
Public-school students eligible for free or reduced-price lunch	40%	**55%**[2023]	+14.4 pp ▲	53%	+1.5 pp ▲	#17	[3]
English learners as a share of public-school students	8.2%	**9.4%**[2022]	+1.2 pp ▲	11%	+1.0 pp ▲	#18	[3]
Charter-school share of public-school enrollment	0.1%	**0.1%**[2023]	0.0 pp	7.6%	+2.2 pp ▲	#44	[3]
High schools offering calculus	73%[2016]	**70%**[2024]	−3.0 pp ▼	51%	−5.8 pp ▼	#7	[14]
High schools offering physics	75%[2016]	**71%**[2024]	−3.8 pp ▼	64%	−2.4 pp ▼	#20	[14]
High schools offering chemistry	83%[2016]	**84%**[2024]	+1.7 pp ▲	81%	+0.7 pp ▲	#23	[14]
High schools offering Algebra II	85%[2016]	**87%**[2024]	+2.7 pp ▲	87%	+1.3 pp ▲	#32	[14]
High schools offering computer science	71%[2021]	**62%**[2024]	−9.1 pp ▼	55%	+2.0 pp ▲	#26	[14]
High schools offering any AP course	75%[2016]	**74%**[2024]	−1.3 pp ▼	56%	0.0 pp	#12	[14]
High schools offering AP computer science	46%[2021]	**40%**[2024]	−5.7 pp ▼	25%	+2.6 pp ▲	#11	[14]

Superscript years mark the year observed; changes since 2015 start from 2015 or the first later year with data. Rank is among the 50 states and DC (1 = highest). "pp" is percentage points. Dollars are 2024 dollars (CPI-U) unless the measure is labeled nominal. Counts and dollars are rounded to three significant figures (K, M, B). Arrow colors show direction only. Source numbers refer to the list of sources that follows the tables.

VIRGINIA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
K-12 PREPARATION (CONTINUED)							
High schools offering dual enrollment	74%[2016]	**79%[2024]**	+5.3 pp ▲	71%	+11.1 pp ▲	#24	[14]
AP course enrollments per 100 high-school students	22.5[2016]	**23.8[2024]**	+1.3 ▲	18.4	+1.0 ▲	#2	[14]
Calculus enrollments per 100 high-school students	5.2[2016]	**4.9[2024]**	−0.2 ▼	3.3	−1.0 ▼	#5	[14]
Physics enrollments per 100 high-school students	9.1[2016]	**7.2[2024]**	−2.0 ▼	9.4	−0.4 ▼	#31	[14]
SAT or ACT takers per 100 high-school students	18.4[2016]	**16.1[2024]**	−2.4 ▼	18.1	−2.5 ▼	#36	[14]
Students in gifted and talented programs per 100 students	12.7[2016]	**14.3[2024]**	+1.6 ▲	7.0	+0.4 ▲	#4	[14]
ENROLLMENT AND INSTITUTIONS							
Pell Grant share of undergraduates	34%	**34%[2025]**	+0.5 pp ▲	36%	−0.8 pp ▼	#26	[15]
Fall enrollment	570K	**575K[2024]**	+0.9% ▲	19.7M	−1.7% ▼	#11	[4]
Share of enrollment at public institutions	69%	**67%[2024]**	−1.8 pp ▼	73%	0.0 pp	#35	[4]
Share of enrollment at for-profit institutions	7.1%	**4.0%[2024]**	−3.1 pp ▼	5.5%	−1.2 pp ▼	#15	[4]
Share of students enrolled part-time	39%	**38%[2024]**	−1.2 pp ▼	39%	+0.7 pp ▲	#31	[4]
Share of students who are graduate students	17%	**22%[2024]**	+4.8 pp ▲	17%	+2.0 pp ▲	#9	[4]
Share of students enrolled exclusively online	20%	**32%[2024]**	+11.3 pp ▲	27%	+12.1 pp ▲	#9	[4]
DEGREES AND FIELDS OF STUDY							
Research doctorates awarded	1.53K	**1.76K[2025]**	+15% ▲	59K	+7.3% ▲	#11	[7]
Science & engineering research doctorates, excluding health	1.01K	**1.26K[2025]**	+25% ▲	44.4K	+14% ▲	#12	[7]
Share of U.S. research doctorates	2.8%	**3.0%[2025]**	+0.2 pp ▲	–	–	#11	[7]
Engineering research doctorates	305	**359[2025]**	+18% ▲	10.9K	+9.9% ▲	#10	[7]
Mathematics & computer science research doctorates	115	**130[2025]**	+13% ▲	5.07K	+33% ▲	#14	[7]
STEM master's degrees per research doctorate	3.9	**5.7[2025]**	+1.8 ▲	6.4	+2.1 ▲	#24	[4]
STEM share of bachelor's	20%	**24%[2025]**	+3.7 pp ▲	24%	+5.3 pp ▲	#21	[4]
Share of bachelor's in computer & information sciences	4.5%	**8.2%[2025]**	+3.7 pp ▲	6.9%	+3.7 pp ▲	#10	[4]
Share of bachelor's in engineering & eng. technology	6.0%	**6.4%[2025]**	+0.4 pp ▲	6.8%	+0.7 pp ▲	#26	[4]
Share of bachelor's in mathematics & statistics	1.1%	**1.0%[2025]**	−0.1 pp ▼	1.2%	0.0 pp	#23	[4]

Superscript years mark the year observed; changes since 2015 start from 2015 or the first later year with data. Rank is among the 50 states and DC (1 = highest). "pp" is percentage points. Dollars are 2024 dollars (CPI-U) unless the measure is labeled nominal. Counts and dollars are rounded to three significant figures (K, M, B). Arrow colors show direction only. Source numbers refer to the list of sources that follows the tables.

VIRGINIA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
DEGREES AND FIELDS OF STUDY (CONTINUED)							
Share of bachelor's in biological & biomedical sciences	5.8%	**5.6%**[2025]	−0.2 pp ▼	6.6%	+0.8 pp ▲	#36	[4]
Share of bachelor's in physical sciences	1.8%	**1.7%**[2025]	−0.2 pp ▼	1.3%	−0.3 pp ▼	#7	[4]
Share of bachelor's in health professions	8.5%	**8.9%**[2025]	+0.4 pp ▲	12%	+1.1 pp ▲	#44	[4]
Share of bachelor's in business	18%	**18%**[2025]	−0.4 pp ▼	19%	−0.1 pp ▼	#34	[4]
STEM share of master's	12%	**19%**[2025]	+6.5 pp ▲	23%	+6.7 pp ▲	#29	[4]
Share of master's in computer & information sciences	4.1%	**9.0%**[2025]	+4.9 pp ▲	10%	+6.3 pp ▲	#20	[4]
Share of master's in engineering & eng. technology	4.6%	**5.0%**[2025]	+0.4 pp ▲	6.7%	−0.2 pp ▼	#32	[4]
Share of master's in mathematics & statistics	0.5%	**0.9%**[2025]	+0.4 pp ▲	1.3%	+0.3 pp ▲	#28	[4]
Share of master's in biological & biomedical sciences	1.2%	**2.2%**[2025]	+1.1 pp ▲	2.5%	+0.6 pp ▲	#32	[4]
Share of master's in physical sciences	0.6%	**0.5%**[2025]	−0.1 pp ▼	0.8%	−0.1 pp ▼	#36	[4]
Share of master's in health professions	8.0%	**13%**[2025]	+4.5 pp ▲	15%	+1.7 pp ▲	#34	[4]
Share of master's in business	22%	**18%**[2025]	−3.9 pp ▼	21%	−3.2 pp ▼	#39	[4]
STEM share of research doctorates	32%	**15%**[2025]	−17.0 pp ▼	39%	−2.2 pp ▼	#50	[4]
Share of research doctorates in computer & information sciences	2.7%	**2.0%**[2025]	−0.7 pp ▼	4.2%	+1.4 pp ▲	#37	[4]
Share of research doctorates in engineering & eng. technology	14%	**6.2%**[2025]	−7.5 pp ▼	14%	−1.0 pp ▼	#48	[4]
Share of research doctorates in mathematics & statistics	1.6%	**0.7%**[2025]	−0.9 pp ▼	2.4%	−0.2 pp ▼	#46	[4]
Share of research doctorates in biological & biomedical sciences	6.1%	**3.2%**[2025]	−2.9 pp ▼	10%	−1.1 pp ▼	#51	[4]
Share of research doctorates in physical sciences	5.8%	**2.1%**[2025]	−3.7 pp ▼	7.1%	−1.1 pp ▼	#50	[4]
Share of research doctorates in health professions	15%	**9.7%**[2025]	−4.9 pp ▼	12%	+3.5 pp ▲	#27	[4]
Share of research doctorates in business	7.1%	**16%**[2025]	+9.1 pp ▲	5.6%	+1.3 pp ▲	#2	[4]
Bachelor's degrees per 100 undergraduates	11.8	**13.4**[2025]	+1.6 ▲	12.3	+1.4 ▲	#17	[4]
INTERNATIONAL STUDENTS							
First-time students from outside the U.S.	2.2%[2016]	**2.0%**[2024]	−0.2 pp ▼	3.1%	+0.1 pp ▲	#35	[4]
STEM master's degrees earned by international students	671	**2.07K**[2025]	3.09× ▲	117K	2.18× ▲	#19	[4]
Share of all U.S. international STEM master's degrees	1.3%	**1.8%**[2025]	+0.5 pp ▲	–	–	#19	[4]

Superscript years mark the year observed; changes since 2015 start from 2015 or the first later year with data. Rank is among the 50 states and DC (1 = highest). "pp" is percentage points. Dollars are 2024 dollars (CPI-U) unless the measure is labeled nominal. Counts and dollars are rounded to three significant figures (K, M, B). Arrow colors show direction only. Source numbers refer to the list of sources that follows the tables.

VIRGINIA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
INTERNATIONAL STUDENTS (CONTINUED)							
International share of undergraduate enrollment	2.3%	**2.7%**[2024]	+0.4 pp ▲	3.0%	−0.3 pp ▼	#24	[4]
International share of graduate enrollment	7.7%	**8.4%**[2024]	+0.7 pp ▲	17%	+2.8 pp ▲	#42	[4]
International share of all bachelor's	2.2%	**2.7%**[2025]	+0.4 pp ▲	4.1%	+0.1 pp ▲	#35	[4]
International share of STEM bachelor's	3.3%	**4.0%**[2025]	+0.7 pp ▲	6.6%	+0.8 pp ▲	#37	[4]
International share of all master's	6.4%	**12%**[2025]	+5.7 pp ▲	22%	+6.2 pp ▲	#32	[4]
International share of STEM master's	23%	**39%**[2025]	+16.4 pp ▲	52%	+9.1 pp ▲	#31	[4]
International share of computer & information sciences master's	28%	**59%**[2025]	+30.9 pp ▲	65%	+10.1 pp ▲	#29	[4]
International share of engineering & eng. technology master's	24%	**28%**[2025]	+4.4 pp ▲	50%	+0.9 pp ▲	#39	[4]
International share of mathematics & statistics master's	34%	**22%**[2025]	−11.9 pp ▼	55%	+6.1 pp ▲	#37	[4]
International share of biological & biomedical sciences master's	8.8%	**6.8%**[2025]	−2.0 pp ▼	23%	+6.2 pp ▲	#45	[4]
International share of physical sciences master's	21%	**29%**[2025]	+8.7 pp ▲	33%	+3.6 pp ▲	#23	[4]
International share of all research doctorates	21%	**9.6%**[2025]	−11.1 pp ▼	25%	−1.0 pp ▼	#49	[4]
International share of STEM research doctorates	48%	**41%**[2025]	−7.3 pp ▼	43%	−0.5 pp ▼	#26	[4]
International share of computer & information sciences research doctorates	60%	**56%**[2025]	−4.8 pp ▼	56%	+2.0 pp ▲	#16	[4]
International share of engineering & eng. technology research doctorates	56%	**49%**[2025]	−7.1 pp ▼	55%	−1.0 pp ▼	#31	[4]
International share of mathematics & statistics research doctorates	53%	**51%**[2025]	−1.5 pp ▼	53%	+2.9 pp ▲	#15	[4]
International share of biological & biomedical sciences research doctorates	26%	**18%**[2025]	−8.3 pp ▼	23%	−4.3 pp ▼	#33	[4]
International share of physical sciences research doctorates	46%	**33%**[2025]	−12.8 pp ▼	38%	−1.1 pp ▼	#30	[4]
Active F-1/M-1 students, all levels	26.4K	**29.8K**[2026]	+13% ▲	1.19M	−0.1% ▼	#13	[20]
Share of U.S. active F-1/M-1 students	2.2%	**2.5%**[2026]	+0.3 pp ▲	–	–	#13	[20]
Share of F-1/M-1 students from India	20%	**22%**[2026]	+1.6 pp ▲	27%	+11.4 pp ▲	#25	[20]
Share of F-1/M-1 students from China	25%	**14%**[2026]	−10.9 pp ▼	18%	−11.7 pp ▼	#18	[20]
Share from the two largest countries of origin	45%	**36%**[2026]	−9.3 pp ▼	45%	−0.3 pp ▼	#29	[20]

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VIRGINIA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
INTERNATIONAL STUDENTS (CONTINUED)							
Countries of citizenship represented	180	**182**[2026]	+1.1% ▲	219	−2.7% ▼	#11	[20]
Active STEM F-1/M-1 students	10.8K	**19.6K**[2026]	+81% ▲	749K	+61% ▲	#13	[20]
Active STEM master's students	4.99K	**10.4K**[2026]	2.09× ▲	393K	+90% ▲	#12	[20]
Active STEM doctoral students	2.57K	**4.59K**[2026]	+78% ▲	158K	+61% ▲	#14	[20]
Share of STEM master's students from India	76%	**56%**[2026]	−19.4 pp ▼	58%	−1.8 pp ▼	#28	[20]
U.S. students studying abroad for credit, from the state's institutions	12.4K[2017]	**11.4K**[2024]	−7.7% ▼	–	–	#10	[1]
Economic contribution of international students, nominal dollars	$758M[2018]	**$894M**[2024]	+18% ▲	$42.9B	+4.6% ▲	#17	[2]
Jobs supported by international students' spending	9.14K[2018]	**7.8K**[2024]	−15% ▼	356K	−22% ▼	#17	[2]
Share of U.S. student and exchange-visitor admissions, by destination	1.8%	**1.8%**[2024]	0.0 pp	–	–	#15	[16]
Share of U.S. F-1 student admissions, by destination	1.6%	**1.5%**[2024]	−0.1 pp ▼	–	–	#14	[16]
Share of U.S. J-1 exchange-visitor admissions, by destination	2.4%	**2.4%**[2024]	0.0 pp	–	–	#13	[16]
J-1 exchange visitors starting a program in the state	11K	**9.06K**[2025]	−17% ▼	275K	−14% ▼	#9	[18]
GRADUATE EDUCATION AND RESEARCH							
Graduate students in science & engineering	15.2K	**14.9K**[2016]	−1.7% ▼	620K	+0.4% ▲	#15	[9]
S&E graduate students on temporary visas	4.2K	**4.38K**[2016]	+4.2% ▲	241K	+2.3% ▲	#17	[9]
Temporary-visa share of S&E graduate students	28%	**29%**[2016]	+1.7 pp ▲	39%	+0.7 pp ▲	#34	[9]
Temporary-visa share of engineering graduate students	41%	**44%**[2016]	+2.4 pp ▲	57%	0.0 pp	#35	[9]
S&E postdoctoral appointees	677	**665**[2016]	−1.8% ▼	45.7K	+1.0% ▲	#20	[9]
Temporary-visa share of S&E postdocs	57%	**55%**[2016]	−1.8 pp ▼	56%	−0.9 pp ▼	#26	[9]
University R&D spending, nominal dollars	$1.41B	**$2.55B**[2024]	+81% ▲	$118B	+71% ▲	#13	[6]
Share of U.S. university R&D	2.1%	**2.2%**[2024]	+0.1 pp ▲	–	–	#13	[6]
Federally funded share of university R&D	52%	**47%**[2024]	−5.8 pp ▼	55%	−0.2 pp ▼	#41	[6]
Business-funded share of university R&D	4.4%	**4.0%**[2024]	−0.3 pp ▼	5.4%	−0.4 pp ▼	#28	[6]
State- and locally funded share of university R&D	5.2%	**7.1%**[2024]	+1.9 pp ▲	5.2%	−0.4 pp ▼	#16	[6]
University R&D in mathematics and computer science, nominal dollars	$58.3M	**$138M**[2024]	2.36× ▲	$5.12B	+97% ▲	#11	[6]

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VIRGINIA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
GRADUATE EDUCATION AND RESEARCH (CONTINUED)							
University R&D in engineering, nominal dollars	$337M	**$546M[2024]**	+62% ▲	$19.3B	+74% ▲	#11	[6]
University R&D personnel, headcount	21.6K	**28.5K[2024]**	+32% ▲	1.09M	+20% ▲	#14	[6]
Federal R&D obligations to the state, nominal dollars	$7.49B	**$15.1B[2024]**	2.02× ▲	$193B	+53% ▲	#3	[8]
Federal R&D obligations to universities in the state, nominal dollars	$458M	**$793M[2024]**	+73% ▲	$44.4B	+64% ▲	#19	[8]
Defense R&D obligations to the state, nominal dollars	$5.78B	**$12.1B[2024]**	2.10× ▲	$97.2B	+60% ▲	#1	[8]
Health (HHS/NIH) R&D obligations to the state, nominal dollars	$364M	**$649M[2024]**	+78% ▲	$48.5B	+62% ▲	#19	[8]
NSF, NIH, NASA and DOE Science grants performed in the state, nominal dollars	$479M	**$808M[2025]**	+69% ▲	$45B	+47% ▲	#18	[19]
Federal R&D contracts performed in the state, nominal dollars	$3.8B	**$6.32B[2026]**	+2,520.0 ▲	$59.3B	+22,105.9 ▲	#2	[19]
COLLEGE COSTS AND OUTCOMES							
Median earnings 10 years after entering college	$43K[2020]	**$44.8K[2022]**	+4.1% ▲	$47.5K	+11% ▲	#36	[15]
Median earnings 1 year after completing	–	**$46.3K[2025]**	–	$47.5K	–	#34	[15]
Median debt of completers, nominal dollars	$20.8K	**$19.5K[2024]**	−6.1% ▼	$17.6K	−4.6% ▼	#18	[15]
STEM bachelor's graduates: median earnings 1 year out	–	**$66.8K[2025]**	–	$68.2K	–	#26	[15]
Non-STEM bachelor's graduates: median earnings 1 year out	–	**$55.5K[2025]**	–	$51.9K	–	#11	[15]
Bachelor's graduates: median earnings 5 years out	–	**$68.8K[2025]**	–	$68.1K	–	#17	[15]
STEM master's graduates: median earnings 1 year out	–	**$86.6K[2025]**	–	$91.6K	–	#25	[15]
Six-year bachelor's graduation rate, 4-year institutions	66%	**68%[2024]**	+2.4 pp ▲	65%	+5.3 pp ▲	#11	[4]
Completion within 150% of normal time, 2-year institutions	30%	**37%[2024]**	+7.0 pp ▲	41%	+12.1 pp ▲	#29	[4]
In-state tuition & fees, public 4-year	$15.6K	**$15.7K[2023]**	+0.7% ▲	$9.71K	−14% ▼	#6	[4]
State & local appropriations per FTE, public institutions	$7.47K	**$11.3K[2024]**	+51% ▲	$11.4K	+22% ▲	#22	[4]
Net tuition share of tuition plus state and local appropriations, public institutions	62%	**50%[2024]**	−11.5 pp ▼	43%	−5.9 pp ▼	#19	[4]
Average full-time faculty salary (9/10-month)	$104K[2016]	**$106K[2024]**	+1.3% ▲	$106K	−4.6% ▼	#16	[4]
WHERE STUDENTS AND GRADUATES GO							

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VIRGINIA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
WHERE STUDENTS AND GRADUATES GO (CONTINUED)							
International graduates on OPT working in the state where they studied	38%	**35%**[2016]	−3.0 pp ▼	44%	−1.8 pp ▼	#26	[21]
OPT retention, bachelor's graduates	51%	**44%**[2016]	−7.4 pp ▼	58%	+0.3 pp ▲	#27	[21]
OPT retention, master's graduates	34%	**31%**[2016]	−2.6 pp ▼	39%	−2.5 pp ▼	#27	[21]
OPT retention, STEM graduates	31%	**31%**[2016]	−0.5 pp ▼	37%	−1.9 pp ▼	#25	[21]
OPT workers employed in the state per OPT graduate of its schools	1.43	**1.45**[2016]	0.0	1.00	0.0	#5	[21]
STEM OPT extensions approved, by state of the school	619	**869**[2016]	+40% ▲	41.2K	+51% ▲	#15	[21]
Bachelor's graduates working in the state 1 year after graduating (PSEO)	62%[2016]	**61%**[2019]	−1.5 pp ▼	72%	−0.1 pp ▼	#21	[11]
Bachelor's graduates working in the state 5 years after graduating (PSEO)	–	**53%**[2016]	–	65%	–	#25	[11]
Bachelor's graduates working in the state 10 years after graduating (PSEO)	–	**50%**[2010]	–	64%	–	#28	[11]
STEM bachelor's graduates working in the state 1 year out (PSEO)	62%[2016]	**62%**[2019]	+0.1 pp ▲	67%	+1.5 pp ▲	#19	[11]
Master's graduates working in the state 1 year out (PSEO)	–	**52%**[2016]	–	66%	–	#28	[11]
Doctoral graduates working in the state 1 year out (PSEO)	–	**36%**[2016]	–	49%	–	#32	[11]
Bachelor's graduates' median earnings 1 year out (PSEO)	$47K[2016]	**$48.8K**[2019]	+3.8% ▲	$48.1K	+2.3% ▲	#13	[11]
Bachelor's graduates' median earnings 5 years out (PSEO)	–	**$68.4K**[2016]	–	$67.3K	–	#11	[11]
STEM bachelor's graduates' median earnings 1 year out (PSEO)	$59.2K[2016]	**$61.6K**[2019]	+4.2% ▲	$59.3K	+0.1% ▲	#7	[11]
Net domestic migration of bachelor's-plus adults, per 1,000 residents	−1.1	**+5.1**[2025]	+6.2 ▲	–	–	#12	[12]
Net domestic migration of graduate-degree holders	−1.2K	**+6.35K**[2025]	+7,546.0 ▲	–	–	#4	[12]
Bachelor's-plus adults who moved in from another state	4.0%	**3.5%**[2025]	−0.5 pp ▼	–	–	#14	[12]
Bachelor's-plus adults who moved out to another state	4.1%	**3.0%**[2025]	−1.1 pp ▼	–	–	#18	[12]
Bachelor's-plus adults who arrived from abroad in the past year	21.9K	**26.4K**[2025]	+21% ▲	–	–	#6	[12]
First-time students who are in-state residents	76%[2016]	**75%**[2024]	−1.5 pp ▼	–	–	#20	[4]
Net student migration (per 100 resident first-time students)	+3[2016]	**+2**[2024]	−1.7 ▼	–	–	#32	[4]

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VIRGINIA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
WHERE STUDENTS AND GRADUATES GO (CONTINUED)							
Recent high school graduates who go to college out of state	18%[2016]	**21%[2022]**	+3.0 pp ▲	22%	+2.5 pp ▲	#30	[3]
WORK VISAS AND GREEN CARDS							
Certified H-1B positions, by worksite	33.3K	**22.4K[2025]**	−33% ▼	914K	−24% ▼	#11	[17]
Share of U.S. certified H-1B positions	2.8%	**2.4%[2025]**	−0.3 pp ▼	–	–	#11	[17]
STEM share of certified H-1B positions	84%	**83%[2025]**	−1.8 pp ▼	76%	−7.8 pp ▼	#8	[17]
Median offered H-1B wage, nominal dollars, nominal dollars	$70K	**$115K[2025]**	+64% ▲	$119K	+66% ▲	#10	[17]
H-1B positions at H-1B-dependent employers	47%	**20%[2025]**	−26.7 pp ▼	22%	−25.7 pp ▼	#23	[17]
Green-card labor certifications	2.26K	**3.46K[2025]**	+53% ▲	138K	+75% ▲	#11	[17]
Share of PERM certifications for citizens of India	61%	**56%[2024]**	−4.5 pp ▼	47%	−10.6 pp ▼	#8	[17]
PERM certifications for workers on F-1 student visas	4.9%	**10%[2024]**	+5.4 pp ▲	8.7%	+4.9 pp ▲	#23	[17]
PERM workers with a master's or doctorate	52%	**57%[2024]**	+4.5 pp ▲	47%	−4.9 pp ▼	#7	[17]
STEM share of PERM certifications	72%	**54%[2025]**	−18.3 pp ▼	49%	−25.8 pp ▼	#10	[17]
Median offered PERM wage, nominal dollars, nominal dollars	$90.8K	**$100K[2025]**	+10% ▲	$100K	+9.9% ▲	#13	[17]
New lawful permanent residents (green cards), by state of residence	27.6K	**37.9K[2024]**	+37% ▲	1.36M	+30% ▲	#7	[16]
Share of U.S. new green-card holders	2.6%	**2.8%[2024]**	+0.1 pp ▲	–	–	#7	[16]
People naturalized as U.S. citizens, by state of residence	18.4K	**24.8K[2024]**	+35% ▲	819K	+12% ▲	#7	[16]
Share of U.S. naturalizations	2.5%	**3.0%[2024]**	+0.5 pp ▲	–	–	#7	[16]
Share of U.S. temporary-worker admissions, by destination	1.1%	**1.0%[2024]**	−0.2 pp ▼	–	–	#15	[16]
Share of U.S. H-1B worker admissions, by destination	2.2%	**1.7%[2024]**	−0.4 pp ▼	–	–	#13	[16]
Share of U.S. L-1 intracompany-transferee admissions, by destination	1.1%	**1.0%[2024]**	−0.1 pp ▼	–	–	#16	[16]
Share of U.S. O-1 extraordinary-ability worker admissions, by destination	0.4%	**0.5%[2024]**	+0.1 pp ▲	–	–	#21	[16]
New H-1B approvals (initial), by employer state	3.13K	**9.42K[2026]**	3.01× ▲	127K	+14% ▲	#4	[13]
Continuing H-1B approvals, by employer state	5.79K	**18.5K[2026]**	3.20× ▲	268K	+52% ▲	#5	[13]
Share of U.S. new H-1B approvals	2.8%	**7.4%[2026]**	+4.6 pp ▲	–	–	#4	[13]

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VIRGINIA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
WORK VISAS AND GREEN CARDS (CONTINUED)							
Denial rate, new H-1B petitions	12%	**2.1%**[2026]	−9.6 pp ▼	3.4%	−3.0 pp ▼	#47	[13]
New H-1B approvals at colleges and schools (NAICS 61)	8.7%	**3.1%**[2026]	−5.6 pp ▼	8.6%	−1.7 pp ▼	#50	[13]
New H-1B approvals in professional, scientific & technical services	71%	**20%**[2026]	−51.0 pp ▼	44%	−12.5 pp ▼	#37	[13]
Five largest employers' share of new H-1B approvals	15%	**66%**[2026]	+51.3 pp ▲	9.1%	−6.7 pp ▼	#4	[13]
STEM WORKFORCE AND INNOVATION							
Workers in STEM occupations	361K	**505K**[2024]	+40% ▲	13.7M	+51% ▲	#7	[12]
STEM workers as a share of all workers	8.6%	**11%**[2024]	+2.6 pp ▲	8.1%	+2.1 pp ▲	#4	[12]
Foreign-born share of STEM workers	21%	**22%**[2024]	+1.2 pp ▲	24%	+1.6 pp ▲	#13	[12]
Foreign-born share of STEM workers with a graduate degree	30%	**33%**[2024]	+3.1 pp ▲	41%	+2.2 pp ▲	#19	[12]
Foreign-born share of STEM workers with a doctorate	36%	**31%**[2024]	−4.8 pp ▼	47%	+0.9 pp ▲	#18	[12]
Noncitizen share of STEM workers	7.7%	**7.5%**[2024]	−0.2 pp ▼	11%	+0.7 pp ▲	#23	[12]
STEM workers born abroad who arrived in the last 10 years	4.9%	**4.7%**[2024]	−0.2 pp ▼	7.5%	+0.3 pp ▲	#27	[12]
Share of foreign-born STEM workers born in India	31%	**31%**[2024]	−0.5 pp ▼	28%	+0.7 pp ▲	#15	[12]
Foreign-born share of residents	12%	**14%**[2025]	+1.8 pp ▲	15%	+1.3 pp ▲	#15	[12]
Foreign-born share of graduate-degree holders, 25+	18%	**21%**[2025]	+3.3 pp ▲	21%	+2.8 pp ▲	#11	[12]
STEM jobs, employer survey	319K	**391K**[2025]	+23% ▲	10.4M	+26% ▲	#5	[10]
STEM jobs concentration vs the U.S., location quotient	1.46	**1.43**[2025]	0.0	1.00	0.0	#5	[10]
Mean annual STEM wage, nominal, nominal dollars	$96.4K	**$127K**[2025]	+32% ▲	$120K	+38% ▲	#8	[10]
Median software developer wage, nominal, nominal dollars	$111K[2019]	**$136K**[2025]	+23% ▲	$136K	+26% ▲	#10	[10]
Software developer jobs	61.6K	**88.3K**[2025]	+43% ▲	1.69M	+48% ▲	#5	[10]
Mean annual wage, all occupations, nominal, nominal dollars	$51.7K	**$74K**[2025]	+43% ▲	$69.8K	+44% ▲	#11	[10]
Patents granted to assignees in the state	1.34K	**3.96K**[2025]	2.94× ▲	135K	−0.6% ▼	#9	[22]
Share of U.S.-assignee patents	1.0%	**2.9%**[2025]	+1.9 pp ▲	–	–	#9	[22]
Share of the state's patents in digital technologies	40%	**55%**[2025]	+15.0 pp ▲	43%	+1.6 pp ▲	#6	[22]
Share of the state's patents in life sciences	13%	**6.0%**[2025]	−6.8 pp ▼	17%	+0.4 pp ▲	#50	[22]

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VIRGINIA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
STEM WORKFORCE AND INNOVATION (CONTINUED)							
Share of the state's patents in semiconductors	1.8%	**1.3%**[2025]	−0.5 pp ▼	2.9%	−1.7 pp ▼	#16	[22]
Women's share of inventors on the state's patents	11%	**12%**[2025]	+1.3 pp ▲	14%	+3.4 pp ▲	#22	[22]

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Sources

[1] Institute of International Education. *Open Doors Report on International Educational Exchange: State Fact Sheets.* Editions 2020–2026. New York: Institute of International Education. https://opendoorsdata.org/. Accessed November 2026.

[2] NAFSA: Association of International Educators. *NAFSA International Student Economic Value Tool*. Academic years 2018–19 to 2024–25. Washington, DC: NAFSA. https://www.nafsa.org/isev. Accessed November 2026.

[3] National Center for Education Statistics. *Digest of Education Statistics*. Editions 1995–2025. Washington, DC: U.S. Department of Education, Institute of Education Sciences. https://nces.ed.gov/programs/digest/. Accessed November 2026.

[4] National Center for Education Statistics. *Integrated Postsecondary Education Data System (IPEDS)*. Survey files, 1980–2025. Washington, DC: U.S. Department of Education, Institute of Education Sciences. https://nces.ed.gov/ipeds/. Accessed November 2026.

[5] National Center for Education Statistics. *National Assessment of Educational Progress (NAEP)*. NAEP Data Service, 1990–2024. Washington, DC: U.S. Department of Education, Institute of Education Sciences. https://www.nationsreportcard.gov/. Accessed November 2026.

[6] National Center for Science and Engineering Statistics. *Higher Education Research and Development Survey*. Data tables, FY2003–2024. Alexandria, VA: National Science Foundation. https://ncses.nsf.gov/surveys/higher-education-research-development. Accessed November 2026.

[7] National Center for Science and Engineering Statistics. *Survey of Earned Doctorates*. Data tables, 1994–2025. Alexandria, VA: National Science Foundation. https://ncses.nsf.gov/surveys/earned-doctorates. Accessed November 2026.

[8] National Center for Science and Engineering Statistics. *Survey of Federal Funds for Research and Development*. Data tables, FY1985–2024. Alexandria, VA: National Science Foundation. https://ncses.nsf.gov/surveys/federal-funds-research-development. Accessed November 2026.

[9] National Center for Science and Engineering Statistics. *Survey of Graduate Students and Postdoctorates in Science and Engineering*. Data tables, 1994–2024. Alexandria, VA: National Science Foundation. https://ncses.nsf.gov/surveys/graduate-students-postdoctorates-s-e. Accessed November 2026.

[10] U.S. Bureau of Labor Statistics. *Occupational Employment and Wage Statistics*. State estimates, 1999–2025. Washington, DC: U.S. Department of Labor. https://www.bls.gov/oes/. Accessed November 2026.

[11] U.S. Census Bureau, Center for Economic Studies. *Post-Secondary Employment Outcomes (PSEO)*. Release R2025. Washington, DC: U.S. Department of Commerce. https://lehd.ces.census.gov/data/pseo_experimental.html. Accessed November 2026.

[12] U.S. Census Bureau. *American Community Survey*. 1-year estimates and Public Use Microdata Sample, 2010–2025. Washington, DC: U.S. Department of Commerce. https://data.census.gov/. Accessed November 2026.

[13] U.S. Citizenship and Immigration Services. *H-1B Employer Data Hub*. Fiscal years 2009–2026. Washington, DC: U.S. Department of Homeland Security. https://www.uscis.gov/tools/reports-and-studies/h-1b-employer-data-hub. Accessed November 2026.

[14] U.S. Department of Education, Office for Civil Rights. *Civil Rights Data Collection*. School years 1980–81 to 2023–24. Washington, DC: U.S. Department of Education. https://civilrightsdata.ed.gov/. Accessed November 2026.

[15] U.S. Department of Education. *College Scorecard*. Institution and field-of-study data, 1996–2025. Washington, DC: U.S. Department of Education. https://collegescorecard.ed.gov/data/. Accessed November 2026.

[16] U.S. Department of Homeland Security, Office of Homeland Security Statistics. *Yearbook of Immigration Statistics*. Fiscal years 1996–2024. Washington, DC: U.S. Department of Homeland Security. https://ohss.dhs.gov/topics/immigration/yearbook. Accessed November 2026.

[17] U.S. Department of Labor, Employment and Training Administration, Office of Foreign Labor Certification. *LCA and PERM Disclosure Data*. Fiscal years 2008–2026. Washington, DC: U.S. Department of Labor. https://www.dol.gov/agencies/eta/foreign-labor/performance. Accessed November 2026.

[18] U.S. Department of State, Bureau of Educational and Cultural Affairs. *BridgeUSA: Exchange Visitor Program Participants by State*. 2015–2025. Washington, DC: U.S. Department of State. https://j1visa.state.gov/. Accessed November 2026.

[19] U.S. Department of the Treasury, Bureau of the Fiscal Service. *USAspending.gov*. Award data, FY2008–2026. Washington, DC: U.S. Department of the Treasury. https://www.usaspending.gov/. Accessed November 2026.

[20] U.S. Immigration and Customs Enforcement, Student and Exchange Visitor Program. *SEVIS by the Numbers: Data Mapping Tool*. Monthly data, 2014–2026. Washington, DC: U.S. Department of Homeland Security. https://studyinthestates.dhs.gov/sevis-data-mapping-tool. Accessed November 2026.

[21] U.S. Immigration and Customs Enforcement. *Student and Exchange Visitor Information System (SEVIS) Data*. Freedom of Information Act releases, 2008–2016. Washington, DC: U.S. Department of Homeland Security. https://www.ice.gov/foia/library. Accessed November 2026.

[22] U.S. Patent and Trademark Office. *PatentsView*. Annualized patent data, grant years 1976–2025. Alexandria, VA: U.S. Patent and Trademark Office. https://patentsview.org/. Accessed November 2026.

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